

Math Guide

Mathematical foundations for Halo 2 and Zcash Orchard

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Abstract

This is the first volume of a four-part series — *Math Guide*, *Crypto Guide*, *Halo 2 Guide*, *Orchard Guide* — developing the cryptography of Zcash’s Orchard protocol from the ground up. Assuming only elementary algebra and calculus, it builds the mathematics the later volumes rely on: sets and modular arithmetic; groups, rings, and fields; finite fields and the number theory of the discrete logarithm; polynomials, roots of unity, and evaluation domains; linear algebra and inner products; elliptic curves and their group law; and the probability, asymptotics, and model of computation needed to state cryptographic claims precisely.

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1 Notation: sets, functions, and relations

This short section fixes the small amount of set-theoretic language we use throughout the series. A reader comfortable with elementary mathematics may skim it and refer back as needed.

1.1 Sets

A *set* is a collection of distinct objects, its *elements*; one writes $x \in A$ for membership and $x \notin A$ for its negation. A set may be given by listing, $\{a, b, c\}$, or by *set-builder* notation $\{x \in A : P(x)\}$ (“the x in A such that $P(x)$ ”). One writes $A \subseteq B$ when every element of A lies in B , and $A = B$ when $A \subseteq B$ and $B \subseteq A$. The usual operations are union $A \cup B$, intersection $A \cap B$, difference $A \setminus B$, and the empty set $\emptyset$. The *Cartesian product* is $A \times B = \{(a, b) : a \in A, b \in B\}$, and A^n is the set of n -tuples over A . For a finite set A , $|A|$ denotes its number of elements. The standard number systems are the naturals $\mathbb{N}$, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$, with $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; we construct the prime field $\mathbb{F}_p$ and the general finite field $\mathbb{F}_q$ in §5.

1.2 Functions

A *function* $f: A \rightarrow B$ assigns to each $a \in A$ a unique value $f(a) \in B$; A is the *domain* and B the *codomain*. It is *injective* (one-to-one) if $f(a) = f(a')$ implies $a = a'$; *surjective* (onto) if every $b \in B$ equals $f(a)$ for some a ; and *bijective* if both. The *composition* of $f: A \rightarrow B$ and $g: B \rightarrow C$ is $(g \circ f)(a) = g(f(a))$. A function has a two-sided *inverse* $f^{-1}: B \rightarrow A$ (with $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$) if and only if it is bijective. For $S \subseteq A$ the *image* is $f(S) = \{f(a) : a \in S\}$, and for $T \subseteq B$ the *preimage* is $f^{-1}(T) = \{a \in A : f(a) \in T\}$ (defined whether or not f is invertible).

1.3 Relations and equivalences

A *binary relation* on A is a subset $R \subseteq A \times A$; one writes $a \sim b$ for $(a, b) \in R$. It is an *equivalence relation* if it is *reflexive* ($a \sim a$), *symmetric* ($a \sim b \Rightarrow b \sim a$), and *transitive* ($a \sim b$ and $b \sim c$ imply $a \sim c$). The *equivalence class* of a is $[a] = \{x \in A : x \sim a\}$; the distinct classes *partition* A into disjoint nonempty pieces whose union is A , and the set of classes is the *quotient* $A/\sim$. We use this construction twice: to form the integers modulo n (§2) and the cosets of a subgroup (§3).

1.4 Logic and proof conventions

We employ the symbols $\forall$ (“for all”), $\exists$ (“there exists”), $\exists!$ (“there exists a unique”), and “:” or “s.t.” for “such that”. Negation swaps the quantifiers: $\neg(\forall x P(x))$ is $\exists x \neg P(x)$, and $\neg(\exists x P(x))$ is $\forall x \neg P(x)$. The order of unlike quantifiers matters: $\forall x \exists y P(x, y)$ (the y may depend on x) is weaker than $\exists y \forall x P(x, y)$ (one y works for all). We prove an implication $P \Rightarrow Q$ directly (assume P , derive Q), by contraposition (assume $\neg Q$, derive $\neg P$), or by contradiction; we prove a statement $\forall x P(x)$ by taking an arbitrary x , and $\exists x P(x)$ by exhibiting a witness. We prove statements about all natural numbers by induction.

2 The integers and modular arithmetic

The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ are the arithmetical bedrock on which every algebraic and cryptographic structure in this monograph ultimately rests. The finite fields underlying elliptic curves, the scalar arithmetic of Halo 2, and the polynomial commitments of Orchard all reduce, in the end, to computations with congruence classes of integers. This section develops the theory of the integers from a small set of assumptions and arrives at the ring $\mathbb{Z}/n\mathbb{Z}$ of integers modulo n , its group of units, and the algorithms that make arithmetic in it effective. We keep the treatment constructive throughout: every existence statement comes, wherever possible, with

an algorithm that produces the object in question, since cryptography requires not merely that inverses and representatives exist but that one can compute them efficiently.

We take the basic algebraic properties of $\mathbb{Z}$ for granted: that it is a commutative ring with the usual addition and multiplication, that it has no zero divisors (if $ab = 0$ then $a = 0$ or $b = 0$), and that it carries a total order $\leq$ compatible with the arithmetic. We assume axiomatically the single deep property that follows.

Theorem 2.1 (Well-ordering principle). *Every nonempty subset $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$ of the natural numbers has a least element: there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

The well-ordering principle is logically equivalent to the principle of mathematical induction; we use both freely. Nearly every existence result below is, at bottom, an application of well-ordering: one exhibits a nonempty set of natural numbers and extracts its minimum.

2.1 Divisibility

Definition 2.2. Let $a, b \in \mathbb{Z}$. One says that a *divides* b , written $a \mid b$, if there exists an integer $c \in \mathbb{Z}$ with $b = ac$. In this case a is called a *divisor* or *factor* of b , and b is a *multiple* of a . If no such c exists one writes $a \nmid b$.

Several immediate consequences of the definition deserve to be recorded; we shall use them constantly and usually without explicit citation.

Proposition 2.3. *Let $a, b, c, d \in \mathbb{Z}$. Then:*

1. $a \mid a$ (*reflexivity*), $1 \mid a$, and $a \mid 0$.
2. If $a \mid b$ and $b \mid c$ then $a \mid c$ (*transitivity*).
3. If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for all $x, y \in \mathbb{Z}$; in particular $a \mid (b \pm c)$.
4. If $a \mid b$ and $b \neq 0$ then $|a| \leq |b|$.
5. If $a \mid b$ and $b \mid a$ then $a = \pm b$.
6. $0 \mid b$ if and only if $b = 0$.

Proof. (1) is immediate: $a = a \cdot 1$, $a = 1 \cdot a$, $0 = a \cdot 0$. For (2), write $b = ae$ and $c = bf$; then $c = a(e f)$, so $a \mid c$. For (3), write $b = ae$ and $c = af$; then $bx + cy = a(ex + fy)$. For (4), write $b = ac$ with $b \neq 0$, so $c \neq 0$ and hence $|c| \geq 1$; then $|b| = |a||c| \geq |a|$. For (5), if either is 0 then by (6) so is the other and the claim holds; otherwise (4) gives $|a| \leq |b|$ and $|b| \leq |a|$, so $|a| = |b|$ and $a = \pm b$. For (6), $0 \mid b$ means $b = 0 \cdot c = 0$; conversely $0 \mid 0$. $\square$

The relation $\mid$ is thus reflexive and transitive; it is a partial order when restricted to the natural numbers (where (5) becomes $a = b$), but on all of $\mathbb{Z}$ it fails to be antisymmetric precisely because of signs. The *units* of $\mathbb{Z}$ —the integers dividing 1—are exactly ± 1 , and two integers that divide each other are called *associates*; in $\mathbb{Z}$ the associates of a are $\pm a$.

2.2 The division algorithm

Divisibility is the special case of exact division. The fundamental fact that makes the arithmetic of $\mathbb{Z}$ tractable is that one can always divide *with remainder*, and that the quotient and remainder are uniquely determined once a convention for the remainder's range is fixed.

Theorem 2.4 (Division algorithm). *Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ with $b > 0$. Then there exist unique integers q (the quotient) and r (the remainder) such that*

$$a = bq + r, \quad 0 \leq r < b.$$

Proof. Existence. Consider the set

$$S = \{a - bk : k \in \mathbb{Z}, a - bk \geq 0\} \subseteq \mathbb{N}.$$

First, $S \neq \emptyset$. If $a \geq 0$ then $a = a - b \cdot 0 \in S$. If $a < 0$ then, since $b \geq 1$, taking $k = a$ gives $a - ba = a(1 - b) \geq 0$ (a product of a nonpositive and a nonpositive factor), so $a - ba \in S$. Thus S is a nonempty subset of $\mathbb{N}$, and by the well-ordering principle (Theorem 2.1) it has a least element $r = a - bq$ for some $q \in \mathbb{Z}$. By construction $r \geq 0$. If $r \geq b$, then $r - b = a - b(q + 1) \geq 0$ would lie in S and be strictly smaller than r , contradicting minimality. Hence $0 \leq r < b$.

Uniqueness. Suppose $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$. Then $b(q - q') = r' - r$, so $b \mid (r' - r)$. But $-b < r' - r < b$, and the only multiple of b strictly between $-b$ and b is 0. Hence $r' = r$, and then $b(q - q') = 0$ with $b \neq 0$ forces $q = q'$. $\square$

Remark 2.5. The hypothesis $b > 0$ is only for definiteness. For general $b \neq 0$ one obtains unique q, r with $a = bq + r$ and $0 \leq r < |b|$, by applying the theorem to $|b|$ and adjusting the sign of q . Note that the remainder convention $0 \leq r < |b|$ differs from the truncation behaviour of many programming languages, where the remainder of a negative dividend may be negative; for the number-theoretic results below the convention above is the appropriate one.

We denote the remainder r produced by the division algorithm by $a \bmod b$, and the quotient q by $\lfloor a/b \rfloor$ when $b > 0$ (the *floor* of the rational number a/b). With this notation, $b \mid a$ holds precisely when $a \bmod b = 0$.

2.3 The greatest common divisor and the Euclidean algorithm

Definition 2.6. A *common divisor* of integers a and b is an integer d with $d \mid a$ and $d \mid b$. The *greatest common divisor* of a and b , not both zero, is the largest such d ; it is denoted $\gcd(a, b)$. By convention $\gcd(0, 0) = 0$. Integers a and b are *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

The greatest common divisor is well defined when $(a, b) \neq (0, 0)$: the set of common divisors is nonempty (it contains 1) and bounded above (every common divisor of, say, the nonzero a has absolute value at most $|a|$ by Proposition 2.3(4)), so it has a largest element, which is positive since 1 is a common divisor. We establish a much stronger characterisation shortly: the gcd is not merely the numerically largest common divisor but is divisible by every common divisor. The key computational engine is the following observation.

Lemma 2.7. Let $a, b \in \mathbb{Z}$. For any $q \in \mathbb{Z}$,

$$\gcd(a, b) = \gcd(b, a - qb).$$

In particular, if $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$, and $\gcd(a, 0) = |a|$.

Proof. It suffices to show the two pairs (a, b) and $(b, a - qb)$ have exactly the same common divisors; equal sets of common divisors have equal greatest elements (and the same coprimality status). If $d \mid a$ and $d \mid b$, then $d \mid (a - qb)$ by Proposition 2.3(3), so d is a common divisor of b and $a - qb$. Conversely, if $d \mid b$ and $d \mid (a - qb)$, then $d \mid ((a - qb) + qb) = a$, so d is a common divisor of a and b . The two sets of common divisors coincide. Finally $\gcd(a, 0) = |a|$ because the common divisors of a and 0 are exactly the divisors of a , the largest of which in absolute value is $|a|$ (and $|a| > 0$ when $a \neq 0$; the case $a = 0$ is the convention $\gcd(0, 0) = 0$). $\square$

Iterating the lemma with the division algorithm yields the oldest nontrivial algorithm in mathematics.

Theorem 2.8 (Euclidean algorithm). Let a, b be integers with $b \neq 0$. Define a sequence of remainders by $r_0 = |a|$, $r_1 = |b|$, and, as long as $r_i \neq 0$, let $r_{i+1} = r_{i-1} \bmod r_i$, so that

$$r_{i-1} = r_i q_i + r_{i+1}, \quad 0 \leq r_{i+1} < r_i.$$

The sequence $r_1 > r_2 > \dots$ of positive remainders is strictly decreasing, so some $r_{n+1} = 0$; the last nonzero remainder satisfies $r_n = \gcd(a, b)$.

Proof. The remainders $r_1, r_2, \dots$ are nonnegative integers and, while nonzero, strictly decreasing because $r_{i+1} = r_{i-1} \bmod r_i < r_i$. A strictly decreasing sequence of nonnegative integers cannot be infinite (again by well-ordering: otherwise the set of values would have no least element), so the algorithm terminates with some $r_{n+1} = 0$ and $r_n > 0$. By Lemma 2.7, each step preserves the gcd:

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_n, r_{n+1}) = \gcd(r_n, 0) = r_n,$$

using $\gcd(|a|, |b|) = \gcd(a, b)$ (signs do not affect common divisors). $\square$

Example 2.9. Compute $\gcd(1071, 462)$:

$$\begin{aligned} 1071 &= 2 \cdot 462 + 147, \\ 462 &= 3 \cdot 147 + 21, \\ 147 &= 7 \cdot 21 + 0. \end{aligned}$$

The last nonzero remainder is 21, so $\gcd(1071, 462) = 21$. Indeed $1071 = 21 \cdot 51$ and $462 = 21 \cdot 22$, with $\gcd(51, 22) = 1$.

Remark 2.10. The Euclidean algorithm is efficient. The number of division steps for inputs $a > b > 0$ is $O(\log b)$; the worst case occurs at consecutive Fibonacci numbers (Lamé’s theorem), where each quotient q_i equals 1 and the remainders shrink as slowly as possible. With schoolbook arithmetic on k -bit inputs the whole computation costs $O(k^2)$ bit operations. This efficiency is what renders gcd-based cryptographic operations—above all modular inversion, below—practical at the field sizes used in Halo 2 and Orchard.

2.4 The Bézout identity and the extended Euclidean algorithm

The Euclidean algorithm computes the gcd, but the back-substitution of its equations yields something more powerful: the gcd is an *integer linear combination* of the inputs. This is among the most useful facts in elementary number theory.

Theorem 2.11 (Bézout’s identity). *Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then there exist integers $u, v \in \mathbb{Z}$ with*

$$au + bv = d.$$

Moreover d is the smallest positive integer expressible in the form $ax + by$ with $x, y \in \mathbb{Z}$, and the set of all such combinations is exactly the set of multiples of d :

$$\{ax + by : x, y \in \mathbb{Z}\} = d\mathbb{Z} = \{kd : k \in \mathbb{Z}\}.$$

Proof. Let $I = \{ax + by : x, y \in \mathbb{Z}\}$. This set contains a and b and is closed under addition and under multiplication by any integer; concretely, if $s = ax_1 + by_1$ and $t = ax_2 + by_2$ lie in I , then for any $m, n \in \mathbb{Z}$ one has $ms + nt = a(mx_1 + nx_2) + b(my_1 + ny_2) \in I$. Since a, b are not both zero, I contains a positive element (one of $\pm a, \pm b$), so the set $I \cap \mathbb{N}_{>0}$ of positive elements of I is nonempty. By well-ordering it has a least element $g = au + bv > 0$.

We claim that g divides every element of I . Indeed, let $c \in I$ and write $c = gq + r$ with $0 \leq r < g$ by the division algorithm. Then $r = c - gq \in I$ because I is closed under the operations just described. By minimality of g among positive elements of I , and $0 \leq r < g$, necessarily $r = 0$. Hence $g \mid c$ for every $c \in I$; in particular $g \mid a$ and $g \mid b$, so g is a common divisor of a and b . Conversely, any common divisor e of a and b divides $au + bv = g$ by Proposition 2.3(3). Therefore g is a common divisor that is divisible by every common divisor; being positive, it is in particular the *greatest* common divisor, so $g = d$. The displayed equation $au + bv = d$ follows, d is the least positive element of I , and $I = g\mathbb{Z} = d\mathbb{Z}$ because every element of I is a multiple of g , as shown, and conversely every multiple $kg = a(ku) + b(kv)$ lies in I . $\square$

Remark 2.12. The proof yields a characterisation of the gcd that is independent of the order $\leq$: $d = \gcd(a, b)$ is, up to sign, the unique common divisor of a and b that is divisible by every common divisor. This is the definition that generalises to arbitrary commutative rings, where “largest” may make no sense but “divisible by every common divisor” still does. We call the subsets $I \subseteq \mathbb{Z}$ closed under addition and under multiplication by arbitrary integers *ideals*; the content of Bézout’s identity is that every ideal of $\mathbb{Z}$ has the form $d\mathbb{Z}$ for a single generator d , i.e. $\mathbb{Z}$ is a *principal ideal domain*. Ideals recur in the construction of quotient rings.

The Bézout coefficients u, v are not unique. If $au + bv = d$, then for every $k \in \mathbb{Z}$,

$$a\left(u + k\frac{b}{d}\right) + b\left(v - k\frac{a}{d}\right) = d,$$

and these are all the solutions. To compute a particular pair (u, v) efficiently, we augment the Euclidean algorithm so that it carries each remainder as an explicit combination of a and b .

Theorem 2.13 (Extended Euclidean algorithm). *There is an algorithm that, on input $a, b \in \mathbb{Z}$ not both zero, outputs $d = \gcd(a, b)$ together with integers u, v satisfying $au + bv = d$, using the same number of division steps as the Euclidean algorithm.*

Proof. Alongside the remainder sequence of Theorem 2.8, maintain two auxiliary sequences (s_i) and (t_i) with the invariant

$$r_i = as_i + bt_i \quad \text{for all } i. \quad (*)$$

Initialise (taking $a, b \geq 0$ for clarity; we restore signs at the end)

$$r_0 = a, \quad s_0 = 1, \quad t_0 = 0; \quad r_1 = b, \quad s_1 = 0, \quad t_1 = 1,$$

so $(*)$ holds for $i = 0, 1$. At step $i \geq 1$, having $r_{i-1} = q_i r_i + r_{i+1}$ from the division algorithm, set

$$s_{i+1} = s_{i-1} - q_i s_i, \quad t_{i+1} = t_{i-1} - q_i t_i.$$

Then

$$as_{i+1} + bt_{i+1} = (as_{i-1} + bt_{i-1}) - q_i(as_i + bt_i) = r_{i-1} - q_i r_i = r_{i+1},$$

so induction preserves $(*)$. When the algorithm terminates with $r_{n+1} = 0$ and $r_n = d = \gcd(a, b)$, the invariant for $i = n$ reads $d = as_n + bt_n$; output $u = s_n, v = t_n$. Termination and the value of d are exactly as in Theorem 2.8, and each step does only a constant amount of extra work. $\square$

In practice we conveniently lay the computation out as a table with columns (r_i, q_i, s_i, t_i) , computing each new row from the previous two.

Example 2.14. Run the extended algorithm on $a = 1071, b = 462$ (continuing Example 2.9). The quotients are $q_1 = 2, q_2 = 3, q_3 = 7$.

i	r_i	s_i	t_i
0	1071	1	0
1	462	0	1
2	147	1	-2
3	21	-3	7
4	0	22	-51

For instance, row 2 is row 0 minus $q_1 = 2$ times row 1: $s_2 = 1 - 2 \cdot 0 = 1, t_2 = 0 - 2 \cdot 1 = -2$, and indeed $1071 \cdot 1 + 462 \cdot (-2) = 1071 - 924 = 147$. The last nonzero remainder is $r_3 = 21 = \gcd(1071, 462)$, with

$$1071 \cdot (-3) + 462 \cdot 7 = -3213 + 3234 = 21.$$

Thus $(u, v) = (-3, 7)$. The final row $(s_4, t_4) = (22, -51) = (b/d, -a/d)$ up to sign records the homogeneous solution: $1071 \cdot 22 + 462 \cdot (-51) = 0$.

A first, decisive application of Bézout's identity is Euclid's lemma, the property of primes that ultimately forces unique factorisation.

Proposition 2.15 (Euclid's lemma). *Let $a, b, c \in \mathbb{Z}$.*

1. *If $c \mid ab$ and $\gcd(c, a) = 1$, then $c \mid b$.*
2. *If $\gcd(a, b) = 1$, $a \mid n$, and $b \mid n$, then $ab \mid n$.*
3. *If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.*

Proof. (1) By Bézout there are u, v with $cu + av = 1$. Multiply by b : $b = cub + avb = c(ub) + (ab)v$. Since $c \mid ab$, c divides both terms on the right, hence $c \mid b$.

(2) Write $n = ak$. Since $b \mid n = ak$ and $\gcd(b, a) = 1$, part (1) gives $b \mid k$, say $k = bm$. Then $n = a(bm) = (ab)m$, so $ab \mid n$.

(3) By Bézout, $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$ for suitable integers. Multiplying,

$$1 = (ax_1 + by_1)(ax_2 + cy_2) = a \underbrace{(ax_1x_2 + x_1cy_2 + by_1x_2)}_{\in \mathbb{Z}} + bc(y_1y_2),$$

which exhibits 1 as an integer combination of a and bc . Hence $\gcd(a, bc)$ divides 1, so it equals 1. $\square$

2.5 Primes and the fundamental theorem of arithmetic

Definition 2.16. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p . An integer $n > 1$ that is not prime is *composite*; thus n is composite iff $n = ab$ for some integers $1 < a, b < n$. The integer 1 is a *unit* and is by convention neither prime nor composite.

The exclusion of 1 from the primes is not arbitrary: it is exactly what makes factorisation into primes unique, since otherwise one could insert arbitrarily many factors of 1. The decisive property of primes is the prime case of Euclid's lemma.

Lemma 2.17 (Euclid's lemma for primes). *Let p be prime and $a, b \in \mathbb{Z}$. If $p \mid ab$ then $p \mid a$ or $p \mid b$. More generally, if $p \mid a_1a_2 \cdots a_k$ then $p \mid a_i$ for some i .*

Proof. The only positive divisors of p are 1 and p , so $\gcd(p, a) \in \{1, p\}$. If $\gcd(p, a) = p$ then $p \mid a$ and the claim holds; otherwise $\gcd(p, a) = 1$, and Proposition 2.15(1) gives $p \mid b$. The general statement follows by induction on k : from $p \mid a_1(a_2 \cdots a_k)$ it follows that $p \mid a_1$ or $p \mid a_2 \cdots a_k$, and the inductive hypothesis handles the latter. $\square$

Theorem 2.18 (Fundamental theorem of arithmetic). *Every integer $n > 1$ can be written as a product of primes,*

$$n = p_1 p_2 \cdots p_k,$$

and this factorisation is unique up to the order of the factors. Equivalently, there is a unique expression

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}, \quad p_1 < p_2 < \cdots < p_r \text{ prime, } e_i \geq 1.$$

Proof. Existence. Suppose, for contradiction, that some integer > 1 is not a product of primes, and let n be the least such (well-ordering). Then n is not prime (a prime is trivially a product of one prime), so $n = ab$ with $1 < a, b < n$. By minimality, each of a, b is a product of primes; concatenating these factorisations expresses n as a product of primes, a contradiction. Hence every $n > 1$ factors into primes.

Uniqueness. Suppose

$$p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$$

are two factorisations of the same integer into primes. The argument proceeds by induction on k . If $k = 0$ the product is the empty product 1, forcing $\ell = 0$ as well (a nonempty product of primes is > 1). For the inductive step, p_1 divides the right-hand product, so by Lemma 2.17 $p_1 \mid q_j$ for some j . Since q_j is prime its only divisors > 1 is itself, so $p_1 = q_j$. Cancelling this common factor (legitimate as $\mathbb{Z}$ has no zero divisors) yields

$$p_2 \cdots p_k = q_1 \cdots \widehat{q_j} \cdots q_\ell,$$

where the hat denotes omission. By the inductive hypothesis these two factorisations agree up to order, with $k - 1 = \ell - 1$; restoring $p_1 = q_j$ shows the original factorisations agree up to order. Collecting equal primes into prime powers gives the canonical form, whose uniqueness is immediate from the uniqueness of the multiset of primes. $\square$

Remark 2.19. Both halves of the theorem are needed and have different characters. Existence is a statement about the order structure (well-ordering) and would hold in any reasonable factorisation; uniqueness rests squarely on Euclid's lemma, hence on Bézout, hence on the division algorithm. There are number systems with a division-like structure failing Euclid's lemma in which factorisation into irreducibles is *not* unique; the integers avoid this pathology precisely because they form a Euclidean domain.

We record two classical consequences, the second of which we revisit when discussing finite fields.

Theorem 2.20 (Euclid). *There are infinitely many primes.*

Proof. Suppose only finitely many primes $p_1, \dots, p_m$ existed. Consider $N = p_1 p_2 \cdots p_m + 1 > 1$. By the fundamental theorem N has a prime divisor p , which must be some p_i . Then $p_i \mid N$ and $p_i \mid p_1 \cdots p_m$, so $p_i \mid (N - p_1 \cdots p_m) = 1$, forcing $p_i = 1$, absurd. Hence the set of primes is infinite. $\square$

Proposition 2.21. *If $a = \prod_p p^{\alpha_p}$ and $b = \prod_p p^{\beta_p}$ are the prime factorisations of positive integers a, b (with finitely many exponents nonzero), then*

$$\gcd(a, b) = \prod_p p^{\min(\alpha_p, \beta_p)}, \quad \text{lcm}(a, b) = \prod_p p^{\max(\alpha_p, \beta_p)},$$

and consequently $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$.

Proof. A positive integer $d = \prod_p p^{\gamma_p}$ divides a iff $\gamma_p \leq \alpha_p$ for every p , by uniqueness of factorisation. Hence d is a common divisor of a and b iff $\gamma_p \leq \min(\alpha_p, \beta_p)$ for all p ; the largest such d takes $\gamma_p = \min(\alpha_p, \beta_p)$, giving the gcd formula. The lcm formula is dual (the least common multiple is the smallest positive common multiple). Adding exponents, $\min(\alpha_p, \beta_p) + \max(\alpha_p, \beta_p) = \alpha_p + \beta_p$ for each p , whence $\gcd(a, b) \text{lcm}(a, b) = \prod_p p^{\alpha_p + \beta_p} = ab$. $\square$

2.6 Congruences and the ring $\mathbb{Z}/n\mathbb{Z}$

The development now passes from divisibility to its quotient, the arithmetic of remainders. Fix an integer $n \geq 1$, the *modulus*.

Definition 2.22. Let $n \geq 1$ and $a, b \in \mathbb{Z}$. The integer a is *congruent to b modulo n* , written

$$a \equiv b \pmod{n},$$

if $n \mid (a - b)$, equivalently if a and b leave the same remainder upon division by n .

The two formulations agree: if $a = nq_1 + r_1$ and $b = nq_2 + r_2$ with $0 \leq r_1, r_2 < n$, then $a - b = n(q_1 - q_2) + (r_1 - r_2)$, and $n \mid (a - b)$ iff $n \mid (r_1 - r_2)$ iff $r_1 = r_2$ (since $|r_1 - r_2| < n$).

Proposition 2.23. Fix $n \geq 1$. Congruence modulo n is an equivalence relation on $\mathbb{Z}$ that is moreover compatible with addition and multiplication: if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then

$$a + b \equiv a' + b' \pmod{n}, \quad ab \equiv a'b' \pmod{n}.$$

Consequently $a^k \equiv (a')^k \pmod{n}$ for all $k \geq 0$, and one may substitute congruent values freely in any polynomial expression with integer coefficients.

Proof. Reflexivity ($n \mid 0$), symmetry ($n \mid (a - b) \Rightarrow n \mid (b - a)$), and transitivity ($n \mid (a - b)$ and $n \mid (b - c)$ give $n \mid (a - c)$ by Proposition 2.3(3)) are immediate. For compatibility, write $a - a' = ns$ and $b - b' = nt$. Then

$$(a + b) - (a' + b') = n(s + t),$$

and

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = n(sb + a't),$$

so both $n \mid ((a + b) - (a' + b'))$ and $n \mid (ab - a'b')$. The statement about powers and polynomials follows by induction from the multiplicative case. $\square$

Because congruence is an equivalence relation, it partitions $\mathbb{Z}$ into disjoint *congruence classes* (or *residue classes*).

Definition 2.24. The *congruence class* of a modulo n is

$$[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod{n}\} = \{a + kn : k \in \mathbb{Z}\} = a + n\mathbb{Z}.$$

We write the set of all congruence classes as

$$\mathbb{Z}/n\mathbb{Z} = \{[a]_n : a \in \mathbb{Z}\}.$$

By the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n-1$, namely its remainder; hence the n classes $[0]_n, [1]_n, \dots, [n-1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$. The set $\{0, 1, \dots, n-1\}$ is the standard *system of representatives*, called the *least nonnegative residues*.

Theorem 2.25. For each $n \geq 1$, the operations

$$[a]_n + [b]_n := [a + b]_n, \quad [a]_n \cdot [b]_n := [ab]_n$$

are well defined and make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring with additive identity $[0]_n$ and multiplicative identity $[1]_n$. It has exactly n elements.

Proof. Well-definedness is precisely the compatibility of Proposition 2.23: the sum and product of classes do not depend on the chosen representatives. Each ring axiom for $\mathbb{Z}/n\mathbb{Z}$ (associativity and commutativity of $+$ and $\cdot$, distributivity, existence of 0 and 1, additive inverses $-[a]_n = [-a]_n$) is inherited directly from the corresponding axiom in $\mathbb{Z}$ by passing to classes; for instance associativity of multiplication reads $([a][b])[c] = [abc] = [a]([b][c])$. The count of exactly n classes appeared above. $\square$

Remark 2.26. Structurally, $\mathbb{Z}/n\mathbb{Z}$ is the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z} = \{kn : k \in \mathbb{Z}\}$ (Remark 2.12). The map $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $\pi(a) = [a]_n$, is a surjective ring homomorphism with kernel $n\mathbb{Z}$; this is the prototype of the quotient constructions used throughout the algebra to come. One often drops the brackets and the subscript, writing a for $[a]_n$ when the modulus is clear, and uses “ $\mathbb{Z}/n\mathbb{Z}$ ” or “ $\mathbb{Z}_n$ ” interchangeably; $\mathbb{Z}_n$ denotes this ring (not the n -adic integers, which do not appear here).

A basic structural question is when $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors—equivalently, when it is a field.

Proposition 2.27. For $n \geq 2$, the ring $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors if and only if n is prime; in that case, as we show below, it is a field, written $\mathbb{F}_p$ when $n = p$.

Proof. If $n = ab$ with $1 < a, b < n$ is composite, then $[a]_n \neq [0]_n \neq [b]_n$ but $[a]_n[b]_n = [n]_n = [0]_n$, so there are zero divisors. Conversely, if $n = p$ is prime and $[a]_p[b]_p = [0]_p$, then $p \mid ab$, so by Lemma 2.17 $p \mid a$ or $p \mid b$, i.e. $[a]_p = [0]_p$ or $[b]_p = [0]_p$. That a finite commutative ring without zero divisors and with $1 \neq 0$ is a field follows from the discussion of units below (Corollary 2.30). $\square$

2.7 Units modulo n and modular inverses

In any ring, the elements with multiplicative inverses are the most important. In $\mathbb{Z}/n\mathbb{Z}$ the Euclidean machinery characterises them and computes them.

Definition 2.28. An element $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a *unit* (or is *invertible*) if there exists $[x]_n$ with $[a]_n[x]_n = [1]_n$, i.e. $ax \equiv 1 \pmod{n}$. Such an $[x]_n$ is the (*modular*) *inverse* of $[a]_n$, written $[a]_n^{-1}$. We write the set of units as

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{ [a]_n : [a]_n \text{ is a unit} \}.$$

Theorem 2.29. Let $n \geq 1$ and $a \in \mathbb{Z}$. The class $[a]_n$ is a unit in $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(a, n) = 1$. When $\gcd(a, n) = 1$, the inverse is unique, and the extended Euclidean algorithm computes a representative: if $au + nv = 1$, then $[a]_n^{-1} = [u]_n$.

Proof. Suppose $\gcd(a, n) = 1$. By Bézout (Theorem 2.11) there are u, v with $au + nv = 1$. Reducing modulo n , $au \equiv 1 \pmod{n}$, so $[u]_n$ is an inverse of $[a]_n$. The extended Euclidean algorithm (Theorem 2.13) produces such a u explicitly.

Conversely, if $[a]_n$ is a unit, say $ax \equiv 1 \pmod{n}$, then $ax - 1 = nk$ for some k , i.e. $ax - nk = 1$. Any common divisor of a and n divides the left side, hence divides 1; thus $\gcd(a, n) = 1$.

Uniqueness: if $ax \equiv 1$ and $ax' \equiv 1 \pmod{n}$, then $x \equiv x(ax') \equiv (xa)x' \equiv x' \pmod{n}$, so $[x]_n = [x']_n$. $\square$

Corollary 2.30. If p is prime then every nonzero class in $\mathbb{Z}/p\mathbb{Z}$ is a unit, so $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a field. More generally, $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.

Proof. If p is prime and $[a]_p \neq [0]_p$ then $p \nmid a$, so $\gcd(a, p) = 1$ (the only positive divisors of p are 1, p), and Theorem 2.29 makes $[a]_p$ a unit. Thus every nonzero element is invertible and $\mathbb{F}_p$ is a field. If n is composite, $n = ab$ with $1 < a, b < n$, then $[a]_n \neq [0]_n$ yet $\gcd(a, n) = a > 1$, so $[a]_n$ is not a unit; a ring with a nonzero non-unit (equivalently, with zero divisors, by Proposition 2.27) is not a field. The case $n = 1$ gives the zero ring, not a field by convention. $\square$

Example 2.31. Consider the computation of $[17]_{43}^{-1}$. The extended Euclidean algorithm on $(43, 17)$:

$$\begin{aligned} 43 &= 2 \cdot 17 + 9, \\ 17 &= 1 \cdot 9 + 8, \\ 9 &= 1 \cdot 8 + 1, \\ 8 &= 8 \cdot 1 + 0, \end{aligned}$$

so $\gcd(43, 17) = 1$. Back-substituting:

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Hence $-5 \cdot 17 \equiv 1 \pmod{43}$, and $[17]_{43}^{-1} = [-5]_{43} = [38]_{43}$. Check: $17 \cdot 38 = 646 = 15 \cdot 43 + 1$, so $17 \cdot 38 \equiv 1 \pmod{43}$.

The set of units forms a group under multiplication, a fact we use heavily in what follows.

Proposition 2.32. *Under multiplication of classes, $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group. Its order is denoted $\varphi(n)$ (Euler’s totient): the number of integers in $\{1, \dots, n\}$ coprime to n .*

Proof. Multiplication of classes is associative and commutative (Theorem 2.25) with identity $[1]_n$. It is closed on units: if $[a]_n, [b]_n$ are units with inverses $[a]_n^{-1}, [b]_n^{-1}$, then $[ab]_n$ has inverse $[b]_n^{-1}[a]_n^{-1}$, since $(ab)(b^{-1}a^{-1}) \equiv 1$. Every unit has an inverse by definition, and that inverse is itself a unit (it is invertible, with inverse the original element). Hence $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group. By Theorem 2.29 its elements are exactly the classes $[a]_n$ with $\gcd(a, n) = 1$, $1 \leq a \leq n$, of which there are $\varphi(n)$ by definition. $\square$

Remark 2.33. The totient $\varphi(n)$ governs the multiplicative order structure modulo n . For prime p , $\varphi(p) = p - 1$, so $\mathbb{F}_p^\times$ has order $p - 1$; for a prime power, $\varphi(p^k) = p^{k-1}(p - 1)$. Lagrange’s theorem (developed in the group-theory section) then gives Euler’s theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$ for $\gcd(a, n) = 1$, and in particular Fermat’s little theorem $a^{p-1} \equiv 1 \pmod{p}$ for $p \nmid a$. These facts underlie modular exponentiation, the cyclic structure of $\mathbb{F}_p^\times$, and the choice of scalar field sizes in the elliptic-curve constructions of Halo 2 and Orchard. The Chinese Remainder Theorem yields the multiplicativity of φ , which then computes $\varphi(n)$ for general n .

3 Groups

The notion of a *group* is the foundational abstraction of modern algebra. A group axiomatises the bare minimum needed to discuss *symmetry* and *invertible composition*: a set together with one associative way of combining its elements, possessing an identity and inverses. This short list of axioms is rich enough to support a deep structure theory, and almost every object encountered later—the integers modulo a prime under addition, the nonzero residues under multiplication, the points of an elliptic curve, the symmetries of a finite field—is a group. This section develops group theory from nothing, proving the central structural results (Lagrange’s theorem, the classification of cyclic groups, the first isomorphism theorem) in full.

3.1 Definition and axioms

Before we can discuss groups, we must make precise the meaning of “combining” two elements of a set.

Definition 3.1. Let S be a set. A *binary operation* on S is a function

$$*: S \times S \rightarrow S, \quad (a, b) \mapsto a * b.$$

The element $a * b$ denotes the image of the pair (a, b) . The defining feature is *closure*: for all $a, b \in S$, the element $a * b$ again lies in S .

Closure is built into the very statement that $*$ is a function *into* S ; we name it explicitly nonetheless because, when we later examine a subset $H \subseteq S$, the question of whether $*$ restricts to a binary operation on H is exactly the question of whether H is closed under $*$.

Definition 3.2. A *group* is a pair $(G, *)$ consisting of a set G and a binary operation $*$ on G satisfying the following three axioms.

(G1) *Associativity.* For all $a, b, c \in G$,

$$(a * b) * c = a * (b * c).$$

(G2) *Identity element.* There exists an element $e \in G$ such that for all $a \in G$,

$$e * a = a * e = a.$$

(G3) *Inverses*. For each $a \in G$ there exists an element $b \in G$ such that

$$a * b = b * a = e,$$

where e is an identity element as in (G2).

A set with a binary operation satisfying only (G1) is called a *semigroup*; if it also satisfies (G2) it is a *monoid*. Thus a group is a monoid in which every element is invertible.

We frequently use the phrase “the group G ”, with the operation understood. The number of elements of G , written $|G|$, is called the *order* of the group; if $|G|$ is finite G is a *finite group*, otherwise an *infinite group*.

The axioms assert the *existence* of an identity and of inverses, but a priori not their uniqueness. Uniqueness is a theorem, and a useful one, since it licenses the notation e and a^{-1} .

Proposition 3.3. *Let $(G, *)$ be a group.*

1. *The identity element is unique.*
2. *Each $a \in G$ has a unique inverse, which is denoted a^{-1} .*
3. (Cancellation.) *For all $a, x, y \in G$, if $a * x = a * y$ then $x = y$, and if $x * a = y * a$ then $x = y$.*
4. *For all $a, b \in G$, $(a^{-1})^{-1} = a$ and $(a * b)^{-1} = b^{-1} * a^{-1}$.*

Proof. (1) Suppose e and e' both satisfy (G2). Then, using that e' is an identity, $e = e * e'$; and using that e is an identity, $e * e' = e'$. Hence $e = e'$.

(2) Suppose b and b' are both inverses of a . Then

$$b = b * e = b * (a * b') = (b * a) * b' = e * b' = b',$$

using (G2), the inverse property of b' , associativity (G1), and the inverse property of b . Thus the inverse is unique.

(3) Suppose $a * x = a * y$. Applying a^{-1} on the left,

$$x = e * x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = e * y = y.$$

The right-cancellation law is symmetric.

(4) Since $a * a^{-1} = a^{-1} * a = e$, the element a is an inverse of a^{-1} ; by uniqueness $(a^{-1})^{-1} = a$. For the second identity, compute

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * e * a^{-1} = a * a^{-1} = e,$$

and symmetrically $(b^{-1} * a^{-1}) * (a * b) = e$. By uniqueness of inverses, $(a * b)^{-1} = b^{-1} * a^{-1}$. $\square$

Remark 3.4. One can weaken the axioms (G2) and (G3): it suffices to demand a *left identity* e (so $e * a = a$ for all a) together with *left inverses* (for each a some b with $b * a = e$). One can then show that the left identity is also a right identity and the left inverses also right inverses, recovering the full axioms. We do not need this refinement here, and we state the symmetric version for clarity.

Notation. Two notational conventions dominate. In *multiplicative* notation one writes the operation as juxtaposition ab (or $a \cdot b$), the identity as 1, and the inverse as a^{-1} ; powers are $a^n = \underbrace{a \cdots a}_n$ for $n \geq 1$, $a^0 = 1$, and $a^{-n} = (a^{-1})^n$. In *additive* notation, reserved almost exclusively for commutative groups, one writes $a + b$, the identity as 0, the inverse as $-a$, and na for the n -fold sum. The usual exponent laws $a^m a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$ hold for all integers m, n (proved by induction together with Proposition 3.3). Henceforth ab usually denotes $a * b$ and we drop the symbol $*$ unless clarity demands it.

3.2 Abelian groups

Definition 3.5. A group $(G, *)$ is *abelian* (or *commutative*) if in addition

$$a * b = b * a \quad \text{for all } a, b \in G.$$

A group that is not abelian we call *non-abelian*.

The term honours Niels Henrik Abel. Commutativity is a strong simplifying assumption: in an abelian group one may freely reorder products, the formula $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$ loses its order-sensitivity, and—as we show below—the structure theory becomes especially clean. Convention reserves additive notation for abelian groups precisely because $+$ connotes commutativity.

3.3 Examples of groups

We now exhibit a generous supply of groups. The reader should check each example against the three axioms; we comment only on the points that are not immediate.

Example 3.6 (Integers under addition). The set $\mathbb{Z}$ of integers with the operation $+$ is an abelian group. Addition is associative and commutative, 0 is the identity, and the inverse of n is $-n$. The same holds for $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$. All of these are infinite. Note $(\mathbb{Z}, \cdot)$ is *not* a group: 2 has no multiplicative inverse in $\mathbb{Z}$.

Example 3.7 (Nonzero elements of a field under multiplication). For $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$, the set of *nonzero* elements, denoted $\mathbb{Q}^\times$, $\mathbb{R}^\times$, $\mathbb{C}^\times$, forms an abelian group under multiplication, with identity 1 and inverse $a^{-1} = 1/a$. We must exclude the element 0 because 0 has no multiplicative inverse. More generally, for any field $\mathbb{F}$ the nonzero elements $\mathbb{F}^\times = \mathbb{F} \setminus \{0\}$ form an abelian group under multiplication; this fact is part of the very definition of a field.

Example 3.8 (Integers modulo n). Fix an integer $n \geq 1$. We call two integers a, b *congruent modulo n* , written $a \equiv b \pmod{n}$, if $n \mid (a - b)$. This is an equivalence relation; its equivalence classes are the *residue classes* $[a] = \{a + kn : k \in \mathbb{Z}\}$, and there are exactly n of them, namely $[0], [1], \dots, [n-1]$. We denote the set of classes $\mathbb{Z}/n\mathbb{Z}$ (or $\mathbb{Z}_n$). Addition of classes obeys the rule

$$[a] + [b] := [a + b].$$

This is *well defined*: if $[a] = [a']$ and $[b] = [b']$, then $n \mid (a - a')$ and $n \mid (b - b')$, so $n \mid ((a + b) - (a' + b'))$, whence $[a + b] = [a' + b']$. With this operation $(\mathbb{Z}/n\mathbb{Z}, +)$ is an abelian group of order n : associativity and commutativity are inherited from $\mathbb{Z}$, the identity is $[0]$, and $-[a] = [-a] = [n - a]$. We typically write the class $[a]$ simply as a when the modulus is clear.

Example 3.9 (Units modulo n). Within $\mathbb{Z}/n\mathbb{Z}$ one may also attempt multiplication: $[a] \cdot [b] := [ab]$, which is well defined by an argument like the one above. This makes $\mathbb{Z}/n\mathbb{Z}$ a commutative monoid with identity $[1]$, but *not* a group, because $[0]$ (and possibly other classes) lack inverses. A class $[a]$ has a multiplicative inverse if and only if $\gcd(a, n) = 1$: indeed, by Bézout's identity there exist integers x, y with $ax + ny = 1$ precisely when $\gcd(a, n) = 1$, and then $[a][x] = [1]$; conversely if $[a][x] = [1]$ then $ax \equiv 1 \pmod{n}$, so $ax - 1$ is a multiple of n and any common divisor of a and n divides 1. The set of invertible classes,

$$(\mathbb{Z}/n\mathbb{Z})^\times := \{[a] : \gcd(a, n) = 1\},$$

is therefore an abelian group under multiplication, the *group of units modulo n* . We denote its order $\varphi(n)$, the *Euler totient*. For example $(\mathbb{Z}/8\mathbb{Z})^\times = \{[1], [3], [5], [7]\}$ has order $\varphi(8) = 4$; one checks $[3]^2 = [9] = [1]$, so every nonidentity element is its own inverse. When $n = p$ is prime, every nonzero class is a unit and $(\mathbb{Z}/p\mathbb{Z})^\times = \mathbb{F}_p^\times$ has order $p - 1$; this is exactly Example 3.7 for the finite field $\mathbb{F}_p$.

Example 3.10 (Trivial group and small groups). The one-element set $\{e\}$ with the only possible operation is a group, the *trivial group*. The group $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ is the smallest nontrivial group. The *Klein four-group* $V = \{e, a, b, c\}$, in which every nonidentity element squares to e and the product of any two distinct nonidentity elements is the third, is abelian of order 4 and is *not* isomorphic to $\mathbb{Z}/4\mathbb{Z}$ (it has no element of order 4).

3.4 Order of an element

Definition 3.11. Let G be a group with identity e and let $a \in G$. The *order* of a , written $\text{ord}(a)$, is the least positive integer n such that $a^n = e$, if such an n exists; in that case a has *finite order*. If no positive power of a equals e , then a has *infinite order* and we write $\text{ord}(a) = \infty$.

The theory of cyclic groups below reconciles the two uses of the word “order”—for an element and for a group: $\text{ord}(a)$ equals the order of the smallest subgroup containing a .

Proposition 3.12. Let $a \in G$ have finite order $d = \text{ord}(a)$. Then for any integer m ,

$$a^m = e \iff d \mid m.$$

Consequently $a^m = a^k$ iff $m \equiv k \pmod{d}$, and the distinct powers of a are exactly $e = a^0, a^1, \dots, a^{d-1}$.

Proof. If $d \mid m$, write $m = dq$; then $a^m = (a^d)^q = e^q = e$. Conversely, suppose $a^m = e$. By the division algorithm write $m = dq + r$ with $0 \leq r < d$. Then

$$e = a^m = (a^d)^q a^r = a^r.$$

Since $0 \leq r < d$ and d is the *least* positive integer with $a^d = e$, it follows that $r = 0$, i.e. $d \mid m$. The remaining claims follow: $a^m = a^k \iff a^{m-k} = e \iff d \mid (m - k)$, and the d elements $a^0, \dots, a^{d-1}$ are pairwise distinct because their exponents are incongruent modulo d . $\square$

Example 3.13. In $(\mathbb{Z}/n\mathbb{Z}, +)$ the order of the element 1 is n , since $\underbrace{1 + \dots + 1}_k = k$ is 0 iff $n \mid k$.

The order of an element a is $n/\gcd(a, n)$ (Corollary 3.23). In $\mathbb{C}^\times$, the element i has order 4: $i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1$. In $\mathbb{Z}$ under addition, every nonzero element has infinite order.

3.5 Subgroups and the subgroup test

Definition 3.14. A subset H of a group $(G, *)$ is a *subgroup*, written $H \leq G$, if H is itself a group under the restriction of $*$ to $H \times H$. Explicitly: H is nonempty, closed under $*$ (so $*$ restricts to a binary operation on H), contains the identity e of G , and contains a^{-1} whenever it contains a .

Every group G has at least two subgroups: the *trivial subgroup* $\{e\}$ and G itself. We call a subgroup other than G *proper*. Note that the identity of a subgroup necessarily coincides with the identity of G (apply cancellation in G to $e_H * e_H = e_H = e_H * e$), and inverses computed in H agree with those in G ; thus the definition is not as redundant as it may appear.

Verifying all the group axioms for a candidate subgroup is wasteful, since associativity is inherited for free. The following test reduces the verification to a single condition.

Proposition 3.15 (One-step subgroup test). Let G be a group and $H \subseteq G$. Then $H \leq G$ if and only if

1. $H \neq \emptyset$, and
2. for all $a, b \in H$, the element $ab^{-1} \in H$.

Proof. ($\Rightarrow$) If H is a subgroup it is nonempty and, given $a, b \in H$, it contains b^{-1} and hence ab^{-1} .

($\Leftarrow$) Assume (1) and (2). Pick any $a \in H$ (possible by (1)). Taking $b = a$ in (2) gives $aa^{-1} = e \in H$. Taking $a = e$ and b arbitrary in H gives $eb^{-1} = b^{-1} \in H$, so H is closed under inverses. Finally, for $a, b \in H$ we now know $b^{-1} \in H$, so applying (2) to the pair (a, b^{-1}) yields $a(b^{-1})^{-1} = ab \in H$; thus H is closed under the operation. Associativity is inherited from G . Hence H is a group, i.e. $H \leq G$. $\square$

Remark 3.16. For a *finite* nonempty subset H closed under the operation, closure alone forces $H \leq G$: given $a \in H$, the powers $a, a^2, a^3, \dots$ all lie in H and cannot all be distinct (as H is finite), so $a^i = a^j$ for some $i < j$, giving $a^{j-i} = e \in H$ and $a^{-1} = a^{j-i-1} \in H$. Thus for finite subsets, closure under multiplication is the only condition one must verify.

Example 3.17. The even integers $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ form a subgroup of $(\mathbb{Z}, +)$; more generally $n\mathbb{Z} \leq \mathbb{Z}$ for every $n \geq 0$, and these are *all* the subgroups of $\mathbb{Z}$ (Proposition 3.26). The special linear group $\mathrm{SL}_n(\mathbb{F}) \leq \mathrm{GL}_n(\mathbb{F})$ since $\det(AB^{-1}) = \det(A)/\det(B) = 1$ when $\det A = \det B = 1$. The set of *rotations* among the symmetries of a regular polygon is a subgroup of the full symmetry group.

Proposition 3.18. *The intersection of any family $\{H_i\}_{i \in I}$ of subgroups of G is a subgroup of G .*

Proof. Let $H = \bigcap_i H_i$. Each H_i contains e , so $e \in H$ and $H \neq \emptyset$. If $a, b \in H$ then $a, b \in H_i$ for every i , so $ab^{-1} \in H_i$ for every i by the subgroup test, whence $ab^{-1} \in H$. By Proposition 3.15, $H \leq G$. $\square$

This lets us define the subgroup “generated” by an arbitrary subset as the smallest subgroup containing it.

Definition 3.19. Let $S \subseteq G$. The *subgroup generated by S* , written $\langle S \rangle$, is the intersection of all subgroups of G that contain S . By Proposition 3.18 it is a subgroup, and it is the smallest subgroup containing S . Concretely, $\langle S \rangle$ consists of all finite products $s_1^{\epsilon_1} \cdots s_k^{\epsilon_k}$ with $s_i \in S$ and $\epsilon_i \in \{+1, -1\}$ (the empty product being e).

3.6 Cyclic groups, generators, and their classification

The simplest groups are those generated by a single element.

Definition 3.20. A group G is *cyclic* if there exists $g \in G$ with $G = \langle g \rangle$, where

$$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$$

is the subgroup generated by g . We call such a g a *generator* of G . Every cyclic group is abelian, since $g^m g^n = g^{m+n} = g^n g^m$.

The description $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$ matches Definition 3.19: the right-hand side is closed under products and inverses and contains g , and any subgroup containing g must contain all its powers.

Proposition 3.21. *Let $G = \langle g \rangle$ be cyclic.*

1. *If g has infinite order, then the powers g^k ($k \in \mathbb{Z}$) are pairwise distinct, so G is infinite and $|G| = \infty$.*
2. *If g has finite order d , then $G = \{e, g, g^2, \dots, g^{d-1}\}$ has exactly d elements, so $|G| = \mathrm{ord}(g) = d$.*

In both cases $|\langle g \rangle| = \mathrm{ord}(g)$.

Proof. (1) If g has infinite order and $g^i = g^j$ with $i \neq j$, then $g^{i-j} = e$ with $i-j \neq 0$, contradicting infinite order; so all powers are distinct. (2) is Proposition 3.12: the distinct powers are exactly $g^0, \dots, g^{d-1}$. $\square$

Example 3.22. $(\mathbb{Z}, +)$ is infinite cyclic, generated by 1 (and also by -1). $(\mathbb{Z}/n\mathbb{Z}, +)$ is cyclic of order n , generated by 1. The group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ is *not* cyclic: every element squares to 1, so no element has order 4. By contrast $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ is cyclic, generated by 2: $2^1 = 2, 2^2 = 4, 2^3 = 3, 2^4 = 1$.

We now determine *which* powers of a generator are themselves generators, and classify the subgroups of a cyclic group.

Corollary 3.23. *Let g have finite order n , and let $k \in \mathbb{Z}$. Then*

$$\text{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

In particular g^k generates $\langle g \rangle$ if and only if $\gcd(n, k) = 1$, so a cyclic group of order n has exactly $\varphi(n)$ generators.

Proof. Let $d = \gcd(n, k)$ and write $k = dk', n = dn'$ with $\gcd(n', k') = 1$. Then $(g^k)^m = g^{km} = e$ iff $n \mid km$, i.e. iff $n' \mid k'm$; since $\gcd(n', k') = 1$, this holds iff $n' \mid m$. The least positive such m is $n' = n/d$, so $\text{ord}(g^k) = n/\gcd(n, k)$. Now g^k generates $\langle g \rangle$ iff $\text{ord}(g^k) = n$ iff $\gcd(n, k) = 1$. The number of k with $1 \leq k \leq n$ and $\gcd(n, k) = 1$ is $\varphi(n)$. $\square$

Theorem 3.24 (Classification of subgroups of cyclic groups). *Let $G = \langle g \rangle$ be cyclic.*

1. *Every subgroup of G is cyclic.*
2. *If G is infinite, its subgroups are exactly $\langle g^m \rangle$ for $m \geq 0$; distinct $m \geq 0$ give distinct subgroups, and $\langle g^m \rangle \cong (\mathbb{Z}, +)$ for $m \geq 1$.*
3. *If $|G| = n$, then for each positive divisor $d \mid n$ there is exactly one subgroup of order d , namely $\langle g^{n/d} \rangle$, and these are all the subgroups. Hence the subgroups of G are in bijection with the positive divisors of n .*

Proof. (1) Let $H \leq G$. If $H = \{e\}$ it is cyclic. Otherwise H contains some g^k with $k \neq 0$; since H is closed under inverses it contains g^{-k} too, so H contains a positive power of g . Let m be the *least* positive integer with $g^m \in H$. We claim that $H = \langle g^m \rangle$. Clearly $\langle g^m \rangle \subseteq H$. Conversely take any $g^t \in H$; by the division algorithm $t = mq + r$ with $0 \leq r < m$, so

$$g^r = g^{t-mq} = g^t(g^m)^{-q} \in H.$$

By minimality of m and $0 \leq r < m$, it follows that $r = 0$, so $g^t = (g^m)^q \in \langle g^m \rangle$. Hence $H = \langle g^m \rangle$ is cyclic.

(2) If G is infinite, then g has infinite order, so the subgroups $\langle g^m \rangle$ ($m \geq 1$) are all infinite cyclic, and $\langle g^0 \rangle = \{e\}$. They are distinct because $\langle g^m \rangle$ has g^m as its least positive power but does not contain g^j for $0 < j < m$. Part (1) shows these are all subgroups.

(3) Suppose $|G| = n$, so $\text{ord}(g) = n$. Let $d \mid n$. The element $g^{n/d}$ has order $\text{ord}(g^{n/d}) = n/\gcd(n, n/d) = n/(n/d) = d$ by Corollary 3.23, so $\langle g^{n/d} \rangle$ is a subgroup of order d . For uniqueness, let $H \leq G$ with $|H| = d$. By (1), $H = \langle g^m \rangle$ for the least positive m with $g^m \in H$; and by Lagrange's theorem (proved below, but a direct argument suffices here) $|H| = \text{ord}(g^m) = n/\gcd(n, m) = d$, so $\gcd(n, m) = n/d$. Writing $m = (n/d)m'$, every element of H is a power of $g^{n/d}$ raised to a power, hence $H \subseteq \langle g^{n/d} \rangle$; comparing orders gives $H = \langle g^{n/d} \rangle$. Thus there is exactly one subgroup of each order $d \mid n$, and by (1) every subgroup arises this way. $\square$

Remark 3.25. Specialised to $G = \mathbb{Z}/n\mathbb{Z}$, part (3) states: for each $d \mid n$ there is a unique subgroup of order d , namely $\langle n/d \rangle = \{0, n/d, 2n/d, \dots\}$, which is itself isomorphic to $\mathbb{Z}/d\mathbb{Z}$. For $n = 12$, the subgroup lattice mirrors the divisor lattice $1 \mid 2 \mid 4, 1 \mid 2 \mid 6, 3 \mid 6, \dots$ of 12.

Proposition 3.26. *Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z}$ for a unique integer $n \geq 0$.*

Proof. This is Theorem 3.24(2) read additively; we give the direct argument. Let $H \leq \mathbb{Z}$. If $H = \{0\}$ then $H = 0\mathbb{Z}$. Otherwise H contains a nonzero element and hence (being closed under negation) a positive one; let n be the least positive element of H . Then $n\mathbb{Z} \subseteq H$. For any $h \in H$, division gives $h = nq + r$ with $0 \leq r < n$, and $r = h - nq \in H$; minimality forces $r = 0$, so $h \in n\mathbb{Z}$. Thus $H = n\mathbb{Z}$. Uniqueness is clear since one recovers n as the least positive element of H . $\square$

3.7 Cosets and Lagrange's theorem

Definition 3.27. Let $H \leq G$ and $a \in G$. The *left coset* of H by a is

$$aH := \{ah : h \in H\},$$

and the *right coset* is $Ha := \{ha : h \in H\}$. We denote the set of left cosets G/H , and the number of left cosets, $[G : H] := |G/H|$, is the *index* of H in G .

Cosets are the blocks of a partition, and they all have the same size. This is the mechanism underlying Lagrange's theorem.

Lemma 3.28. *Let $H \leq G$. Then:*

1. $a \in aH$ for every a ; the cosets cover G .
2. Two left cosets are either equal or disjoint: $aH = bH$ if and only if $a^{-1}b \in H$, and otherwise $aH \cap bH = \emptyset$.
3. Every left coset has the same cardinality as H ; the map $h \mapsto ah$ is a bijection $H \rightarrow aH$.

The same statements hold for right cosets, with $ba^{-1} \in H$ in (2).

Proof. (1) $a = ae \in aH$ since $e \in H$. Hence the union of all cosets is G .

(2) Suppose $aH \cap bH \neq \emptyset$, say $ah_1 = bh_2$ with $h_1, h_2 \in H$. Then $a^{-1}b = h_1h_2^{-1} \in H$. Conversely if $a^{-1}b = h \in H$, then $b = ah$, so for any $h' \in H$, $bh' = a(hh') \in aH$, giving $bH \subseteq aH$; by symmetry ($b^{-1}a = h^{-1} \in H$) we obtain $aH \subseteq bH$, so $aH = bH$. Thus two cosets intersect only if they coincide; equivalently, distinct cosets are disjoint.

(3) The map $\lambda_a : H \rightarrow aH$, $h \mapsto ah$, is surjective by definition of aH and injective by left-cancellation ($ah = ah' \Rightarrow h = h'$). Hence $|aH| = |H|$. $\square$

Theorem 3.29 (Lagrange). *Let G be a finite group and $H \leq G$. Then $|H|$ divides $|G|$, and*

$$|G| = [G : H] |H|.$$

Proof. By Lemma 3.28, the distinct left cosets of H partition G into $[G : H]$ blocks, each of cardinality $|H|$. Summing the sizes of the blocks,

$$|G| = \sum_{\text{distinct cosets } aH} |aH| = [G : H] \cdot |H|.$$

In particular $|H|$ divides $|G|$. $\square$

The corollaries are immediate but far-reaching.

Corollary 3.30. *In a finite group G , the order of every element divides $|G|$.*

Proof. For $a \in G$, the cyclic subgroup $\langle a \rangle$ has order $\text{ord}(a)$ by Proposition 3.21. By Lagrange, $\text{ord}(a) = |\langle a \rangle|$ divides $|G|$. $\square$

Corollary 3.31. *If G is finite of order N , then $a^N = e$ for every $a \in G$.*

Proof. Let $d = \text{ord}(a)$. By Corollary 3.30, $d \mid N$, say $N = dm$. Then $a^N = (a^d)^m = e^m = e$. $\square$

Corollary 3.32 (Euler, Fermat). *For $\gcd(a, n) = 1$ one has $a^{\varphi(n)} \equiv 1 \pmod{n}$. In particular, for a prime p and any a not divisible by p , $a^{p-1} \equiv 1 \pmod{p}$, and $a^p \equiv a \pmod{p}$ for all a .*

Proof. Apply Corollary 3.31 to the group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$: the class $[a]$ satisfies $[a]^{\varphi(n)} = [1]$. For $n = p$ prime, $\varphi(p) = p - 1$, yielding Fermat's little theorem; multiplying $a^{p-1} \equiv 1$ by a gives $a^p \equiv a$, which holds even when $p \mid a$ since both sides are then 0. $\square$

Corollary 3.33. *Every group of prime order p is cyclic, hence isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and has no proper nontrivial subgroups.*

Proof. Let $|G| = p$ and pick $a \neq e$. Then $\langle a \rangle$ has order > 1 dividing p , so $|\langle a \rangle| = p = |G|$, forcing $\langle a \rangle = G$. Thus G is cyclic. Any subgroup has order dividing p , hence order 1 or p . The isomorphism with $\mathbb{Z}/p\mathbb{Z}$ follows from the classification of cyclic groups (Theorem 3.43). $\square$

Remark 3.34. The converse of Lagrange's theorem is *false* in general: a divisor d of $|G|$ need not be the order of any subgroup. The smallest counterexample is the alternating group A_4 of order 12, which has no subgroup of order 6. Partial converses do hold—Cauchy's theorem guarantees a subgroup (indeed an element) of order p for every prime $p \mid |G|$, and Sylow's theorems control subgroups of prime-power order—but they are not needed here.

3.8 Homomorphisms, kernels, and images

We compare groups by means of maps that respect their operations.

Definition 3.35. Let $(G, *)$ and $(G', \circ)$ be groups. A *group homomorphism* is a function $\varphi: G \rightarrow G'$ such that

$$\varphi(a * b) = \varphi(a) \circ \varphi(b) \quad \text{for all } a, b \in G.$$

We denote the set of all homomorphisms $G \rightarrow G'$ by $\text{Hom}(G, G')$.

Proposition 3.36. *Let $\varphi: G \rightarrow G'$ be a homomorphism, with identities $e \in G$, $e' \in G'$. Then*

1. $\varphi(e) = e'$;
2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for all a ;
3. $\varphi(a^n) = \varphi(a)^n$ for all $n \in \mathbb{Z}$;
4. the image $\varphi(G)$ is a subgroup of G' .

Proof. (1) From $\varphi(e) = \varphi(e * e) = \varphi(e) \circ \varphi(e)$; cancelling $\varphi(e)$ in G' gives $e' = \varphi(e)$.

(2) $\varphi(a) \circ \varphi(a^{-1}) = \varphi(a * a^{-1}) = \varphi(e) = e'$, and likewise on the other side, so $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.

(3) For $n \geq 0$ by induction using the homomorphism property; for $n < 0$ combine with (2).

(4) Nonempty since $e' = \varphi(e) \in \varphi(G)$. If $x = \varphi(a)$, $y = \varphi(b)$ lie in $\varphi(G)$, then $xy^{-1} = \varphi(a)\varphi(b)^{-1} = \varphi(ab^{-1}) \in \varphi(G)$. By the subgroup test, $\varphi(G) \leq G'$. $\square$

Definition 3.37. The *kernel* of a homomorphism $\varphi: G \rightarrow G'$ is

$$\ker \varphi := \{a \in G : \varphi(a) = e'\} = \varphi^{-1}(\{e'\}),$$

and its *image* is $\text{im } \varphi := \varphi(G) = \{\varphi(a) : a \in G\}$.

Proposition 3.38. *Let $\varphi: G \rightarrow G'$ be a homomorphism. Then $\ker \varphi \leq G$, and φ is injective if and only if $\ker \varphi = \{e\}$.*

Proof. $\ker \varphi$ is nonempty ($e \in \ker \varphi$ by Proposition 3.36(1)). If $a, b \in \ker \varphi$, then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e'(e')^{-1} = e'$, so $ab^{-1} \in \ker \varphi$; thus $\ker \varphi \leq G$.

If φ is injective, then $\varphi(a) = e' = \varphi(e)$ forces $a = e$, so $\ker \varphi = \{e\}$. Conversely, suppose $\ker \varphi = \{e\}$ and $\varphi(a) = \varphi(b)$. Then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e'$, so $ab^{-1} \in \ker \varphi = \{e\}$, giving $ab^{-1} = e$, i.e. $a = b$. Hence φ is injective. $\square$

The kernel is not merely a subgroup; it is a subgroup of a particular kind, namely one *closed under conjugation*. This property is precisely what renders the quotient construction valid.

Definition 3.39. A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gng^{-1} \in N$ for all $g \in G$ and $n \in N$; equivalently $gN = Ng$ for all g (left and right cosets coincide), equivalently $gNg^{-1} = N$ for all g .

In an abelian group *every* subgroup is normal, since $gng^{-1} = gg^{-1}n = n$.

Proposition 3.40. *The kernel of any homomorphism $\varphi: G \rightarrow G'$ is a normal subgroup of G .*

Proof. By the above, $\ker \varphi \leq G$. For $g \in G$ and $n \in \ker \varphi$,

$$\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g)e'\varphi(g)^{-1} = e',$$

so $gng^{-1} \in \ker \varphi$. Hence $\ker \varphi \trianglelefteq G$. $\square$

3.9 Isomorphisms

Definition 3.41. A homomorphism $\varphi: G \rightarrow G'$ that is bijective is an *isomorphism*; we then write $G \cong G'$ and call G, G' *isomorphic*. An isomorphism from G to itself is an *automorphism*. A homomorphism is a *monomorphism* if injective and an *epimorphism* if surjective.

Proposition 3.42. *If $\varphi: G \rightarrow G'$ is an isomorphism, so is its set-theoretic inverse $\varphi^{-1}: G' \rightarrow G$. The relation $\cong$ is an equivalence relation on any collection of groups.*

Proof. Let $x, y \in G'$, say $x = \varphi(a)$, $y = \varphi(b)$. Then $\varphi^{-1}(xy) = \varphi^{-1}(\varphi(a)\varphi(b)) = \varphi^{-1}(\varphi(ab)) = ab = \varphi^{-1}(x)\varphi^{-1}(y)$, so φ^{-1} is a homomorphism, and it is bijective. Reflexivity of $\cong$ uses the identity automorphism, symmetry uses φ^{-1} , and transitivity uses that a composite of isomorphisms is an isomorphism (the composite of homomorphisms is a homomorphism, and the composite of bijections is a bijection). $\square$

Isomorphic groups are “the same” for all group-theoretic purposes: every property expressible in terms of the operation transfers across an isomorphism. For this reason we speak of classifying groups *up to isomorphism*.

Theorem 3.43 (Classification of cyclic groups). *Every cyclic group is isomorphic to exactly one of the following:*

1. *the infinite cyclic group $(\mathbb{Z}, +)$, if it is infinite;*
2. *the finite cyclic group $(\mathbb{Z}/n\mathbb{Z}, +)$, if it has order n .*

In particular two cyclic groups are isomorphic if and only if they have the same order.

Proof. Let $G = \langle g \rangle$. Define $\varphi: \mathbb{Z} \rightarrow G$ by $\varphi(k) = g^k$. This is a homomorphism from $(\mathbb{Z}, +)$: $\varphi(k + \ell) = g^{k+\ell} = g^k g^\ell = \varphi(k)\varphi(\ell)$. It is surjective since G consists exactly of the powers of g .

If g has infinite order, φ is injective by Proposition 3.21(1) (distinct exponents give distinct powers), so φ is an isomorphism $\mathbb{Z} \cong G$.

If g has order n , then $\ker \varphi = \{k : g^k = e\} = n\mathbb{Z}$ by Proposition 3.12. The induced map $\bar{\varphi}: \mathbb{Z}/n\mathbb{Z} \rightarrow G$, $[k] \mapsto g^k$, is well defined (if $[k] = [\ell]$ then $n \mid k - \ell$, so $g^k = g^\ell$), a homomorphism, surjective, and injective (if $g^k = e$ then $n \mid k$, so $[k] = [0]$). Hence $\mathbb{Z}/n\mathbb{Z} \cong G$. Finally, $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$ are pairwise non-isomorphic for distinct n since isomorphic groups have equal order. $\square$

4 Rings and fields

The preceding development treated groups: sets equipped with a single associative binary operation possessing an identity and inverses. Most of the structures of genuine interest in cryptography, however, support *two* interacting operations, an addition and a multiplication, that together obey the familiar arithmetic laws of $\mathbb{Z}$. The integers, the rationals, the residues modulo n , polynomials, and matrices are all of this kind. The abstract structure that captures these examples is the *ring*, and the most important special case — the one in which division by every nonzero element is possible — is the *field*. Fields are the natural arena for linear algebra and for the theory of polynomials, and the finite fields $\mathbb{F}_p$ and $\mathbb{F}_q$ are the computational substrate of essentially all of the elliptic-curve and proof-system machinery developed later in this monograph. This section builds the theory from the ground up.

Throughout, we assume only the theory of abelian groups developed earlier. We write the additive structure of a ring additively, with identity element 0, and the multiplicative structure multiplicatively, with (when it exists) identity element 1.

4.1 Rings and commutative rings

Definition 4.1. A *ring* is a set R equipped with two binary operations, *addition* $+: R \times R \rightarrow R$ and *multiplication* $\cdot: R \times R \rightarrow R$, satisfying the following axioms.

1. **Additive abelian group.** $(R, +)$ is an abelian group; that is, addition is associative and commutative, there is an element $0 \in R$ with $a + 0 = a$ for all a , and every $a \in R$ has an additive inverse $-a$ with $a + (-a) = 0$.
2. **Multiplicative associativity.** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$.
3. **Multiplicative identity.** There is an element $1 \in R$ with $1 \cdot a = a \cdot 1 = a$ for all $a \in R$.
4. **Distributivity.** For all $a, b, c \in R$,

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

A ring R is *commutative* if in addition $a \cdot b = b \cdot a$ for all $a, b \in R$.

We customarily write ab for $a \cdot b$, adopting the convention that multiplication binds more tightly than addition, so that $ab + ac$ means $(ab) + (ac)$. The element 0 is the *additive identity* or *zero*, and 1 is the *multiplicative identity* or *unity*.

Remark 4.2. Conventions in the literature vary. Some authors do not require a multiplicative identity 1, reserving the term *ring with unity* for the structure of Definition 4.1 and using *rng* (“ring without the *i* of identity”) for the weaker notion. In this monograph *every ring has a 1*, every ring homomorphism preserves it, and “ring” will frequently mean “commutative ring” in the examples that matter to us; we always state commutativity explicitly when a result depends on it.

Before we produce examples, we record the elementary arithmetic consequences of the axioms. These are the rules we use constantly and without comment; it is worth establishing once that they all follow from distributivity.

Proposition 4.3. *Let R be a ring and let $a, b \in R$. Then:*

1. $0 \cdot a = a \cdot 0 = 0$;
2. $(-a)b = a(-b) = -(ab)$;
3. $(-a)(-b) = ab$;

4. more generally, for integers m, n (interpreting na as the n -fold sum in the additive group),
 $(ma)(nb) = (mn)(ab)$.

Proof. (1) Since $0 + 0 = 0$, distributivity gives $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$. Adding $-(0 \cdot a)$ to both sides yields $0 = 0 \cdot a$. The argument for $a \cdot 0$ is symmetric.

(2) One computes $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$ by distributivity and part (1). Hence $(-a)b$ is the additive inverse of ab , i.e. $(-a)b = -(ab)$. The identity $a(-b) = -(ab)$ is symmetric.

(3) Applying (2) twice, $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$, where the last step uses that negation is an involution in the abelian group $(R, +)$.

(4) This follows from (2) and repeated distributivity (a routine induction on $|m|$ and $|n|$); we omit the bookkeeping. $\square$

Remark 4.4. If $1 = 0$ in a ring R , then for every $a \in R$ one has $a = 1 \cdot a = 0 \cdot a = 0$ by Proposition 4.3(1), so $R = \{0\}$. This single-element structure is the *zero ring* (or *trivial ring*); it is a perfectly legitimate ring. Whenever we wish to exclude it we impose the assumption $1 \neq 0$, and indeed the definition of a field below builds in the standing assumption $1 \neq 0$.

Example 4.5. We list the basic examples; the reader should verify the axioms in each case.

1. **The integers** $\mathbb{Z}$ with ordinary addition and multiplication form a commutative ring. This is the prototype, and much of ring theory is an attempt to isolate which features of $\mathbb{Z}$ survive in greater generality.
2. **The rationals** $\mathbb{Q}$, **the reals** $\mathbb{R}$, and **the complexes** $\mathbb{C}$ are commutative rings (indeed fields, see §4.4).
3. **The integers modulo n** , $\mathbb{Z}/n\mathbb{Z}$. For an integer $n \geq 1$, the set of residue classes $\{0, 1, \dots, n-1\}$ with addition and multiplication induced from $\mathbb{Z}$ is a commutative ring, the additive group being the cyclic group of order n . One writes $\mathbb{Z}/n\mathbb{Z}$ (or $\mathbb{Z}_n$ when no confusion with n -adic integers can arise). The case $n = 1$ gives the zero ring.
4. **Polynomial rings.** If R is a commutative ring, the set $R[x]$ of polynomials $\sum_{i=0}^d a_i x^i$ with coefficients $a_i \in R$, under the usual addition and multiplication, is a commutative ring. A later section treats polynomials systematically; here we note only that $R[x]$ is again a commutative ring with the same 0 and 1 as R (viewed as constant polynomials).
5. **Matrix rings.** For $n \geq 1$ the set $M_n(R)$ of $n \times n$ matrices over a ring R , with matrix addition and matrix multiplication, is a ring. For $n \geq 2$ it is *not* commutative even when R is: for instance $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ while $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. This is the first important example of a noncommutative ring.
6. **Product rings.** If R and S are rings, the Cartesian product $R \times S$ with componentwise operations $(r, s) + (r', s') = (r + r', s + s')$ and $(r, s)(r', s') = (rr', ss')$ is a ring, with zero $(0, 0)$ and identity $(1, 1)$. More generally any finite product $R_1 \times \dots \times R_k$ is a ring.
7. **The zero ring** $\{0\}$ of Remark 4.4.

Remark 4.6. The Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$ and, more generally, rings of the form $\mathbb{Z}[\alpha]$ for an algebraic number α , are commutative rings that sit strictly between $\mathbb{Z}$ and the fields containing them. Such rings are fundamental to algebraic number theory and reappear in the study of the endomorphism rings of elliptic curves. We mention them now only to widen the reader's stock of examples.

4.2 Units, zero divisors, and integral domains

The integers $\mathbb{Z}$ enjoy two properties beyond the bare ring axioms that will guide the rest of this section: a product of nonzero integers is nonzero (*cancellation*), and the only integers by which one may divide within $\mathbb{Z}$ are ± 1 . We now abstract both.

Definition 4.7. Let R be a ring. An element $u \in R$ is a *unit* (or is *invertible*) if there exists $v \in R$ with $uv = vu = 1$. The element v is then unique (if $uv = vu = 1$ and $uv' = v'u = 1$ then $v = v(uv') = (vu)v' = v'$), is called the *inverse* of u , and is written u^{-1} . We denote the set of units of R by $R^\times$.

Proposition 4.8. For any ring R , the set $R^\times$ of units is a group under multiplication, the group of units (or multiplicative group) of R .

Proof. The identity 1 is a unit (it is its own inverse), so $1 \in R^\times$. If $u, w \in R^\times$ with inverses u^{-1}, w^{-1} , then $(uw)(w^{-1}u^{-1}) = u(ww^{-1})u^{-1} = uu^{-1} = 1$ and symmetrically $(w^{-1}u^{-1})(uw) = 1$, so $uw \in R^\times$ with $(uw)^{-1} = w^{-1}u^{-1}$; thus $R^\times$ is closed under multiplication. Associativity is inherited from R . Finally every $u \in R^\times$ has the inverse $u^{-1} \in R^\times$ (whose inverse is u). Hence $R^\times$ is a group. $\square$

Example 4.9. 1. $\mathbb{Z}^\times = \{1, -1\}$.

2. $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, and likewise $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$, $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. In a field, every nonzero element is a unit; this is the defining feature, taken up in §4.4.

3. In $\mathbb{Z}/n\mathbb{Z}$, the class $\bar{a}$ is a unit if and only if $\gcd(a, n) = 1$. Indeed, $\bar{a}$ is a unit iff there is b with $ab \equiv 1 \pmod{n}$, i.e. $ab + kn = 1$ for some integer k , which by Bézout's identity holds iff $\gcd(a, n) = 1$. Thus $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the residues coprime to n ; its order is $\varphi(n)$, Euler's totient.

4. $M_n(R)^\times$ is the group of invertible matrices. When R is a field this is the *general linear group* GL_n , consisting of the matrices of nonzero determinant.

We now isolate the failure of cancellation.

Definition 4.10. Let R be a ring. A nonzero element $a \in R$ is a *left zero divisor* if there exists a nonzero $b \in R$ with $ab = 0$; *right zero divisor* is defined symmetrically with $ba = 0$. In a commutative ring the two notions coincide and one simply says *zero divisor*. An element that is neither a left nor a right zero divisor (and is nonzero) is called *regular* or *cancellable*.

The following observation justifies the terminology, making precise the sense in which zero divisors obstruct “dividing out”.

Proposition 4.11. Let R be a ring and $a \in R$ a nonzero element. Then a is not a left zero divisor if and only if it is left-cancellable: $ab = ac$ implies $b = c$ for all $b, c \in R$.

Proof. Suppose a is not a left zero divisor and $ab = ac$. Then $a(b - c) = ab - ac = 0$ by distributivity and Proposition 4.3(2). Since a is not a left zero divisor and $a \neq 0$, the factor $b - c$ must be 0, i.e. $b = c$. Conversely, if a is a left zero divisor, say $ab = 0$ with $b \neq 0$, then $ab = a \cdot 0$ but $b \neq 0$, so cancellation fails. $\square$

Remark 4.12. A unit is never a zero divisor: if u is a unit and $ub = 0$, then $b = u^{-1}(ub) = u^{-1} \cdot 0 = 0$. The converse fails in general — in $\mathbb{Z}$ every nonzero element is a non-zero-divisor, but only ± 1 are units — though it does hold in finite rings, as Theorem 4.28 establishes.

Example 4.13. 1. In $\mathbb{Z}/6\mathbb{Z}$, $\bar{2} \cdot \bar{3} = \bar{6} = \bar{0}$, so $\bar{2}$ and $\bar{3}$ are zero divisors. More generally $\bar{a} \neq \bar{0}$ is a zero divisor in $\mathbb{Z}/n\mathbb{Z}$ precisely when $\gcd(a, n) > 1$; thus in $\mathbb{Z}/n\mathbb{Z}$ every nonzero element is either a unit or a zero divisor (Theorem 4.28 revisits this dichotomy).

2. In the product ring $R \times S$ with both factors nonzero, $(1, 0)(0, 1) = (0, 0)$, so product rings always have zero divisors.
3. In $M_2(\mathbb{R})$, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, so singular nonzero matrices are zero divisors.

The rings with no zero divisors are precisely those in which the cancellation law of $\mathbb{Z}$ holds; in the commutative case they receive a name.

Definition 4.14. An *integral domain* (or simply a *domain*) is a commutative ring R with $1 \neq 0$ and no zero divisors: for all $a, b \in R$, $ab = 0$ implies $a = 0$ or $b = 0$.

Equivalently, by Proposition 4.11, an integral domain is a nonzero commutative ring in which every nonzero element is cancellable.

Example 4.15. $\mathbb{Z}$ is an integral domain — this is the example that names the concept. Every field is an integral domain (Proposition 4.27). The ring $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime, by Example 4.13(1) together with the fact that for $n = p$ prime, $ab \equiv 0 \pmod{p}$ forces $p \mid a$ or $p \mid b$ (Euclid's lemma). If R is an integral domain then so is the polynomial ring $R[x]$: the leading coefficient of a product of nonzero polynomials is the product of the leading coefficients, hence nonzero. By contrast $\mathbb{Z}/6\mathbb{Z}$, product rings, and $M_n(R)$ for $n \geq 2$ are not domains.

4.3 Ideals and quotient rings

The formation of quotients of rings — the ring-theoretic analogue of the quotient groups built from normal subgroups — requires the appropriate notion of substructure by which to quotient. For rings this is the *ideal*. We treat this material briefly, as we need it chiefly to construct $\mathbb{Z}/n\mathbb{Z}$ conceptually and, later, the finite fields $\mathbb{F}_q$; a fuller treatment belongs to a course in commutative algebra.

Definition 4.16. Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* if

1. I is an additive subgroup of $(R, +)$: $0 \in I$, and $a - b \in I$ whenever $a, b \in I$; and
2. I is closed under multiplication by arbitrary ring elements: $ra \in I$ whenever $r \in R$ and $a \in I$ (the *absorption* property).

In a noncommutative ring one distinguishes left, right, and two-sided ideals; for the present purposes commutative rings suffice, where all three coincide. Note that an ideal is almost never a subring: if an ideal I contains 1, then by absorption $r = r \cdot 1 \in I$ for every $r \in R$, so $I = R$. Thus the only ideal containing a unit is R itself.

Example 4.17. 1. In any commutative ring R , the subsets $\{0\}$ and R are ideals, the *trivial* and *improper* ideals respectively.

2. For a fixed $a \in R$, the set $(a) := aR = \{ar : r \in R\}$ is an ideal, the *principal ideal generated by a* . In $\mathbb{Z}$ the principal ideal $(n) = n\mathbb{Z}$ consists of all multiples of n . In fact *every* ideal of $\mathbb{Z}$ is principal: an ideal $I \neq \{0\}$ contains a least positive element n , and the division algorithm shows $I = n\mathbb{Z}$ (any $a \in I$ gives $a = qn + r$ with $0 \leq r < n$, whence $r = a - qn \in I$ forces $r = 0$). A domain in which every ideal is principal is a *principal ideal domain*.
3. More generally, for $a_1, \dots, a_k \in R$, the set $(a_1, \dots, a_k) = \{\sum_i r_i a_i : r_i \in R\}$ is the ideal *generated by the a_i* .

The significance of ideals is that they are exactly the subsets by which one may quotient.

Definition 4.18. Let I be an ideal of a commutative ring R . Since $(I, +)$ is a subgroup of the abelian group $(R, +)$, it is normal, and we may form the quotient group R/I of additive cosets $a + I$. We define multiplication of cosets by

$$(a + I)(b + I) := ab + I.$$

Proposition 4.19. *With the operations of Definition 4.18, R/I is a commutative ring, the quotient ring of R by I , with zero $0 + I = I$ and identity $1 + I$. In particular the multiplication is well defined.*

Proof. We must check that the product of cosets does not depend on the chosen representatives. Suppose $a + I = a' + I$ and $b + I = b' + I$, so that $a' = a + s$ and $b' = b + t$ with $s, t \in I$. Then

$$a'b' = (a + s)(b + t) = ab + at + sb + st.$$

The three terms at , sb , and st all lie in I : $at \in I$ and $st \in I$ by absorption ($t \in I$), and $sb \in I$ by absorption ($s \in I$). Hence $a'b' - ab \in I$, i.e. $a'b' + I = ab + I$, and multiplication is well defined. The ring axioms for R/I are then inherited coset-wise from those of R : for instance distributivity reads $(a + I)((b + I) + (c + I)) = (a(b + c)) + I = (ab + ac) + I = (a + I)(b + I) + (a + I)(c + I)$. Commutativity, associativity, and the identity laws follow identically. $\square$

Example 4.20. Taking $R = \mathbb{Z}$ and $I = n\mathbb{Z}$, we recover the quotient ring $\mathbb{Z}/n\mathbb{Z}$ of Example 4.5(3) as a quotient ring: its elements are the cosets $a + n\mathbb{Z}$, i.e. the residue classes modulo n , and the coset operations are exactly residue arithmetic. This is the conceptual origin of modular arithmetic, and it generalises: quotients of polynomial rings $\mathbb{F}_p[x]/(f)$ by an irreducible polynomial f will furnish the finite fields $\mathbb{F}_q$ with $q = p^k$, as developed in a later section.

Two special classes of ideals govern the interaction between properties of I and properties of the quotient R/I .

Definition 4.21. Let $I \neq R$ be a (proper) ideal of a commutative ring R .

- I is *prime* if $ab \in I$ implies $a \in I$ or $b \in I$.
- I is *maximal* if there is no ideal J with $I \subsetneq J \subsetneq R$.

Proposition 4.22. *Let I be a proper ideal of a commutative ring R . Then:*

1. R/I is an integral domain if and only if I is prime;
2. R/I is a field if and only if I is maximal.

Consequently every maximal ideal is prime.

Proof. (1) The quotient R/I has $1 \neq 0$ iff $I \neq R$, which holds by hypothesis. For $a, b \in R$, the equation $(a + I)(b + I) = 0 + I$ means $ab \in I$, while $a + I = 0 + I$ means $a \in I$. Thus “ R/I has no zero divisors” ($ab \in I \Rightarrow a \in I$ or $b \in I$) is verbatim the statement that I is prime.

(2) We defer the full proof to §4.4, where we characterise fields via their ideals (Proposition 4.31); the equivalence is then immediate, since the ideals of R/I correspond to the ideals of R containing I . Granting (2), every field is a domain (Proposition 4.27), so a maximal ideal makes R/I a field, hence a domain, hence is prime. $\square$

Example 4.23. In $\mathbb{Z}$, the ideal $(n) = n\mathbb{Z}$ is prime iff $n = 0$ or n is prime, and maximal iff n is prime. Indeed $\mathbb{Z}/p\mathbb{Z}$ is a field for p prime (§4.4), so (p) is maximal; $\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}$ is a domain but not a field, so (0) is prime but not maximal. This is the prototype of the slogan “maximal $\Rightarrow$ prime, but not conversely.”

4.4 Fields

We now introduce the central object of this section.

Definition 4.24. A *field* is a commutative ring F in which $1 \neq 0$ and every nonzero element is a unit. Equivalently, F is a set with two operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group with identity 0 , $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1 , and multiplication distributes over addition.

Unwinding the definition: in a field one may add, subtract, and multiply as in any commutative ring, and additionally *divide by any nonzero element*, with $a/b := ab^{-1}$. The group $(F \setminus \{0\}, \cdot) = F^\times$ is the *multiplicative group* of the field.

Remark 4.25. Dropping the requirement that multiplication be commutative while retaining the demand that every nonzero element have a multiplicative inverse yields a *division ring* (or *skew field*). The quaternions $\mathbb{H}$ are the standard example. By a theorem of Wedderburn, *every finite division ring is a field* (commutativity is automatic in the finite case); we do not require this fact here, but it harmonises with Theorem 4.28.

Example 4.26. 1. $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are fields; $\mathbb{Z}$ is not, since 2 has no inverse in $\mathbb{Z}$.

2. For p prime, $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ is a field, the *prime field* of characteristic p . We prove this in two ways below (Proposition 4.30 and, as a special case, Theorem 4.28). These fields, and their finite extensions $\mathbb{F}_q$ with $q = p^k$, are fundamental to the cryptography developed later.
3. The field $\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ and, more generally, number fields, are fields obtained by adjoining algebraic elements to $\mathbb{Q}$.
4. The field of rational functions $F(x) = \{f/g : f, g \in F[x], g \neq 0\}$ over a field F is a field; it is the field of fractions of $F[x]$.

Proposition 4.27. *Every field is an integral domain.*

Proof. Let F be a field; by definition $1 \neq 0$ and F is commutative. Suppose $ab = 0$ with $a \neq 0$. Since a is a unit, multiply by a^{-1} on the left: $b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0$. Hence $a = 0$ or $b = 0$, so F has no zero divisors. $\square$

The converse is false ($\mathbb{Z}$ is a domain but not a field), but it becomes true under finiteness — a fact of central importance for finite fields.

Theorem 4.28. *Every finite integral domain is a field.*

Proof. Let R be a finite integral domain and let $a \in R$ with $a \neq 0$; we must produce a multiplicative inverse for a . Consider the “multiplication by a ” map

$$\mu_a : R \rightarrow R, \quad \mu_a(x) = ax.$$

This map is injective: if $ax = ay$ then $a(x - y) = 0$, and since $a \neq 0$ and R is a domain, $x - y = 0$, i.e. $x = y$ (this is cancellation, Proposition 4.11). An injective map from a finite set to itself is necessarily surjective. Hence μ_a is onto, so there exists $b \in R$ with $ab = \mu_a(b) = 1$. Because R is commutative, $ba = 1$ as well, so $b = a^{-1}$. As a was an arbitrary nonzero element, every nonzero element is a unit, and R is a field. $\square$

Corollary 4.29. *For a positive integer n , the ring $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.*

Proof. If n is prime then $\mathbb{Z}/n\mathbb{Z}$ is a finite integral domain (Example 4.15), hence a field by Theorem 4.28. If $n = 1$ then $\mathbb{Z}/n\mathbb{Z}$ is the zero ring, not a field. If n is composite, say $n = ab$ with $1 < a, b < n$, then $\bar{a}\bar{b} = \bar{0}$ with $\bar{a}, \bar{b} \neq \bar{0}$, so $\mathbb{Z}/n\mathbb{Z}$ has zero divisors and is not even a domain, let alone a field. $\square$

A second, more arithmetic proof that $\mathbb{F}_p$ is a field follows, as the technique (Bézout) is itself useful and constructs inverses explicitly.

Proposition 4.30. *For p prime, every nonzero class $\bar{a} \in \mathbb{Z}/p\mathbb{Z}$ has a multiplicative inverse, computable from the extended Euclidean algorithm.*

Proof. Let $\bar{a} \neq \bar{0}$, so $p \nmid a$. Since p is prime, $\gcd(a, p) = 1$. By Bézout's identity there are integers u, v with $ua + vp = 1$; the extended Euclidean algorithm produces them. Reducing modulo p gives $\bar{u}\bar{a} = \bar{1}$, so $\bar{u} = \bar{a}^{-1}$. $\square$

We record the characterisation of fields in terms of ideals last, completing the deferred part of Proposition 4.22.

Proposition 4.31. *A commutative ring R with $1 \neq 0$ is a field if and only if its only ideals are $\{0\}$ and R .*

Proof. ($\Rightarrow$) Let F be a field and $I \subseteq F$ an ideal with $I \neq \{0\}$. Pick $a \in I$, $a \neq 0$. Then a is a unit, so $1 = a^{-1}a \in I$ by absorption, whence $I = F$. Thus the only ideals are $\{0\}$ and F .

($\Leftarrow$) Suppose the only ideals of R are $\{0\}$ and R , and let $a \in R$ be nonzero. The principal ideal $(a) = aR$ is not $\{0\}$ (it contains $a = a \cdot 1$), so $(a) = R$. In particular $1 \in (a)$, so $1 = ab$ for some $b \in R$; since R is commutative, $b = a^{-1}$. Hence every nonzero element is a unit and R is a field. $\square$

This establishes the equivalence “ I maximal $\iff R/I$ a field” of Proposition 4.22(2): the ideals of R/I correspond bijectively, and inclusion-preservingly, to the ideals of R containing I (a standard correspondence we assume here without proof), so R/I has only the two trivial ideals exactly when there is no ideal strictly between I and R .

4.5 The characteristic of a ring

There is a canonical way in which a copy of $\mathbb{Z}$ maps into any ring, and the size of its image is a fundamental invariant.

Definition 4.32. Let R be a ring. For an integer n and the identity $1 \in R$, write $n \cdot 1$ for the n -fold sum $1 + 1 + \cdots + 1$ (n terms) if $n > 0$, the element 0 if $n = 0$, and $-((-n) \cdot 1)$ if $n < 0$. The *characteristic* of R , written $\text{char}(R)$, is the least positive integer n such that $n \cdot 1 = 0$, if such an n exists; otherwise $\text{char}(R) = 0$.

Equivalently, the map $\iota: \mathbb{Z} \rightarrow R$ given by $\iota(n) = n \cdot 1$ is a ring homomorphism, and $\text{char}(R)$ is the nonnegative generator of its kernel, an ideal of $\mathbb{Z}$. Thus $\text{char}(R) = 0$ when ι is injective (a copy of $\mathbb{Z}$ sits inside R), and $\text{char}(R) = n$ when the kernel is $n\mathbb{Z}$.

Example 4.33. $\text{char}(\mathbb{Z}) = \text{char}(\mathbb{Q}) = \text{char}(\mathbb{R}) = \text{char}(\mathbb{C}) = 0$. For the residue rings, $\text{char}(\mathbb{Z}/n\mathbb{Z}) = n$, since $n \cdot \bar{1} = \bar{n} = \bar{0}$ and no smaller positive multiple of $\bar{1}$ vanishes. In particular $\text{char}(\mathbb{F}_p) = p$.

Theorem 4.34. *The characteristic of an integral domain (in particular of a field) is either 0 or a prime number.*

Proof. Let R be an integral domain and suppose $\text{char}(R) = n > 0$. First, $n \neq 1$: if $n = 1$ then $1 = 1 \cdot 1 = 0$, forcing R to be the zero ring, contrary to $1 \neq 0$ in a domain. Suppose, for contradiction, that n is composite, $n = ab$ with $1 < a, b < n$. Then in R ,

$$0 = n \cdot 1 = (ab) \cdot 1 = (a \cdot 1)(b \cdot 1),$$

using the distributive bookkeeping of Proposition 4.3(4). Since R is a domain, either $a \cdot 1 = 0$ or $b \cdot 1 = 0$. But $0 < a < n$ and $0 < b < n$, contradicting the minimality of $n = \text{char}(R)$. Hence n has no nontrivial factorisation, i.e. n is prime. $\square$

Remark 4.35. A field F of characteristic 0 contains a copy of $\mathbb{Z}$ (via ι) and hence, dividing, a copy of $\mathbb{Q}$; a field of characteristic p contains a copy of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. In either case this smallest subfield is the *prime subfield* of F . Every field is therefore an extension either of $\mathbb{Q}$ or of some $\mathbb{F}_p$. This dichotomy — characteristic zero versus characteristic p — pervades the subject; the fields $\mathbb{F}_q$ underlying the cryptography treated here all have prime characteristic p , where the “freshman’s dream” identity of the next proposition holds.

Proposition 4.36 (Frobenius / the freshman's dream). *Let R be a commutative ring of prime characteristic p . Then for all $a, b \in R$,*

$$(a + b)^p = a^p + b^p.$$

Consequently the map $\phi: R \rightarrow R$, $\phi(a) = a^p$, is a ring homomorphism, the Frobenius endomorphism.

Proof. By the binomial theorem (valid in any commutative ring),

$$(a + b)^p = \sum_{k=0}^p \binom{p}{k} a^k b^{p-k}.$$

For $0 < k < p$, the binomial coefficient $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ is divisible by p : the numerator $p!$ contains the factor p , while the denominator $k!(p-k)!$, being a product of integers all strictly less than p , is coprime to p ; since $\binom{p}{k}$ is an integer, p must divide it. In R , where $p \cdot 1 = 0$, every such middle term $\binom{p}{k} a^k b^{p-k}$ therefore vanishes. Only $k = 0$ and $k = p$ survive, giving $(a + b)^p = a^p + b^p$. That ϕ preserves products is the commutative-ring identity $(ab)^p = a^p b^p$, and $\phi(1) = 1$; together with the additive identity just proved, ϕ is a ring homomorphism. $\square$

5 Finite fields

A *field* is a set in which one can add, subtract, multiply, and divide by anything nonzero, with all the usual algebraic laws holding. The familiar examples $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are all infinite. This section develops the theory of *finite* fields: fields with only finitely many elements. These objects are the arithmetic substrate of essentially all modern public-key cryptography, and in particular of the elliptic-curve constructions underlying Halo 2 and Zcash Orchard. There the coordinates of curve points live in a prime field $\mathbb{F}_p$, while the scalars that multiply those points live in a second prime field $\mathbb{F}_q$; understanding both, and the relationship between them, begins with understanding finite fields in general.

We assume the reader has met the definition of a field, of a commutative ring with identity, and of a group. We recall the bare statements required as needed, so that the section is self-contained, but the emphasis is on the structure theory specific to the finite case.

5.1 The prime field $\mathbb{F}_p$ and modular arithmetic

Fix an integer $m \geq 1$. Two integers $a, b \in \mathbb{Z}$ are *congruent modulo m* , written $a \equiv b \pmod{m}$, if $m \mid (a - b)$, that is, if a and b leave the same remainder upon division by m . Congruence modulo m is an equivalence relation: it is reflexive ($m \mid 0$), symmetric ($m \mid (a - b)$ iff $m \mid (b - a)$), and transitive (if $m \mid (a - b)$ and $m \mid (b - c)$ then $m \mid (a - c)$). The equivalence class of a is

$$\bar{a} = a + m\mathbb{Z} = \{a + mk : k \in \mathbb{Z}\},$$

called the *residue class* of a modulo m . There are exactly m such classes, represented by $0, 1, \dots, m - 1$.

Definition 5.1 (The ring $\mathbb{Z}/m\mathbb{Z}$). Let $\mathbb{Z}/m\mathbb{Z}$ denote the set of residue classes modulo m . Define

$$\bar{a} + \bar{b} = \overline{a + b}, \quad \bar{a} \cdot \bar{b} = \overline{ab}.$$

These operations are well defined: if $a \equiv a' \pmod{m}$ and $b \equiv b' \pmod{m}$, then $m \mid (a - a')$ and $m \mid (b - b')$, whence $m \mid ((a + b) - (a' + b'))$ and $m \mid (ab - a'b') = (a - a')b + a'(b - b')$, so the sums and products of classes do not depend on the chosen representatives. With these operations $\mathbb{Z}/m\mathbb{Z}$ is a commutative ring with additive identity $\bar{0}$ and multiplicative identity $\bar{1}$.

The arithmetic of $\mathbb{Z}/m\mathbb{Z}$ is exactly integer arithmetic followed by reduction modulo m : to add or multiply two classes, add or multiply any representatives and take the remainder mod m . What distinguishes finite fields is the question of *division*, which the following notion governs.

Definition 5.2 (Unit). An element $\bar{a}$ of a commutative ring R with identity is a *unit* (or *invertible*) if there exists $\bar{b} \in R$ with $\bar{a}\bar{b} = \bar{1}$. A *field* is a commutative ring with $\bar{0} \neq \bar{1}$ in which every nonzero element is a unit.

Proposition 5.3 (Units of $\mathbb{Z}/m\mathbb{Z}$). *The class $\bar{a} \in \mathbb{Z}/m\mathbb{Z}$ is a unit if and only if $\gcd(a, m) = 1$.*

Proof. Suppose $\gcd(a, m) = 1$. By Bézout's identity there exist integers u, v with $ua + vm = 1$. Reducing modulo m gives $\bar{u}\bar{a} = \bar{1}$, so $\bar{a}$ is a unit with inverse $\bar{u}$. Conversely, if $\bar{a}\bar{b} = \bar{1}$ then $ab \equiv 1 \pmod{m}$, i.e. $ab - 1 = mk$ for some $k \in \mathbb{Z}$, so $ab - mk = 1$. Any common divisor of a and m then divides 1, forcing $\gcd(a, m) = 1$. $\square$

Theorem 5.4 (The prime field $\mathbb{F}_p$). *The ring $\mathbb{Z}/m\mathbb{Z}$ is a field if and only if m is prime. When p is prime we write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ and call it the prime field of order p .*

Proof. If $m = 1$ then $\bar{0} = \bar{1}$ and $\mathbb{Z}/m\mathbb{Z}$ has only one element, which is not a field. Suppose $m \geq 2$ is composite, say $m = ab$ with $1 < a, b < m$. Then $\bar{a} \neq \bar{0}$ but $\gcd(a, m) = a > 1$, so by Proposition 5.3 $\bar{a}$ is not a unit; hence $\mathbb{Z}/m\mathbb{Z}$ is not a field. Conversely, if $m = p$ is prime, then every $a \in \{1, \dots, p-1\}$ satisfies $\gcd(a, p) = 1$, so every nonzero class is a unit by Proposition 5.3, and $\bar{0} \neq \bar{1}$ since $p \geq 2$. Thus $\mathbb{F}_p$ is a field. $\square$

There are now two practical ways to compute the inverse of a nonzero $\bar{a} \in \mathbb{F}_p$.

Inversion via the extended Euclidean algorithm. The Euclidean algorithm computes $\gcd(a, p)$ by repeated division with remainder; its *extended* form additionally tracks Bézout coefficients u, v with $ua + vp = \gcd(a, p)$. Since $\gcd(a, p) = 1$ for $0 < a < p$, the algorithm returns u with $ua \equiv 1 \pmod{p}$, and $\bar{a}^{-1} = \bar{u}$. Concretely, set $(r_0, u_0) = (p, 0)$ and $(r_1, u_1) = (a, 1)$; at each step write $r_{i-1} = q_i r_i + r_{i+1}$ with $0 \leq r_{i+1} < r_i$ and set $u_{i+1} = u_{i-1} - q_i u_i$. When $r_{n+1} = 0$ the last nonzero remainder $r_n = \gcd(a, p) = 1$ and $u_n a \equiv 1 \pmod{p}$. This runs in $O(\log p)$ division steps and is the standard method in practice.

Example 5.5. In $\mathbb{F}_{17}$, invert $\bar{5}$. Apply the extended algorithm to $(17, 5)$: $17 = 3 \cdot 5 + 2$, $5 = 2 \cdot 2 + 1$, $2 = 2 \cdot 1$. Back-substitute: $1 = 5 - 2 \cdot 2 = 5 - 2(17 - 3 \cdot 5) = 7 \cdot 5 - 2 \cdot 17$. Hence $7 \cdot 5 \equiv 1 \pmod{17}$, so $\bar{5}^{-1} = \bar{7}$. Indeed $35 = 2 \cdot 17 + 1$.

Inversion via Fermat's little theorem. The second method rests on a fundamental congruence.

Theorem 5.6 (Fermat's little theorem). *Let p be prime. For every $a \in \mathbb{Z}$ with $p \nmid a$ we have $a^{p-1} \equiv 1 \pmod{p}$, and consequently $a^p \equiv a \pmod{p}$ for all a .*

Proof. Consider the map $x \mapsto \bar{a}x$ on the set $\mathbb{F}_p^\times = \mathbb{F}_p \setminus \{\bar{0}\}$ of nonzero classes. Because $\bar{a}$ is a unit (Theorem 5.4), this map is a bijection of $\mathbb{F}_p^\times$ onto itself: it is injective since $\bar{a}x = \bar{a}x'$ implies $x = x'$ after multiplying by $\bar{a}^{-1}$, and a self-injection of a finite set is a bijection. Therefore the products of all elements of $\mathbb{F}_p^\times$ before and after multiplication agree:

$$\prod_{x \in \mathbb{F}_p^\times} x = \prod_{x \in \mathbb{F}_p^\times} (\bar{a}x) = \bar{a}^{p-1} \prod_{x \in \mathbb{F}_p^\times} x.$$

The product $\prod_x x$ is a unit (a product of units), so we may cancel it to obtain $\bar{a}^{p-1} = \bar{1}$, i.e. $a^{p-1} \equiv 1 \pmod{p}$. Multiplying by a yields $a^p \equiv a$, which also holds trivially when $p \mid a$. $\square$

For nonzero $\bar{a} \in \mathbb{F}_p$, Theorem 5.6 gives $\bar{a} \cdot \bar{a}^{p-2} = \bar{a}^{p-1} = \bar{1}$, so

$$\bar{a}^{-1} = \bar{a}^{p-2}.$$

Fast exponentiation (repeated squaring) computes this inverse with $O(\log p)$ modular multiplications. It is constant-time-friendly — the sequence of operations does not branch on the secret a — which is why cryptographic implementations where side-channel resistance matters often use this formula rather than extended Euclid.

Remark 5.7. In $\mathbb{F}_p$ one has $\underbrace{\bar{1} + \cdots + \bar{1}}_p = \bar{p} = \bar{0}$. The smallest positive integer n with $n \cdot \bar{1} = \bar{0}$ in a ring is its *characteristic*; for $\mathbb{F}_p$ the characteristic is p . We show below that every finite field has prime characteristic.

5.2 Characteristic, prime subfield, and order

Let F be any field. The map $\varphi: \mathbb{Z} \rightarrow F$, $\varphi(n) = n \cdot 1_F$ (meaning 1_F added to itself n times, with the usual sign conventions), is a ring homomorphism. Its image is the smallest subring containing 1_F .

Definition 5.8 (Characteristic). The *characteristic* of F , denoted $\text{char } F$, is the nonnegative generator of the kernel $\ker \varphi = \{n \in \mathbb{Z} : n \cdot 1_F = 0\}$. Equivalently, it is 0 if φ is injective, and otherwise the least positive n with $n \cdot 1_F = 0$.

Proposition 5.9. *The characteristic of a field is either 0 or a prime number. A finite field has prime characteristic.*

Proof. Suppose $\text{char } F = n > 0$ and $n = ab$ with $1 \leq a, b \leq n$. Then $(a \cdot 1_F)(b \cdot 1_F) = (ab) \cdot 1_F = n \cdot 1_F = 0$. A field has no zero divisors, so $a \cdot 1_F = 0$ or $b \cdot 1_F = 0$; by minimality of n this forces $a = n$ or $b = n$. Hence n has no nontrivial factorization and is prime. If F is finite, the map $\varphi: \mathbb{Z} \rightarrow F$ cannot be injective (its domain is infinite), so $\ker \varphi \neq \{0\}$ and $\text{char } F > 0$; thus it is prime. $\square$

Definition 5.10 (Prime subfield). The *prime subfield* of F is the smallest subfield of F , namely the intersection of all subfields, equivalently the subfield generated by 1_F .

Proposition 5.11. *If $\text{char } F = p > 0$, the prime subfield of F is isomorphic to $\mathbb{F}_p$. If $\text{char } F = 0$, it is isomorphic to $\mathbb{Q}$.*

Proof. Consider $\varphi: \mathbb{Z} \rightarrow F$ as above. When $\text{char } F = p$, the kernel is $p\mathbb{Z}$, so φ factors through an injective ring homomorphism $\bar{\varphi}: \mathbb{Z}/p\mathbb{Z} \rightarrow F$, whose image is a subfield isomorphic to $\mathbb{F}_p$. Any subfield contains 1_F and hence this image, so it is the prime subfield. When $\text{char } F = 0$, φ is injective and extends to an embedding of $\mathbb{Q}$ by sending $a/b \mapsto (a \cdot 1_F)(b \cdot 1_F)^{-1}$; its image is the prime subfield. $\square$

Thus every finite field F is a vector space over its prime subfield $\mathbb{F}_p$. Since F is finite it is finite-dimensional; if $\dim_{\mathbb{F}_p} F = k$, then choosing a basis $e_1, \dots, e_k$ shows that every element of F is uniquely $c_1 e_1 + \cdots + c_k e_k$ with $c_i \in \mathbb{F}_p$, so F has exactly p^k elements.

Theorem 5.12 (Order of a finite field is a prime power). *Every finite field has cardinality p^k for some prime p and integer $k \geq 1$, where $p = \text{char } F$.*

Proof. By Proposition 5.9, $p = \text{char } F$ is prime, and by Proposition 5.11 the prime subfield is $\mathbb{F}_p$. As just observed, F is a finite-dimensional $\mathbb{F}_p$ -vector space, say of dimension $k \geq 1$, and counting coordinate vectors gives $|F| = p^k$. $\square$

Henceforth write $q = p^k$ and denote by $\mathbb{F}_q$ a field with q elements; the next subsections justify the use of the definite article by proving existence and uniqueness.

5.3 The Frobenius map and the freshman's dream

A recurring phenomenon of prime characteristic is that the p -th power map respects addition.

Lemma 5.13 (Freshman's dream). *Let R be a commutative ring of prime characteristic p . Then for all $a, b \in R$,*

$$(a + b)^p = a^p + b^p, \quad \text{and more generally} \quad (a + b)^{p^n} = a^{p^n} + b^{p^n}.$$

Proof. The binomial theorem gives $(a + b)^p = \sum_{i=0}^p \binom{p}{i} a^i b^{p-i}$. For $0 < i < p$, the integer $\binom{p}{i} = \frac{p!}{i!(p-i)!}$ is divisible by p : the numerator contains the factor p , while the denominator, being a product of integers all smaller than p , is coprime to p . Since $p \cdot 1_R = 0$, every middle term vanishes in R , leaving $(a + b)^p = a^p + b^p$. The general case follows by induction on n , applying the case $n = 1$ to $a^{p^{n-1}}$ and $b^{p^{n-1}}$. $\square$

Definition 5.14 (Frobenius endomorphism). Let F be a field of characteristic p . The *Frobenius map* is

$$\text{Frob}: F \rightarrow F, \quad \text{Frob}(x) = x^p.$$

Proposition 5.15. *The Frobenius map of a field F of characteristic p is a ring homomorphism fixing the prime subfield $\mathbb{F}_p$ pointwise. If F is finite it is an automorphism of F .*

Proof. It preserves products since $(xy)^p = x^p y^p$ by commutativity, and sums by Lemma 5.13; it sends $1 \mapsto 1$. Thus it is a ring homomorphism, and any nonzero ring homomorphism of fields is injective (its kernel is an ideal not equal to all of F , hence $\{0\}$). On $\mathbb{F}_p$, Fermat's little theorem (Theorem 5.6) gives $x^p = x$ for all x , so Frob fixes $\mathbb{F}_p$. When F is finite, an injective self-map of a finite set is bijective, so Frob is an automorphism. $\square$

5.4 The multiplicative group of a finite field is cyclic

For a field F write $F^\times = F \setminus \{0\}$ for its multiplicative group. When F is finite, $F^\times$ has a strikingly simple structure: it is cyclic, that is, generated by a single element. This fact underlies discrete-logarithm cryptography and the construction of finite fields with prescribed roots of unity used in Halo 2. We give a complete, self-contained proof below.

We first recall the order of a group element. In a group G , the *order* $\text{ord}(g)$ of g is the least positive integer n with $g^n = e$ (the identity), if one exists. In a finite group every element has finite order dividing $|G|$, by Lagrange's theorem applied to the cyclic subgroup $\langle g \rangle$.

Lemma 5.16 (Polynomials over a field have few roots). *A nonzero polynomial f of degree d with coefficients in a field F has at most d roots in F , counted without multiplicity.*

Proof. We induct on d . A nonzero constant ($d = 0$) has no roots. If $d \geq 1$ and f has no root, the claim holds. Otherwise let $\alpha \in F$ be a root. The division algorithm for polynomials over a field writes $f(X) = (X - \alpha)g(X) + r$ with r a constant; evaluating at α gives $r = f(\alpha) = 0$, so $f(X) = (X - \alpha)g(X)$ with $\deg g = d - 1$. If $\beta \neq \alpha$ is any other root, then $0 = f(\beta) = (\beta - \alpha)g(\beta)$; since $\beta - \alpha \neq 0$ and F has no zero divisors, $g(\beta) = 0$. By induction g has at most $d - 1$ roots, so f has at most d roots. $\square$

Lemma 5.17. *Let G be a finite abelian group and let M be the maximum of the orders of its elements. Then the order of every element of G divides M .*

Proof. Let g have maximal order M , and let $h \in G$ have order n . We must show $n \mid M$. Suppose not; then there is a prime ℓ with $\ell^a \parallel M$ and $\ell^b \parallel n$ (meaning ℓ^a is the exact power of ℓ dividing each) and $b > a$. Consider $g' = g^{\ell^a}$, of order M/ℓ^a (coprime to ℓ), and $h' = h^{\ell^b}$, of order ℓ^b . Since $\text{ord}(g')$ and $\text{ord}(h')$ are coprime and G is abelian, the order of $g'h'$ is the product $\text{ord}(g')\text{ord}(h') = (M/\ell^a)\ell^b = M\ell^{b-a} > M$, contradicting the maximality of M . (We used that in

an abelian group, if $\text{ord}(x)$ and $\text{ord}(y)$ are coprime then $\text{ord}(xy) = \text{ord}(x)\text{ord}(y)$: from $(xy)^k = e$ one deduces x^k and y^{-k} lie in $\langle x \rangle \cap \langle y \rangle = \{e\}$, forcing both $\text{ord}(x)$ and $\text{ord}(y)$ to divide k , hence $\text{ord}(x)\text{ord}(y) \mid k$; conversely $(xy)^{\text{ord}(x)\text{ord}(y)} = e$.) Therefore $n \mid M$. $\square$

Theorem 5.18 (Multiplicative group of a finite field is cyclic). *For any finite field F with q elements, the group $F^\times$ is cyclic of order $q - 1$.*

Proof. The group $G = F^\times$ is abelian of order $q - 1$. Let M be the maximal order of an element of G ; by Lagrange $M \mid q - 1$, so $M \leq q - 1$. By Lemma 5.17, every element $x \in G$ has order dividing M , hence satisfies $x^M = 1$. Thus all $q - 1$ elements of G are roots of the polynomial $X^M - 1 \in F[X]$. By Lemma 5.16 this polynomial has at most M roots, so $q - 1 \leq M$. Combining, $M = q - 1$, and an element of order $q - 1 = |G|$ generates G . Hence G is cyclic. $\square$

Definition 5.19 (Primitive root). We call a generator of $\mathbb{F}_q^\times$ a *primitive root* (or *primitive element*) of $\mathbb{F}_q$. Equivalently, g is primitive iff $\text{ord}(g) = q - 1$.

Example 5.20. In $\mathbb{F}_7^\times$, of order 6, test $g = 3$: the powers are $3^1 = 3$, $3^2 = 2$, $3^3 = 6$, $3^4 = 4$, $3^5 = 5$, $3^6 = 1 \pmod{7}$, running through all of $\{1, \dots, 6\}$. So 3 is a primitive root. By contrast 2 has order 3 (2, 4, 1) and is not primitive. The number of primitive roots of $\mathbb{F}_q$ is $\varphi(q - 1)$, where φ is Euler's totient, since a cyclic group of order N has $\varphi(N)$ generators.

Corollary 5.21 (Roots of unity). *Let F be a finite field with q elements and let $n \mid q - 1$. Then $F^\times$ contains exactly n elements x with $x^n = 1$, and they form a cyclic subgroup of order n . Conversely, if $n \nmid q - 1$, the only solution of $x^n = 1$ shared with the whole group structure is governed by $\gcd(n, q - 1)$: there are exactly $\gcd(n, q - 1)$ such x .*

Proof. Let g be a primitive root, so $F^\times = \langle g \rangle$ is cyclic of order $q - 1$. Writing $x = g^j$, the equation $x^n = 1$ becomes $g^{jn} = 1$, i.e. $(q - 1) \mid jn$, i.e. $\frac{q-1}{d} \mid j$ where $d = \gcd(n, q - 1)$. The values $j \in \{0, 1, \dots, q - 2\}$ satisfying this are $j = \frac{q-1}{d} \cdot t$ for $t = 0, \dots, d - 1$, exactly d of them, and they form the cyclic subgroup generated by $g^{(q-1)/d}$. When $n \mid q - 1$ we have $d = n$. $\square$

This corollary is precisely what renders a curve such as Pallas or Vesta suitable for Halo 2: one chooses q so that $q - 1$ is divisible by a large power of 2, guaranteeing a multiplicative subgroup of 2-power order on which the number-theoretic transform, and hence efficient polynomial arithmetic, operates.

5.5 Prime fields in elliptic-curve cryptography

This section closes by indicating how the prime field $\mathbb{F}_p$ — the simplest finite field — enters elliptic-curve cryptography, the setting of Halo 2 and Zcash Orchard. We develop the geometry of elliptic curves in full later; here we only locate the relevant fields.

An elliptic curve over $\mathbb{F}_p$ (with $p > 3$) typically takes the short Weierstrass form, the set of solutions

$$E(\mathbb{F}_p) = \{ (x, y) \in \mathbb{F}_p \times \mathbb{F}_p : y^2 = x^3 + ax + b \} \cup \{\mathcal{O}\},$$

together with a distinguished point $\mathcal{O}$ at infinity, where $a, b \in \mathbb{F}_p$ and $4a^3 + 27b^2 \neq 0$. All the arithmetic defining the group law — the computation of slopes $\lambda = (y_2 - y_1)(x_2 - x_1)^{-1}$, squaring, and the inversions required to add and double points — proceeds in $\mathbb{F}_p$ using exactly the operations of Subsection 5.1. We call this $\mathbb{F}_p$ the *base field* (or *coordinate field*) of the curve.

The set $E(\mathbb{F}_p)$ forms a finite abelian group under the chord-and-tangent addition law. Its order $N = |E(\mathbb{F}_p)|$ satisfies the Hasse bound $|N - (p + 1)| \leq 2\sqrt{p}$. For cryptography one selects a curve whose group has a large prime factor r ; the subgroup of order r is where keys and signatures reside. The scalars acting on this subgroup live modulo r , and because r is prime, these scalars themselves form a field.

Definition 5.22 (Base field and scalar field). For an elliptic curve used in cryptography with a prime-order subgroup of order r :

- the *base field* $\mathbb{F}_p$ is the field over which the curve is defined and in which point coordinates lie;
- the *scalar field* $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ is the field of scalars by which one multiplies points, i.e. the ring of integers modulo the subgroup order r .

Both are prime fields, but in general $p \neq r$.

Remark 5.23 (Two fields, and the Pasta cycle). Scalar multiplication $P \mapsto sP$ on a point P of order r depends only on $s \bmod r$, because $rP = \mathcal{O}$; the exponents thus lie in $\mathbb{F}_r$, while the coordinates of sP lie in $\mathbb{F}_p$. A typical elliptic-curve operation therefore manipulates two distinct prime fields simultaneously. Halo 2’s Pallas and Vesta curves (the “Pasta” cycle) exploit this deliberately: they form a *cycle of curves*, meaning the base field of one is the scalar field of the other and vice versa,

$$|E_{\text{Pallas}}| = p_{\text{Vesta}}, \quad |E_{\text{Vesta}}| = p_{\text{Pallas}},$$

so that the scalar field of Pallas equals the base field of Vesta. This permits a proof system to express the verifier of one curve’s arithmetic efficiently inside the native field of the other, enabling the recursive proof composition at the heart of Halo 2. One chooses both p_{Pallas} and p_{Vesta} to be $\equiv 1 \pmod{2^{32}}$, so that by Corollary 5.21 each field has a multiplicative subgroup of order 2^{32} , furnishing the high-order roots of unity needed for fast (number-theoretic-transform-based) polynomial multiplication.

Remark 5.24 (The role of cyclicity and Frobenius in cryptography). Two structural facts from this section recur throughout the cryptographic development. First, Theorem 5.18 (cyclicity of $\mathbb{F}_q^\times$) underlies the discrete-logarithm problem and guarantees the existence of the roots of unity used in polynomial commitment schemes. Second, the Frobenius endomorphism (Definition 5.14) is not merely an abstraction: on elliptic curves over $\mathbb{F}_p$ it induces an endomorphism whose characteristic polynomial $T^2 - tT + p$ encodes the curve order $N = p + 1 - t$ via the Hasse trace t , tying the count $|E(\mathbb{F}_p)|$ directly to the field-theoretic Frobenius studied here. We take up these connections in the elliptic-curve sections that follow.

6 Number theory for cryptography

Modern public-key cryptography resides almost entirely inside finite cyclic groups and finite fields, and the security of the schemes used in Halo 2 and Zcash Orchard rests on number-theoretic facts about such structures. This section develops, from elementary first principles, the classical number theory of the integers modulo n : Fermat's little theorem and Euler's generalisation, the structure of the multiplicative group and the notion of a primitive root, probabilistic primality testing, the theory of quadratic residues together with the Legendre and Jacobi symbols and quadratic reciprocity, algorithms for extracting square roots modulo a prime, and finally the discrete logarithm problem and the reason the *prime order* of a group is so important for cryptographic hardness.

Throughout, n and m denote positive integers and p, q denote primes. We write $a \mid b$ to mean “ a divides b ”, i.e. $b = ac$ for some integer c , and $a \equiv b \pmod{n}$ to mean $n \mid (a - b)$. We assume the reader has met the integers $\mathbb{Z}$, the greatest common divisor $\gcd(a, b)$, the Euclidean algorithm, and the fact that $\gcd(a, b)$ admits an integer combination $\gcd(a, b) = ua + vb$ (Bézout's identity). We also use freely the ring $\mathbb{Z}/n\mathbb{Z}$ of integers modulo n and its prime field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ when p is prime.

6.1 The integers modulo n and units

Recall that $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$ is a commutative ring under addition and multiplication modulo n . The objects that can be inverted multiplicatively are exactly those coprime to n .

Definition 6.1 (Unit). An element $a \in \mathbb{Z}/n\mathbb{Z}$ is a *unit* (or is *invertible*) if there exists $b \in \mathbb{Z}/n\mathbb{Z}$ with $ab \equiv 1 \pmod{n}$. We write $(\mathbb{Z}/n\mathbb{Z})^\times$ for the set of units.

Proposition 6.2. *An element $a \in \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if $\gcd(a, n) = 1$. Consequently $(\mathbb{Z}/n\mathbb{Z})^\times$ is a finite abelian group under multiplication modulo n .*

Proof. If $\gcd(a, n) = 1$, Bézout gives integers u, v with $ua + vn = 1$, so $ua \equiv 1 \pmod{n}$ and $u \bmod n$ is an inverse of a . Conversely, if $ab \equiv 1 \pmod{n}$ then $ab - 1 = kn$ for some integer k , so $ab - kn = 1$; any common divisor of a and n divides the left-hand side and hence divides 1, forcing $\gcd(a, n) = 1$. Closure under multiplication holds because $\gcd(ab, n) = 1$ whenever $\gcd(a, n) = \gcd(b, n) = 1$; the identity is 1; inverses exist by definition; associativity and commutativity are inherited from $\mathbb{Z}$. Finiteness is clear. $\square$

In particular, when p is prime every nonzero residue is coprime to p , so $(\mathbb{Z}/p\mathbb{Z})^\times = \mathbb{F}_p^\times = \{1, 2, \dots, p-1\}$ is a group of order $p-1$, and $\mathbb{F}_p$ is a field.

6.2 Euler's totient function

Definition 6.3 (Euler totient). *Euler's totient function $\varphi(n)$ is the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$. Equivalently, $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.*

Example 6.4. We have $\varphi(1) = 1$, $\varphi(p) = p-1$ for prime p , and $\varphi(12) = |\{1, 5, 7, 11\}| = 4$.

Computing φ requires two facts: its value on prime powers, and its behaviour on coprime factors.

Proposition 6.5 (Totient of a prime power). *For a prime p and integer $k \geq 1$, $\varphi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right)$.*

Proof. Among $1, \dots, p^k$ the integers *not* coprime to p^k are exactly the multiples of p , namely $p, 2p, \dots, p^{k-1} \cdot p$, of which there are p^{k-1} . Subtracting gives $p^k - p^{k-1}$. $\square$

Theorem 6.6 (Chinese Remainder Theorem). *If $\gcd(m, n) = 1$, then the map*

$$\mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \quad a \bmod mn \longmapsto (a \bmod m, a \bmod n)$$

is an isomorphism of rings. It restricts to a group isomorphism $(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$.

Proof. The map is a well-defined ring homomorphism since reduction modulo m and modulo n are. It is injective: if $a \equiv 0 \pmod{m}$ and $a \equiv 0 \pmod{n}$ with $\gcd(m, n) = 1$, then $mn \mid a$, so $a \equiv 0 \pmod{mn}$; thus the kernel is trivial. Both sides have mn elements, so injectivity forces bijectivity. A ring isomorphism carries units to units, and (u, v) is a unit of the product ring iff u, v are units in their factors; this gives the multiplicative statement. $\square$

Corollary 6.7 (Multiplicativity and the product formula). *If $\gcd(m, n) = 1$ then $\varphi(mn) = \varphi(m)\varphi(n)$. Hence if $n = \prod_i p_i^{k_i}$ is the prime factorisation of n , then*

$$\varphi(n) = \prod_i \left(p_i^{k_i} - p_i^{k_i-1} \right) = n \prod_{p|n} \left(1 - \frac{1}{p} \right).$$

Proof. The cardinality of a product of groups is the product of cardinalities, so Theorem 6.6 gives $\varphi(mn) = \varphi(m)\varphi(n)$. Applying this across the distinct prime powers in the factorisation and using Proposition 6.5 yields the product formula. $\square$

Proposition 6.8 (Gauss's divisor-sum identity). *For every $n \geq 1$, $\sum_{d|n} \varphi(d) = n$.*

Proof. Partition $\{1, 2, \dots, n\}$ according to the value $g = \gcd(a, n)$. For a fixed divisor $d \mid n$, the integers a with $\gcd(a, n) = d$ are exactly $a = db$ with $1 \leq b \leq n/d$ and $\gcd(b, n/d) = 1$; there are $\varphi(n/d)$ of these. Hence $n = \sum_{d|n} \varphi(n/d)$, and as d ranges over divisors of n so does n/d , giving $\sum_{d|n} \varphi(d) = n$. $\square$

6.3 Fermat's little theorem and Euler's theorem

The single most-used fact in this section is Lagrange's theorem from group theory: in a finite group G , the order of every element divides $|G|$. We recall it in the form required and then specialise it.

Lemma 6.9. *Let G be a finite abelian group of order N . Then $g^N = e$ (the identity) for every $g \in G$.*

Proof. Let $g \in G$. Consider the map $\mu_g : G \rightarrow G$, $x \mapsto gx$. It is a bijection (its inverse is $x \mapsto g^{-1}x$), so as x runs over all of G so does gx . Multiplying all elements together,

$$\prod_{x \in G} x = \prod_{x \in G} (gx) = g^N \prod_{x \in G} x,$$

where we used commutativity to pull out the N copies of g . Cancelling the (invertible) product $\prod_x x$ from both sides gives $g^N = e$. $\square$

Theorem 6.10 (Euler's theorem). *If $\gcd(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. The hypothesis says $a \bmod n \in (\mathbb{Z}/n\mathbb{Z})^\times$, a finite abelian group of order $\varphi(n)$ by Definition 6.3. Apply Lemma 6.9 with $G = (\mathbb{Z}/n\mathbb{Z})^\times$ and $N = \varphi(n)$. $\square$

Corollary 6.11 (Fermat's little theorem). *Let p be prime. If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. For every integer a , $a^p \equiv a \pmod{p}$.*

Proof. Take $n = p$ in Theorem 6.10, using $\varphi(p) = p - 1$, to get the first statement. Multiplying $a^{p-1} \equiv 1$ by a yields $a^p \equiv a$ when $p \nmid a$; and when $p \mid a$ both sides are $\equiv 0$, so $a^p \equiv a \pmod{p}$ in all cases. $\square$

Remark 6.12. Euler's theorem underlies RSA: if $ed \equiv 1 \pmod{\varphi(n)}$ then $(a^e)^d = a^{1+k\varphi(n)} \equiv a \pmod{n}$ whenever $\gcd(a, n) = 1$. For the elliptic-curve cryptography of Orchard one instead exploits Fermat's little theorem in the field $\mathbb{F}_p$: it shows every nonzero element satisfies $x^{p-1} = 1$, so one can compute inversion as $x^{-1} = x^{p-2}$, and it underlies the Euler criterion for square roots developed below.

6.4 Multiplicative order and primitive roots

Definition 6.13 (Multiplicative order). Let $\gcd(a, n) = 1$. The *multiplicative order* of a modulo n , written $\text{ord}_n(a)$, is the least positive integer k with $a^k \equiv 1 \pmod{n}$. (Such a k exists because $a^{\varphi(n)} \equiv 1$ by Theorem 6.10.)

Lemma 6.14. *With $\gcd(a, n) = 1$, $a^m \equiv 1 \pmod{n}$ if and only if $\text{ord}_n(a) \mid m$. In particular $\text{ord}_n(a) \mid \varphi(n)$.*

Proof. Let $k = \text{ord}_n(a)$. If $k \mid m$, write $m = kq$; then $a^m = (a^k)^q \equiv 1$. Conversely, suppose $a^m \equiv 1$. Divide with remainder: $m = kq + r$ with $0 \leq r < k$. Then $1 \equiv a^m = (a^k)^q a^r \equiv a^r$, and minimality of k forces $r = 0$, i.e. $k \mid m$. Taking $m = \varphi(n)$ gives the last claim. $\square$

Lemma 6.15. *If $\gcd(a, n) = 1$ and $k = \text{ord}_n(a)$, then $\text{ord}_n(a^j) = \frac{k}{\gcd(j, k)}$.*

Proof. Let $d = \gcd(j, k)$. We have $(a^j)^m \equiv 1$ iff $k \mid jm$ (by Lemma 6.14) iff $\frac{k}{d} \mid \frac{j}{d}m$. Since $\gcd(\frac{k}{d}, \frac{j}{d}) = 1$, this holds iff $\frac{k}{d} \mid m$. The least such positive m is k/d . $\square$

Definition 6.16 (Primitive root). A *primitive root* modulo n is an element g with $\text{ord}_n(g) = \varphi(n)$; equivalently, g generates $(\mathbb{Z}/n\mathbb{Z})^\times$ as a cyclic group, so that $\{g^0, g^1, \dots, g^{\varphi(n)-1}\}$ lists all the units.

Primitive roots need not exist for every n (for instance $(\mathbb{Z}/8\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has no element of order 4). The decisive case for cryptography is $n = p$ prime, where the multiplicative group is always cyclic. We prove this now; it constitutes the structural heart of the theory.

Lemma 6.17 (A polynomial of degree d over a field has at most d roots). *Let F be a field and $f \in F[x]$ a nonzero polynomial of degree d . Then f has at most d roots in F .*

Proof. By induction on d . For $d = 0$, f is a nonzero constant and has no roots. For $d \geq 1$, if f has no root we are done; otherwise let r be a root. Polynomial division gives $f(x) = (x - r)q(x) + c$ with c constant; evaluating at r gives $c = f(r) = 0$, so $f(x) = (x - r)q(x)$ with $\deg q = d - 1$. Any root $s \neq r$ satisfies $(s - r)q(s) = 0$; since F has no zero divisors and $s - r \neq 0$, we get $q(s) = 0$. By induction q has at most $d - 1$ roots, so f has at most d roots. $\square$

Theorem 6.18 ($\mathbb{F}_p^\times$ is cyclic). *For every prime p , the group $\mathbb{F}_p^\times = (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$. In particular primitive roots modulo p exist.*

Proof. Let $N = p - 1 = |\mathbb{F}_p^\times|$. For each divisor $d \mid N$, let $\psi(d)$ be the number of elements of $\mathbb{F}_p^\times$ of order exactly d . Every element has some order dividing N (Lemma 6.14), so

$$\sum_{d \mid N} \psi(d) = N. \quad (1)$$

We claim that for each $d \mid N$, either $\psi(d) = 0$ or $\psi(d) = \varphi(d)$. Suppose $\psi(d) > 0$, so there is some element a of order d . The d distinct powers $a^0, a^1, \dots, a^{d-1}$ are all roots of $x^d - 1$ in $\mathbb{F}_p$. But $x^d - 1$ has at most d roots by Lemma 6.17, so these powers are *all* the roots, and every element of order

d is among them. Among the powers a^j , the ones of order exactly d are those with $\gcd(j, d) = 1$ by Lemma 6.15, and there are $\varphi(d)$ of those. Hence $\psi(d) = \varphi(d)$ whenever $\psi(d) > 0$.

Therefore $\psi(d) \leq \varphi(d)$ for all $d \mid N$. Summing and using Proposition 6.8,

$$N = \sum_{d \mid N} \psi(d) \leq \sum_{d \mid N} \varphi(d) = N.$$

Equality throughout forces $\psi(d) = \varphi(d)$ for every $d \mid N$. In particular $\psi(N) = \varphi(N) \geq 1$, so there is an element of order N , i.e. a primitive root, and $\mathbb{F}_p^\times$ is cyclic. $\square$

Remark 6.19. The same argument shows that *any* finite subgroup of the multiplicative group of *any* field is cyclic. This is why the prime-order subgroups used in pairing-based and elliptic-curve cryptography behave like $\mathbb{Z}/q\mathbb{Z}$ under addition once one fixes a generator: a cyclic group of order q is isomorphic to $(\mathbb{Z}/q\mathbb{Z}, +)$ via $g^k \leftrightarrow k$. The full classification (which we will not need) states that primitive roots modulo n exist exactly when $n \in \{1, 2, 4, p^k, 2p^k\}$ for an odd prime p .

Example 6.20. Modulo 7, $\varphi(7) = 6$. Powers of 3: 3, 2, 6, 4, 5, 1, so $\text{ord}_7(3) = 6$ and 3 is a primitive root. By Lemma 6.15 the other primitive roots are 3^j with $\gcd(j, 6) = 1$, namely $3^1 = 3$ and $3^5 = 5$; indeed there are $\varphi(6) = 2$ of them.

6.5 Quadratic residues and the Euler criterion

We now turn to squares modulo a prime, which control whether a curve point with a given x -coordinate exists, and which we need to compute square roots in $\mathbb{F}_p$.

Definition 6.21 (Quadratic residue). Let p be an odd prime and a an integer with $p \nmid a$. We say a is a *quadratic residue* modulo p (a QR) if the congruence $x^2 \equiv a \pmod{p}$ has a solution, and a *quadratic non-residue* (a QNR) otherwise.

Proposition 6.22 (Exactly half are residues). *For an odd prime p , exactly $\frac{p-1}{2}$ of the nonzero residues modulo p are quadratic residues and $\frac{p-1}{2}$ are non-residues. The squaring map $x \mapsto x^2$ is exactly two-to-one on $\mathbb{F}_p^\times$.*

Proof. Consider $\sigma : \mathbb{F}_p^\times \rightarrow \mathbb{F}_p^\times$, $\sigma(x) = x^2$. Its image is exactly the set of QRs. For b in the image, $\sigma(x) = \sigma(y)$ iff $x^2 = y^2$ iff $(x - y)(x + y) = 0$ iff $y = \pm x$; since p is odd, $x \neq -x$, so each fibre has exactly two elements. Hence $|\text{im } \sigma| = \frac{p-1}{2}$, and the complement has the same size. $\square$

Theorem 6.23 (Euler's criterion). *Let p be an odd prime and $p \nmid a$. Then*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 1 \pmod{p} & \text{if } a \text{ is a quadratic residue,} \\ -1 \pmod{p} & \text{if } a \text{ is a quadratic non-residue.} \end{cases}$$

Proof. By Fermat, $a^{p-1} \equiv 1$, so $(a^{(p-1)/2})^2 \equiv 1$, and since the only square roots of 1 modulo a prime are ± 1 , we have $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$.

If a is a QR, say $a \equiv x^2$, then $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$ by Fermat. Thus all $\frac{p-1}{2}$ quadratic residues are roots of the polynomial $t^{(p-1)/2} - 1$ over $\mathbb{F}_p$. By Lemma 6.17 this polynomial has at most $\frac{p-1}{2}$ roots, so the QRs are *exactly* its roots. Any QNR a is therefore not a root, yet still satisfies $a^{(p-1)/2} \equiv \pm 1$; hence $a^{(p-1)/2} \equiv -1$. $\square$

6.6 Square roots modulo a prime

Deciding that a is a residue is one matter; *producing* an x with $x^2 \equiv a$ is another, and we need it, for instance, to recover a curve point's y -coordinate from its x -coordinate (point decompression). The difficulty depends on $p \bmod 4$.

Proposition 6.24 (The $p \equiv 3 \pmod{4}$ shortcut). *Let $p \equiv 3 \pmod{4}$ be prime and let a be a quadratic residue modulo p with $p \nmid a$. Then*

$$x \equiv a^{\frac{p+1}{4}} \pmod{p}$$

is a square root of a ; the two square roots are $\pm x$.

Proof. Since $p \equiv 3 \pmod{4}$, $\frac{p+1}{4}$ is a positive integer. Compute

$$x^2 = a^{\frac{p+1}{2}} = a^{\frac{p-1}{2}} \cdot a \equiv \left(\frac{a}{p}\right) \cdot a \equiv 1 \cdot a = a \pmod{p},$$

using Euler's criterion and the assumption that a is a residue, so $\left(\frac{a}{p}\right) = 1$. The two roots are $\pm x$ by Proposition 6.22. $\square$

Remark 6.25. This shortcut is a single modular exponentiation and is the reason cryptographers often choose primes with $p \equiv 3 \pmod{4}$. When $p \equiv 1 \pmod{4}$ (as for the Pallas and Vesta base fields of the Pasta cycle used in Halo 2, whose moduli are $\equiv 1 \pmod{4}$), no such closed form exists and one resorts to Tonelli–Shanks.

When $p \equiv 1 \pmod{4}$ one writes $p - 1 = 2^s d$ with d odd and $s \geq 2$. The Tonelli–Shanks algorithm operates inside the cyclic 2-power-torsion subgroup of $\mathbb{F}_p^\times$.

Theorem 6.26 (Tonelli–Shanks). *Let p be an odd prime, $p - 1 = 2^s d$ with d odd, and let a be a quadratic residue with $p \nmid a$. One can compute a square root of a modulo p using $O(s^2 + \log p)$ multiplications plus a few modular exponentiations, given any quadratic non-residue z modulo p .*

Sketch. Fix a non-residue z (found by trying random elements; by Proposition 6.22 a random element is a non-residue with probability $\frac{1}{2}$, and one checks $\left(\frac{z}{p}\right)$ in $O((\log p)^2)$ time). Set $c = z^d$; this is a generator of the unique cyclic subgroup H of order 2^s in $\mathbb{F}_p^\times$, since $\text{ord}_p(z^d) = 2^s / \gcd(d', \cdot)$ works out to 2^s for a non-residue. Initialise

$$x = a^{\frac{d+1}{2}}, \quad t = a^d, \quad m = s.$$

Then $x^2 = a \cdot t$, so x is a square root of a up to the factor $t \in H$; moreover $t^{2^{s-1}} = a^{(p-1)/2} = 1$, so t lies in H and has 2-power order. The algorithm repeatedly drives t to 1: at each step find the least i with $t^{2^i} = 1$; then $b = c^{2^{m-i-1}}$ satisfies $b^2 \in H$ and replacing

$$x \leftarrow xb, \quad t \leftarrow tb^2, \quad c \leftarrow b^2, \quad m \leftarrow i$$

strictly decreases the order of t while preserving the invariant $x^2 = at$. Since the order of t is a strictly decreasing power of 2, after at most s iterations $t = 1$ and $x^2 = a$. Correctness rests on the cyclic structure of H (Theorem 6.18); the running-time bound counts the at most s iterations, each costing $O(s)$ squarings to locate i . $\square$

Remark 6.27. In every case the square roots come in a pair $\pm x$ (Proposition 6.22), so to pin down a canonical root one fixes a convention, e.g. “choose the root whose least nonnegative representative is even”, as elliptic-curve point encodings do. If a is *not* a residue, all these algorithms detect failure (e.g. the candidate fails the final check $x^2 \equiv a$).

6.7 The discrete logarithm problem

The hardness underpinning elliptic-curve cryptography is the difficulty of *inverting* exponentiation in a finite cyclic group. We phrase it here for a generic cyclic group, since in Orchard the group is the group of points of an elliptic curve, written additively, of large prime order.

Definition 6.28 (Discrete logarithm problem). Let $G = \langle g \rangle$ be a finite cyclic group of order N , written multiplicatively, with generator g . Given $h \in G$, the *discrete logarithm* of h to base g , written $\log_g h$, is the unique residue $x \in \mathbb{Z}/N\mathbb{Z}$ with $g^x = h$. The *discrete logarithm problem* (DLP) is to compute x from (G, g, h) .

The map $x \mapsto g^x$ is a group isomorphism $(\mathbb{Z}/N\mathbb{Z}, +) \xrightarrow{\sim} (G, \cdot)$ (it is a homomorphism with trivial kernel because g has order N), so $\log_g$ is genuinely a well-defined inverse. Computing g^x from x is fast (repeated squaring, $O(\log N)$ operations); the DLP asks for the reverse, for which no efficient algorithm is known in well-chosen groups.

Remark 6.29 (Generic complexity and group choice). Two “generic” algorithms that operate in any group of order N , using only the group operation, solve the DLP in $O(\sqrt{N})$ time: *baby-step giant-step* (deterministic, $O(\sqrt{N})$ time and space) and *Pollard’s rho* (randomised, $O(\sqrt{N})$ time and $O(1)$ space). Shoup’s theorem shows $\Omega(\sqrt{N})$ is optimal for generic algorithms when N is prime. Thus to reach a λ -bit security level one requires $N \approx 2^{2\lambda}$; this is why the Pallas/Vesta scalar fields have ≈ 255 -bit prime order, targeting ≈ 128 -bit security. For the integers modulo p under multiplication, sub-exponential index-calculus attacks exist, which is why elliptic-curve groups (where no such attack is known) permit much smaller parameters.

The next result makes the phrase “well-chosen” precise: if the group order N is *composite* with small prime factors, the DLP reduces to several DLPs in small subgroups, and is therefore easy.

Theorem 6.30 (Pohlig–Hellman reduction). *Let $G = \langle g \rangle$ be cyclic of order $N = \prod_{i=1}^r q_i^{e_i}$. One can compute the discrete logarithm in G by solving the discrete logarithm in each subgroup of prime order q_i (a bounded number of times) and combining the answers with the Chinese Remainder Theorem. If the largest prime factor of N is $q_{\max}$, the total cost is*

$$O\left(\sum_i e_i (\log N + \sqrt{q_i})\right)$$

group operations, dominated by $O(\sqrt{q_{\max}})$.

Sketch. The reduction proceeds in two stages. (a) *CRT across coprime factors.* By Lemma 6.14 and the structure of cyclic groups, G decomposes as the internal direct product of its Sylow subgroups G_i of order $q_i^{e_i}$. Knowing $x = \log_g h$ modulo each $q_i^{e_i}$ determines x modulo $N = \prod q_i^{e_i}$ uniquely, by the Chinese Remainder Theorem (Theorem 6.6, applied to the integers $\mathbb{Z}/N\mathbb{Z}$). To find $x \bmod q_i^{e_i}$ one projects: set $g_i = g^{N/q_i^{e_i}}$ and $h_i = h^{N/q_i^{e_i}}$; then g_i has order $q_i^{e_i}$ and $\log_{g_i} h_i \equiv x \pmod{q_i^{e_i}}$.

(b) *Lifting a prime power digit by digit.* Within a group of order q^e , write the unknown exponent in base q as $x \equiv x_0 + x_1 q + \cdots + x_{e-1} q^{e-1} \pmod{q^e}$ with $0 \leq x_j < q$. Raising the relevant elements to the power q^{e-1} lands inside the order- q subgroup and isolates x_0 as a single discrete log in a group of order q ; subtracting off its contribution and repeating isolates x_1 , and so on. Each digit costs one DLP in the order- q subgroup, solvable in $O(\sqrt{q})$ by baby-step giant-step or Pollard rho, plus $O(\log N)$ for the exponentiations. Summing over the e_i digits of each of the r primes yields the stated bound. $\square$

Corollary 6.31 (Why prime order matters). *If N has a small prime factor q , then by Theorem 6.30 an attacker can recover the discrete logarithm modulo that factor in $O(\sqrt{q})$ time, leaking part of the secret x . Security therefore requires N to have a large prime factor; the safest and simplest choice is to make N itself a large prime, so that G has no proper nontrivial subgroups and Pohlig–Hellman offers no reduction at all. This is precisely why the elliptic-curve groups used in Halo 2 and Orchard have large prime order.*

Remark 6.32 (Cofactors and subgroup attacks). In practice an elliptic curve over $\mathbb{F}_p$ has order $h \cdot q$ where q is a large prime and h is a small *cofactor*; cryptographic operations stay confined to the prime-order- q subgroup. If protocols neglect to check that incoming points lie in this subgroup (or fail to clear the cofactor), an adversary can supply a low-order point and use Pohlig–Hellman in the small factor to extract information modulo h – a real-world “small-subgroup attack”. The Pasta curves of Halo 2 have cofactor $h = 1$, i.e. *prime* order, which eliminates this class of attacks at the group level. This is the cryptographic payoff of all the number theory developed above: the multiplicative and additive structure of $\mathbb{Z}/q\mathbb{Z}$ for a large prime q is what gives the scheme both its functionality (a field of scalars) and its security (no Pohlig–Hellman reduction).

7 Polynomials over a field

Throughout this section, $\mathbb{F}$ denotes a field. Polynomials over a field are among the central objects of the entire development: they are the engine behind finite fields, error-correcting codes, and — most importantly for the present purposes — the polynomial commitment schemes and arithmetization that underlie Halo 2 and Zcash Orchard. The single most exploited fact in that entire edifice is deceptively elementary: *a nonzero polynomial of degree d over a field has at most d roots*. From it we extract Lagrange interpolation and the Schwartz–Zippel principle, the probabilistic heart of every interactive proof we eventually construct. This section develops all of this from the definition of a polynomial, assuming only that the reader knows what a field is (a commutative ring in which every nonzero element has a multiplicative inverse).

7.1 The polynomial ring $\mathbb{F}[X]$

We begin with the formal definition. The crucial pedagogical point — to which the development returns at length in Subsection 7.9 — is that a polynomial is *not* a function. It is a formal expression, a finite list of coefficients, that one may *evaluate* to obtain a function, but which carries more information than that function in general.

Definition 7.1 (Polynomial). A *polynomial* over $\mathbb{F}$ in the indeterminate X is a formal expression

$$f = a_0 + a_1X + a_2X^2 + \cdots + a_nX^n = \sum_{i=0}^n a_iX^i,$$

where $n \in \mathbb{N}$ (taking $\mathbb{N} = \{0, 1, 2, \dots\}$) and the *coefficients* $a_0, a_1, \dots, a_n$ lie in $\mathbb{F}$. Formally, we may identify f with the sequence $(a_0, a_1, a_2, \dots)$ of its coefficients, in which only finitely many entries are nonzero. Two polynomials are *equal* precisely when their coefficient sequences agree term by term, i.e. $\sum_i a_iX^i = \sum_i b_iX^i$ if and only if $a_i = b_i$ for every $i \in \mathbb{N}$. We call the symbol X an *indeterminate* or *formal variable*; it is not an element of $\mathbb{F}$ and stands for nothing in particular — it is merely a placeholder that organizes the coefficients. We denote the set of all polynomials over $\mathbb{F}$ in X by $\mathbb{F}[X]$.

Remark 7.2. The sequence model is the cleanest way to render Definition 7.1 fully rigorous: $\mathbb{F}[X]$ is the set of all functions $a: \mathbb{N} \rightarrow \mathbb{F}$ that are *eventually zero* (there is some N with $a(i) = 0$ for all $i > N$). The expression $\sum_i a_iX^i$ is then merely suggestive notation for the sequence $(a_0, a_1, \dots)$. We use the two points of view interchangeably. The notation is convenient because, as we show below, the arithmetic of these sequences is exactly the arithmetic the notation suggests.

We now equip the ring $\mathbb{F}[X]$ with addition and multiplication. Let $f = \sum_i a_iX^i$ and $g = \sum_i b_iX^i$, where we pad the shorter coefficient sequence with zeros so that both run over the same index set.

Definition 7.3 (Ring operations on $\mathbb{F}[X]$). The *sum* and *product* of $f = \sum_i a_iX^i$ and $g = \sum_i b_iX^i$ are

$$f + g = \sum_{i=0}^{\infty} (a_i + b_i) X^i, \quad f \cdot g = \sum_{k=0}^{\infty} c_k X^k, \quad \text{where} \quad c_k = \sum_{i+j=k} a_i b_j = \sum_{i=0}^k a_i b_{k-i}.$$

The coefficient c_k is the *convolution* of the two coefficient sequences. Both sums are finite: only finitely many a_i and b_j are nonzero, so only finitely many c_k are nonzero, and the result is again a polynomial.

The product formula is exactly what we obtain by multiplying out $(\sum_i a_iX^i)(\sum_j b_jX^j)$ and collecting like powers of X , using the rule $X^i \cdot X^j = X^{i+j}$. Stating it as a convolution makes no appeal to “substituting a value for X ”; it is a purely formal manipulation of coefficient sequences.

Theorem 7.4 ($\mathbb{F}[X]$ is a commutative ring). *With the operations of Definition 7.3, the set $\mathbb{F}[X]$ is a commutative ring with the following distinguished elements:*

- the zero polynomial 0, the sequence $(0, 0, 0, \dots)$, which is the additive identity;
- the constant polynomial 1, the sequence $(1, 0, 0, \dots)$, which is the multiplicative identity.

Moreover, the map $c \mapsto (c, 0, 0, \dots)$ embeds $\mathbb{F}$ into $\mathbb{F}[X]$ as a subring (the constant polynomials), so that $\mathbb{F} \subseteq \mathbb{F}[X]$.

Proof. We verify the ring axioms. *Additive structure.* Addition is defined coordinatewise, so it inherits commutativity, associativity, the identity $(0, 0, \dots)$, and inverses $(-a_0, -a_1, \dots)$ directly from the additive group of $\mathbb{F}$. Thus $(\mathbb{F}[X], +)$ is an abelian group.

Commutativity of multiplication. With $f \cdot g = \sum_k (\sum_{i+j=k} a_i b_j) X^k$, swapping the roles of f and g replaces $a_i b_j$ by $b_i a_j$; reindexing the inner sum $(i, j) \mapsto (j, i)$ and using commutativity of $\mathbb{F}$ shows $\sum_{i+j=k} a_i b_j = \sum_{i+j=k} b_i a_j$. Hence $f \cdot g = g \cdot f$.

Associativity of multiplication. For f, g, h with coefficients a_i, b_j, c_ℓ , the coefficient of X^m in $(fg)h$ is

$$\sum_{k+\ell=m} \left(\sum_{i+j=k} a_i b_j \right) c_\ell = \sum_{i+j+\ell=m} a_i b_j c_\ell,$$

and the same triple sum arises symmetrically as the coefficient of X^m in $f(gh)$. The two agree, so $(fg)h = f(gh)$.

Distributivity. The coefficient of X^k in $f(g+h)$ is $\sum_{i+j=k} a_i (b_j + c_j) = \sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$, which is the coefficient of X^k in $fg + fh$; this uses distributivity in $\mathbb{F}$. Hence $f(g+h) = fg + fh$, and the other distributive law follows by commutativity.

Identities. The sequence $(0, 0, \dots)$ is plainly additively neutral. For $1 = (1, 0, 0, \dots)$, the convolution gives $(1 \cdot f)_k = \sum_{i+j=k} \mathbf{1}_{[i=0]} a_j = a_k$, so $1 \cdot f = f$.

The embedding. The map $\iota: c \mapsto (c, 0, 0, \dots)$ sends sums to sums and, since $(c, 0, \dots) \cdot (d, 0, \dots)$ has k -th coefficient $\sum_{i+j=k} \iota(c)_i \iota(d)_j = cd \mathbf{1}_{[k=0]}$, products to products; it is injective because distinct constants give distinct sequences. Thus ι is an injective ring homomorphism, and $\mathbb{F}[X]$ is a commutative ring containing a copy of $\mathbb{F}$. $\square$

Remark 7.5. Multiplication of a polynomial $f = \sum_i a_i X^i$ by a constant $c \in \mathbb{F}$ (viewed as the constant polynomial) is the *scalar multiple* $cf = \sum_i (c a_i) X^i$. Together with addition this makes $\mathbb{F}[X]$ a vector space over $\mathbb{F}$ — indeed an infinite-dimensional one, with basis $\{1, X, X^2, X^3, \dots\}$. We rely on this linear structure constantly, for example when counting polynomials of bounded degree.

7.2 Degree and leading coefficients

Definition 7.6 (Degree, leading coefficient, monic). Let $f = \sum_i a_i X^i \in \mathbb{F}[X]$ be nonzero. Its *degree* $\deg f$ is the largest index n for which $a_n \neq 0$. The corresponding coefficient a_n is the *leading coefficient* of f , written $\text{lc}(f)$, and $a_n X^n$ is the *leading term*. The coefficient a_0 is the *constant term*. A nonzero polynomial is *monic* if its leading coefficient is 1. A polynomial of degree 0 is a nonzero constant; the nonzero constants together with the rule below are the only *constant polynomials* of finite degree. By convention we set $\deg 0 = -\infty$, with the arithmetic conventions $-\infty < n$, $-\infty + n = -\infty$, and $-\infty + (-\infty) = -\infty$ for every $n \in \mathbb{Z}$. We call polynomials of degree 1, 2, 3 *linear*, *quadratic*, *cubic* respectively.

The $-\infty$ convention for the zero polynomial is not pedantry: it makes the degree formulas below hold without exceptions, which streamlines every later induction on degree.

Proposition 7.7 (Degree of sums and products). *For all $f, g \in \mathbb{F}[X]$,*

$$\deg(f + g) \leq \max\{\deg f, \deg g\}, \quad (2)$$

$$\deg(f \cdot g) = \deg f + \deg g. \quad (3)$$

Equality holds in (2) whenever $\deg f \neq \deg g$.

Proof. If either f or g is zero, both claims reduce to the $-\infty$ conventions, so assume $f, g \neq 0$ with $\deg f = m$, $\deg g = n$, and leading coefficients $a_m, b_n \neq 0$.

For the sum, the coefficient of X^k in $f + g$ is $a_k + b_k$, which is 0 for every $k > \max\{m, n\}$; hence $\deg(f + g) \leq \max\{m, n\}$. If $m \neq n$, say $m > n$, then the coefficient of X^m in $f + g$ equals $a_m + 0 = a_m \neq 0$, giving $\deg(f + g) = m = \max\{m, n\}$. (When $m = n$, cancellation $a_m + b_m = 0$ can occur, lowering the degree; this is why (2) is only an inequality in general.)

For the product, the coefficient of X^{m+n} is $c_{m+n} = \sum_{i+j=m+n} a_i b_j$. Every term with $i > m$ has $a_i = 0$, and every term with $i < m$ forces $j = m+n-i > n$, so $b_j = 0$; thus the only surviving term is $i = m, j = n$, giving $c_{m+n} = a_m b_n$. For $k > m+n$ the same argument shows every term vanishes, so $c_k = 0$. Since $\mathbb{F}$ is a field it has no zero divisors, so $a_m b_n \neq 0$; hence $\deg(fg) = m+n$ and the leading coefficient of fg is $a_m b_n$. $\square$

The product formula (3) is where the field hypothesis first earns its keep: it required $a_m b_n \neq 0$, i.e. the absence of zero divisors. This single fact has sweeping consequences.

Corollary 7.8 ($\mathbb{F}[X]$ is an integral domain). *The ring $\mathbb{F}[X]$ has no zero divisors: if $fg = 0$ then $f = 0$ or $g = 0$. Equivalently, $\mathbb{F}[X]$ is an integral domain.*

Proof. If both f and g were nonzero, then $\deg(fg) = \deg f + \deg g \geq 0$ by (3), so $fg \neq 0$. Contrapositively, $fg = 0$ forces $f = 0$ or $g = 0$. $\square$

Corollary 7.9 (Cancellation). *If $f, g, h \in \mathbb{F}[X]$ with $h \neq 0$ and $fh = gh$, then $f = g$.*

Proof. From $fh = gh$ it follows that $(f - g)h = 0$; since $h \neq 0$, Corollary 7.8 forces $f - g = 0$. $\square$

Proposition 7.10 (Units of $\mathbb{F}[X]$). *A polynomial $f \in \mathbb{F}[X]$ is a unit (has a multiplicative inverse in $\mathbb{F}[X]$) if and only if it is a nonzero constant. Thus the units of $\mathbb{F}[X]$ are exactly the nonzero elements of $\mathbb{F}$.*

Proof. A nonzero constant $c \in \mathbb{F}^\times$ has inverse $c^{-1} \in \mathbb{F} \subseteq \mathbb{F}[X]$. Conversely, if $fg = 1$ then by (3), $\deg f + \deg g = \deg 1 = 0$; since degrees of nonzero polynomials are nonnegative, $\deg f = \deg g = 0$, so f is a nonzero constant. $\square$

Definition 7.11 (Associates). Two polynomials $f, g \in \mathbb{F}[X]$ are *associates* if $f = ug$ for some unit $u \in \mathbb{F}^\times$, i.e. if they differ only by a nonzero scalar factor.

Every nonzero polynomial f has exactly one monic associate, obtained by dividing through by its leading coefficient: $\text{lc}(f)^{-1}f$. Choosing the monic representative is the standard way to pin down "the" gcd or "the" irreducible factor uniquely, and we adopt it below.

7.3 The division algorithm

The arithmetic of $\mathbb{F}[X]$ parallels that of the integers $\mathbb{Z}$ with remarkable fidelity, and the source of the parallel is a division-with-remainder statement. Here the degree plays the role that absolute value plays in $\mathbb{Z}$.

Theorem 7.12 (Division algorithm). *Let $f, g \in \mathbb{F}[X]$ with $g \neq 0$. Then there exist unique polynomials $q, r \in \mathbb{F}[X]$ such that*

$$f = qg + r, \quad \text{with} \quad \deg r < \deg g.$$

We call q the quotient and r the remainder; we write $r = f \bmod g$ and $q = f \operatorname{div} g$.

Proof. Existence. Fix $g \neq 0$ and induct on $\deg f$. If $\deg f < \deg g$ (which includes the case $f = 0$, where $\deg f = -\infty$), take $q = 0$ and $r = f$; then $f = 0 \cdot g + f$ with $\deg r = \deg f < \deg g$.

Otherwise $m := \deg f \geq \deg g =: n$. Write the leading terms as $a_m X^m$ and $b_n X^n$, with $b_n \neq 0$. Consider

$$f_1 = f - \frac{a_m}{b_n} X^{m-n} g.$$

The subtracted term has degree m and leading coefficient $(a_m/b_n)b_n = a_m$, so it cancels the leading term of f ; hence $\deg f_1 < m$ (or $f_1 = 0$). By the induction hypothesis applied to f_1 , there are q_1, r with $f_1 = q_1 g + r$ and $\deg r < \deg g$. Then

$$f = f_1 + \frac{a_m}{b_n} X^{m-n} g = \left(q_1 + \frac{a_m}{b_n} X^{m-n} \right) g + r,$$

so $q = q_1 + (a_m/b_n)X^{m-n}$ and r work. The field hypothesis enters visibly here: division by the leading coefficient b_n requires b_n^{-1} to exist.

Uniqueness. Suppose $f = qg + r = q'g + r'$ with $\deg r, \deg r' < \deg g$. Subtracting, $(q - q')g = r' - r$. If $q \neq q'$, then $\deg((q - q')g) = \deg(q - q') + \deg g \geq \deg g$ by (3); but $\deg(r' - r) \leq \max\{\deg r', \deg r\} < \deg g$ by (2). These contradict each other, so $q = q'$, and then $r = r'$. $\square$

Remark 7.13 (The underlying algorithm). The existence proof is constructive: it is precisely *polynomial long division*. We repeatedly eliminate the current leading term by subtracting an appropriate monomial multiple of g ; each step lowers the degree, so the process terminates after at most $\deg f - \deg g + 1$ steps. For example, dividing $f = X^3 + 2X + 1$ by $g = X^2 + 1$ over $\mathbb{Q}$: first subtract $X \cdot g = X^3 + X$ to obtain $X + 1$, whose degree 1 is below $\deg g = 2$; hence $q = X$, $r = X + 1$, and indeed $X^3 + 2X + 1 = X(X^2 + 1) + (X + 1)$.

Definition 7.14 (Divisibility). We say g *divides* f , written $g \mid f$, if there is some $q \in \mathbb{F}[X]$ with $f = qg$; equivalently (for $g \neq 0$), if the remainder $f \bmod g$ is 0. In that case g is a *divisor* or *factor* of f .

7.4 Roots and the factor theorem

We now make precise the substitution of an element of $\mathbb{F}$ for the indeterminate. This is the first place where a polynomial gives rise to a function.

Definition 7.15 (Evaluation and roots). For $f = \sum_{i=0}^n a_i X^i \in \mathbb{F}[X]$ and $\alpha \in \mathbb{F}$, the *evaluation* of f at α is the field element

$$f(\alpha) = \sum_{i=0}^n a_i \alpha^i \in \mathbb{F},$$

which substitutes α for the indeterminate X . An element $\alpha \in \mathbb{F}$ is a *root* (or *zero*) of f if $f(\alpha) = 0$.

Proposition 7.16 (Evaluation is a ring homomorphism). Fix $\alpha \in \mathbb{F}$. The evaluation map $\text{ev}_\alpha: \mathbb{F}[X] \rightarrow \mathbb{F}$, $f \mapsto f(\alpha)$, satisfies, for all $f, g \in \mathbb{F}[X]$,

$$(f + g)(\alpha) = f(\alpha) + g(\alpha), \quad (f \cdot g)(\alpha) = f(\alpha)g(\alpha), \quad 1(\alpha) = 1.$$

That is, ev_α is a ring homomorphism.

Proof. Additivity is immediate from coordinatewise addition: $\sum_i (a_i + b_i) \alpha^i = \sum_i a_i \alpha^i + \sum_i b_i \alpha^i$. For multiplicativity, the product fg has k -th coefficient $c_k = \sum_{i+j=k} a_i b_j$, so

$$(fg)(\alpha) = \sum_k c_k \alpha^k = \sum_k \sum_{i+j=k} a_i b_j \alpha^{i+j} = \left(\sum_i a_i \alpha^i \right) \left(\sum_j b_j \alpha^j \right) = f(\alpha)g(\alpha),$$

where the middle step expands the product of the two sums and groups terms by $k = i + j$. Finally $1(\alpha) = 1$. This is the formal reason evaluation respects the algebra: the convolution defining polynomial multiplication collapses to ordinary multiplication in $\mathbb{F}$ because $\alpha^i \alpha^j = \alpha^{i+j}$. $\square$

The link between roots and divisibility is the factor theorem, an immediate but powerful consequence of the division algorithm.

Theorem 7.17 (Remainder theorem and factor theorem). *Let $f \in \mathbb{F}[X]$ and $\alpha \in \mathbb{F}$.*

1. (Remainder theorem.) *The remainder of f upon division by the linear polynomial $X - \alpha$ equals the constant $f(\alpha)$:*

$$f = q(X - \alpha) + f(\alpha) \quad \text{for some } q \in \mathbb{F}[X].$$

2. (Factor theorem.) *α is a root of f if and only if $(X - \alpha) \mid f$.*

Proof. (1) Divide f by $X - \alpha$. Since $\deg(X - \alpha) = 1$, the remainder r has degree < 1 , hence is a constant $r \in \mathbb{F}$. Write $f = q(X - \alpha) + r$ and evaluate at α using Proposition 7.16:

$$f(\alpha) = q(\alpha)(\alpha - \alpha) + r = q(\alpha) \cdot 0 + r = r.$$

Thus $r = f(\alpha)$.

- (2) By part (1), $(X - \alpha) \mid f$ means the remainder $f(\alpha)$ is 0, i.e. α is a root. □

This single theorem yields the central counting bound of the section.

Theorem 7.18 (At most d roots). *A nonzero polynomial $f \in \mathbb{F}[X]$ of degree d has at most d distinct roots in $\mathbb{F}$.*

Proof. Induct on $d = \deg f$. If $d = 0$, then f is a nonzero constant, which is never 0, so it has $0 \leq d$ roots. Suppose $d \geq 1$ and the claim holds for all smaller degrees. If f has no root in $\mathbb{F}$, then $0 \leq d$ and the claim holds. Otherwise let α be a root. By the factor theorem, $f = (X - \alpha)q$ for some q with $\deg q = d - 1$ (by (3)). Now let $\beta \neq \alpha$ be any root of f . Then

$$0 = f(\beta) = (\beta - \alpha)q(\beta),$$

and since $\beta - \alpha \neq 0$ and $\mathbb{F}$ has no zero divisors, it follows that $q(\beta) = 0$. Thus every root of f other than α is a root of q . By the induction hypothesis q has at most $d - 1$ distinct roots, so f has at most $(d - 1) + 1 = d$ distinct roots. □

Remark 7.19. Both the field hypothesis (no zero divisors) and the degree bookkeeping are essential. Over a ring with zero divisors the bound fails: in $\mathbb{Z}/8\mathbb{Z}$ the quadratic $X^2 - 1$ has the four roots 1, 3, 5, 7. This is why the polynomial methods underlying Halo 2 require a field — and indeed a finite field $\mathbb{F}_p$ or $\mathbb{F}_q$ — beneath them.

Corollary 7.20 (Few points determine a low-degree polynomial). *Let $f, g \in \mathbb{F}[X]$ both have degree at most d . If $f(\alpha) = g(\alpha)$ for $d + 1$ distinct values $\alpha \in \mathbb{F}$, then $f = g$ as polynomials.*

Proof. The difference $h := f - g$ has $\deg h \leq \max\{\deg f, \deg g\} \leq d$ and vanishes at $d + 1$ distinct points. If h were nonzero it would have degree at most d and yet $d + 1$ distinct roots, contradicting Theorem 7.18. Hence $h = 0$, i.e. $f = g$. □

Corollary 7.20 is the deterministic skeleton of the Schwartz–Zippel principle and of Lagrange interpolation; we develop both below.

7.5 Greatest common divisors and the Euclidean algorithm

Because $\mathbb{F}[X]$ supports division with remainder, the entire theory of greatest common divisors from elementary number theory transfers verbatim, with degree replacing magnitude.

Definition 7.21 (Greatest common divisor). Let $f, g \in \mathbb{F}[X]$, not both zero. A *greatest common divisor* of f and g is a polynomial $d \in \mathbb{F}[X]$ such that

1. $d \mid f$ and $d \mid g$ (it is a common divisor), and
2. every common divisor c of f and g satisfies $c \mid d$ (it is divisible by every other common divisor).

A gcd is determined only up to multiplication by a unit; we single out the unique *monic* gcd and denote it $\gcd(f, g)$. When $\gcd(f, g) = 1$ we say f and g are *coprime* or *relatively prime*.

The existence of such a d , its uniqueness up to associates, and its expression as an $\mathbb{F}[X]$ -linear combination of f and g constitute the content of the next two results. The Euclidean algorithm delivers all of this algorithmically.

Lemma 7.22 (Euclidean step). Let $f, g \in \mathbb{F}[X]$ with $g \neq 0$, and write $f = qg + r$ as in the division algorithm. Then the pair (f, g) and the pair (g, r) have exactly the same common divisors; in particular their gcds coincide.

Proof. If $c \mid f$ and $c \mid g$, then $c \mid (f - qg) = r$, so c is a common divisor of g and r . Conversely, if $c \mid g$ and $c \mid r$, then $c \mid (qg + r) = f$, so c is a common divisor of f and g . The two sets of common divisors are equal, hence so are their maximal elements under divisibility. $\square$

Theorem 7.23 (Euclidean algorithm and Bézout's identity). Let $f, g \in \mathbb{F}[X]$, not both zero. Then $\gcd(f, g)$ exists, is unique as a monic polynomial, and there exist $s, t \in \mathbb{F}[X]$ with

$$\gcd(f, g) = s f + t g. \quad (4)$$

Concretely, define a sequence by $r_0 = f$, $r_1 = g$, and, as long as $r_i \neq 0$,

$$r_{i-1} = q_i r_i + r_{i+1}, \quad \deg r_{i+1} < \deg r_i.$$

The degrees strictly decrease, so some $r_{N+1} = 0$; the last nonzero remainder r_N , made monic, is $\gcd(f, g)$.

Proof. Termination and the gcd property. If $g = 0$ the gcd is the monic associate of f ; so assume $g \neq 0$. The remainders satisfy $\deg r_1 > \deg r_2 > \dots \geq 0$ (a strictly decreasing sequence of nonnegative integers once we pass r_1), so the algorithm halts with some $r_{N+1} = 0$ and $r_N \neq 0$. Applying Lemma 7.22 repeatedly,

$$\{\text{common divisors of } (f, g)\} = \{\text{common divisors of } (r_1, r_2)\} = \dots = \{\text{common divisors of } (r_N, 0)\}.$$

The common divisors of $(r_N, 0)$ are exactly the divisors of r_N (since everything divides 0), whose greatest element under divisibility is r_N itself. Hence r_N is a gcd of f and g : it divides both, and any common divisor divides it. Making it monic gives the distinguished $\gcd(f, g)$.

Uniqueness. If d, d' are both monic gcds, then $d \mid d'$ and $d' \mid d$ by property (2), so $d' = ud$ for a unit u ; comparing leading coefficients of two monic polynomials forces $u = 1$, whence $d = d'$.

Bézout. We show by induction on the index i that each r_i is an $\mathbb{F}[X]$ -linear combination of f and g . Indeed $r_0 = f$ and $r_1 = g$ are such combinations, and the recurrence $r_{i+1} = r_{i-1} - q_i r_i$ expresses r_{i+1} as a combination once r_{i-1} and r_i are. Hence $r_N = s_0 f + t_0 g$ for some s_0, t_0 ; dividing through by $\text{lc}(r_N)$ to make it monic yields (4). $\square$

Remark 7.24 (Extended Euclidean algorithm). Carrying the linear-combination coefficients along with the remainders — the *extended Euclidean algorithm* — produces s and t explicitly alongside the gcd. This is precisely how one inverts elements in a field extension $\mathbb{F}[X]/(m)$: if $\gcd(f, m) = 1$ then $sf + tm = 1$, so s is the inverse of f modulo m . Finite fields $\mathbb{F}_q$, which carry the elliptic curves of Orchard, arise this way, and field inversion via the extended Euclidean algorithm is a basic operation of their arithmetic.

Corollary 7.25 (Euclid’s lemma in $\mathbb{F}[X]$). *Let $p, f, g \in \mathbb{F}[X]$.*

1. *If $\gcd(p, f) = 1$ and $p \mid fg$, then $p \mid g$.*
2. *If p is irreducible (Definition 7.26) and $p \mid fg$, then $p \mid f$ or $p \mid g$.*

Proof. (1) By Bézout there are s, t with $sp + tf = 1$. Multiply by g : $spg + tfg = g$. Now p divides spg trivially and divides tfg because $p \mid fg$; hence p divides their sum g .

(2) If $p \nmid f$, then $\gcd(p, f)$ is a monic divisor of p that is not the monic associate of p ; as p is irreducible its only monic divisors are 1 and its own monic associate, so $\gcd(p, f) = 1$. By part (1), $p \mid g$. $\square$

7.6 Irreducible polynomials and unique factorization

Definition 7.26 (Irreducible polynomial). A polynomial $p \in \mathbb{F}[X]$ is *irreducible* (over $\mathbb{F}$) if $\deg p \geq 1$ and p admits no factorization as a product $p = gh$ with both $\deg g \geq 1$ and $\deg h \geq 1$. Equivalently, p is nonconstant and its only factorizations are the trivial ones $p = (\text{unit}) \cdot (\text{associate of } p)$. A nonconstant polynomial that is not irreducible is *reducible*.

Remark 7.27 (Irreducibility is relative to the field). Irreducibility depends crucially on $\mathbb{F}$. The polynomial $X^2 + 1$ is irreducible over $\mathbb{R}$ (it has no real root, and a real quadratic factors iff it has a root), but over $\mathbb{C}$ it splits as $(X - i)(X + i)$, and over $\mathbb{F} = \mathbb{F}_2$ it factors as $(X + 1)^2$. The degree-1 polynomials are irreducible over every field. For later use, observe that a polynomial of degree 2 or 3 is irreducible over $\mathbb{F}$ if and only if it has no root in $\mathbb{F}$: any nontrivial factorization of a quadratic or cubic must include a linear factor, which by the factor theorem supplies a root. This rootlessness test fails for degree ≥ 4 , where a factorization into two irreducible quadratics has no roots yet is reducible.

Irreducible polynomials are the “primes” of $\mathbb{F}[X]$. We now show that $\mathbb{F}[X]$ admits unique factorization into them, exactly as $\mathbb{Z}$ factors into primes.

Theorem 7.28 ($\mathbb{F}[X]$ is a unique factorization domain). *Every nonconstant polynomial $f \in \mathbb{F}[X]$ factors as*

$$f = c p_1 p_2 \cdots p_k, \quad (5)$$

where $c = \text{lc}(f) \in \mathbb{F}^\times$ is a constant and $p_1, \dots, p_k$ are monic irreducible polynomials. This factorization is unique up to the order of the factors: if also $f = c' q_1 \cdots q_\ell$ with $c' \in \mathbb{F}^\times$ and each q_j monic irreducible, then $c = c'$, $k = \ell$, and the q_j are a reordering of the p_i .

Proof. Existence. Induct on $\deg f \geq 1$. If f is irreducible, write $f = \text{lc}(f) \cdot (\text{lc}(f)^{-1} f)$; the second factor is monic irreducible, giving $k = 1$. Otherwise $f = gh$ with $1 \leq \deg g, \deg h < \deg f$. By the induction hypothesis g and h each factor as a constant times monic irreducibles; multiplying these factorizations and collecting the two leading constants into a single $c \in \mathbb{F}^\times$ gives a factorization of f of the required form. (The overall constant must be $\text{lc}(f)$, since the product of the monic irreducibles is monic and so contributes leading coefficient 1.)

Uniqueness. Suppose

$$c p_1 \cdots p_k = c' q_1 \cdots q_\ell.$$

Comparing leading coefficients, both sides are products of monic polynomials scaled by c resp. c' , so $c = c' = \text{lc}(f)$; cancel them. The multisets $\{p_i\}$ and $\{q_j\}$ now coincide, by induction on k .

The irreducible p_1 divides $q_1 \cdots q_\ell$, so by the second part of Corollary 7.25 (Euclid's lemma) p_1 divides some q_j ; reindex so $j = 1$. Since q_1 is irreducible its only monic divisor of positive degree is itself, and p_1 is monic of positive degree dividing it, so $p_1 = q_1$. Cancel $p_1 = q_1$ (legitimate by Corollary 7.9) to obtain $p_2 \cdots p_k = q_2 \cdots q_\ell$. By the induction hypothesis the remaining factors agree up to order, and in particular $k = \ell$. This completes the induction. $\square$

Remark 7.29. The structural reason underlying Theorem 7.28 is that $\mathbb{F}[X]$ is a *Euclidean domain* (via the degree function and the division algorithm), hence a *principal ideal domain*: every ideal of $\mathbb{F}[X]$ is of the form $(g) = \{qg : q \in \mathbb{F}[X]\}$ for a single generator g , namely a gcd of any generating set. Bézout's identity (4) is the concrete shadow of this fact. Euclidean $\Rightarrow$ PID $\Rightarrow$ UFD is the standard chain of implications in ring theory; we have just walked it explicitly for $\mathbb{F}[X]$.

7.7 Lagrange interpolation

Corollary 7.20 established that a polynomial of degree at most d is determined by its values at $d + 1$ points. Lagrange interpolation is the constructive converse: *any* prescribed values at $d + 1$ distinct points arise from a *unique* such polynomial. This is the algebraic backbone of secret sharing, of Reed–Solomon codes, and of the polynomial encodings in Halo 2.

Theorem 7.30 (Lagrange interpolation). *Let $x_0, x_1, \dots, x_d \in \mathbb{F}$ be $d + 1$ distinct points, and let $y_0, y_1, \dots, y_d \in \mathbb{F}$ be arbitrary target values. Then there exists a unique polynomial $f \in \mathbb{F}[X]$ with $\deg f \leq d$ and $f(x_i) = y_i$ for all $i = 0, 1, \dots, d$. The explicit formula*

$$f(X) = \sum_{i=0}^d y_i \ell_i(X), \quad \text{where} \quad \ell_i(X) = \prod_{\substack{0 \leq j \leq d \\ j \neq i}} \frac{X - x_j}{x_i - x_j} \quad (6)$$

gives it. We call the polynomials ℓ_i the Lagrange basis polynomials for the nodes $x_0, \dots, x_d$.

Proof. The basis polynomials are well defined. Each denominator $\prod_{j \neq i} (x_i - x_j)$ is a product of nonzero elements of $\mathbb{F}$ (distinctness of the nodes), hence is a nonzero element of $\mathbb{F}$ and is invertible. Thus each ℓ_i is a genuine polynomial of degree exactly d (a product of d linear factors).

The interpolation property. Evaluate ℓ_i at a node x_m . If $m = i$, every factor is $(x_i - x_j)/(x_i - x_j) = 1$, so $\ell_i(x_i) = 1$. If $m \neq i$, then the factor with $j = m$ has numerator $x_m - x_m = 0$, so $\ell_i(x_m) = 0$. In short,

$$\ell_i(x_m) = \delta_{im} = \begin{cases} 1 & m = i, \\ 0 & m \neq i. \end{cases} \quad (7)$$

Therefore $f(x_m) = \sum_i y_i \ell_i(x_m) = \sum_i y_i \delta_{im} = y_m$ for each m , as required, and $\deg f \leq \max_i \deg \ell_i = d$.

Uniqueness. If f and $\tilde{f}$ both have degree at most d and agree at the $d + 1$ distinct points $x_0, \dots, x_d$, then by Corollary 7.20 they are equal as polynomials. Hence the interpolant is unique. $\square$

Example 7.31 (A small interpolation). Over $\mathbb{Q}$, find the polynomial of degree ≤ 2 with $f(0) = 1$, $f(1) = 3$, $f(2) = 7$. The basis polynomials for nodes 0, 1, 2 are

$$\ell_0 = \frac{(X-1)(X-2)}{(0-1)(0-2)} = \frac{(X-1)(X-2)}{2}, \quad \ell_1 = \frac{(X-0)(X-2)}{(1-0)(1-2)} = -X(X-2), \quad \ell_2 = \frac{(X-0)(X-1)}{(2-0)(2-1)} = \frac{X(X-1)}{2}.$$

Then $f = 1 \cdot \ell_0 + 3 \cdot \ell_1 + 7 \cdot \ell_2 = X^2 + X + 1$, and one verifies $f(0) = 1$, $f(1) = 3$, $f(2) = 7$.

Remark 7.32 (The evaluation isomorphism). Fix $d + 1$ distinct nodes $x_0, \dots, x_d$ and let $V = \{f \in \mathbb{F}[X] : \deg f \leq d\}$, an $(d+1)$ -dimensional $\mathbb{F}$ -vector space (basis $1, X, \dots, X^d$). The *evaluation map*

$$\text{ev}: V \rightarrow \mathbb{F}^{d+1}, \quad f \mapsto (f(x_0), \dots, f(x_d)),$$

is $\mathbb{F}$ -linear. Surjectivity is exactly the existence half of Lagrange interpolation, and injectivity is exactly the uniqueness half; together they say ev is a *linear isomorphism* $V \cong \mathbb{F}^{d+1}$. The Lagrange basis $\{\ell_i\}$ is precisely the basis of V dual to the standard basis of $\mathbb{F}^{d+1}$, by (7). This isomorphism — switching between a polynomial’s *coefficient representation* and its *evaluation representation* — is the conceptual core of the fast (FFT-based) polynomial arithmetic and the evaluation-domain encodings used throughout Halo 2.

Remark 7.33 (The vanishing polynomial). The product of all the linear factors over the node set,

$$Z(X) = \prod_{i=0}^d (X - x_i),$$

is the unique monic polynomial of degree $d+1$ that vanishes at every node; we call it the *vanishing polynomial* of the set $\{x_0, \dots, x_d\}$. A polynomial g vanishes on the whole node set if and only if $Z \mid g$ — one direction is the factor theorem applied repeatedly (using that the distinct $X - x_i$ are pairwise coprime), the other is immediate. The pair (Lagrange basis, vanishing polynomial) is exactly the data a proof system manipulates when it argues that a committed polynomial agrees with prescribed values on an evaluation domain.

7.8 The formal derivative and multiplicities

Roots can occur “with multiplicity,” and to detect repeated roots without any appeal to limits we introduce a purely algebraic derivative.

Definition 7.34 (Multiplicity of a root). Let $f \in \mathbb{F}[X]$ be nonzero and $\alpha \in \mathbb{F}$ a root. The *multiplicity* of α (as a root of f) is the largest integer $m \geq 1$ such that $(X - \alpha)^m \mid f$. We write $m = \text{mult}_\alpha(f)$. A root of multiplicity 1 is *simple*; a root of multiplicity ≥ 2 is *repeated* (or *multiple*).

This is well defined: the powers $(X - \alpha)^m$ have strictly increasing degree m , so only finitely many can divide the fixed polynomial f ; the largest such m exists. Equivalently, m is determined by writing $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$.

Proposition 7.35 (Counting roots with multiplicity). *If $f \in \mathbb{F}[X]$ is nonzero of degree d with distinct roots $\alpha_1, \dots, \alpha_r \in \mathbb{F}$ of multiplicities $m_1, \dots, m_r$, then*

$$f = (X - \alpha_1)^{m_1} \dots (X - \alpha_r)^{m_r} g$$

for some $g \in \mathbb{F}[X]$ with no roots in $\mathbb{F}$, and $\sum_{i=1}^r m_i \leq d$. In particular the number of roots counted with multiplicity is at most d , with equality iff f splits into linear factors over $\mathbb{F}$.

Proof. The distinct linear polynomials $X - \alpha_i$ are pairwise coprime (their gcd is 1, as a common divisor would be a nonzero constant by degree). Hence if $(X - \alpha_i)^{m_i} \mid f$ for each i , their product divides f : this follows by induction using that if coprime a, b both divide f then $ab \mid f$ (write $f = au = bv$; since $\gcd(a, b) = 1$, Bézout gives $sa + tb = 1$, so $f = saf + tbv = sa(bv) + tb(au) = ab(sv + tu)$). Thus $f = \prod_i (X - \alpha_i)^{m_i} g$; by maximality of each m_i , $g(\alpha_i) \neq 0$, and g has no other roots either (any root of g would be a root of f , hence some α_i). Taking degrees, $\sum_i m_i + \deg g = d$, so $\sum_i m_i \leq d$, with equality iff $\deg g = 0$, i.e. g is a nonzero constant and f is a product of linear factors. $\square$

Definition 7.36 (Formal derivative). The *formal derivative* is the map $D: \mathbb{F}[X] \rightarrow \mathbb{F}[X]$ defined on $f = \sum_{i=0}^n a_i X^i$ by

$$f'(X) = Df = \sum_{i=1}^n i a_i X^{i-1} = a_1 + 2a_2 X + 3a_3 X^2 + \dots + n a_n X^{n-1},$$

where $i a_i$ means a_i added to itself i times in $\mathbb{F}$ (the image of the integer i under $\mathbb{Z} \rightarrow \mathbb{F}$). No limits are involved; this is a formal, algebraic definition that makes sense over any field.

Proposition 7.37 (Rules of formal differentiation). *For all $f, g \in \mathbb{F}[X]$ and $c \in \mathbb{F}$:*

1. (Linearity.) $(f + g)' = f' + g'$ and $(cf)' = cf'$.
2. (Product rule / Leibniz.) $(fg)' = f'g + fg'$.
3. (Power rule.) $((X - \alpha)^m)' = m(X - \alpha)^{m-1}$ for every $\alpha \in \mathbb{F}$ and $m \geq 1$.

Proof. (1) is immediate from the coordinatewise, linear definition of D .

(2) By linearity it suffices to prove the product rule on monomials $f = X^a$, $g = X^b$, since both sides are bilinear in (f, g) and every polynomial is a linear combination of monomials. For monomials, $(X^a X^b)' = (X^{a+b})' = (a + b)X^{a+b-1}$, while $f'g + fg' = aX^{a-1}X^b + X^a bX^{b-1} = (a + b)X^{a+b-1}$. The two agree. Extending by bilinearity: writing $f = \sum_a u_a X^a$, $g = \sum_b v_b X^b$, we have $(fg)' = \sum_{a,b} u_a v_b (X^a X^b)' = \sum_{a,b} u_a v_b (X^a)' X^b + \sum_{a,b} u_a v_b X^a (X^b)' = f'g + fg'$.

(3) Induct on m using the product rule. For $m = 1$, $(X - \alpha)' = 1 = 1 \cdot (X - \alpha)^0$. Assuming the claim for m ,

$$((X - \alpha)^{m+1})' = ((X - \alpha)^m (X - \alpha))' = m(X - \alpha)^{m-1}(X - \alpha) + (X - \alpha)^m \cdot 1 = (m+1)(X - \alpha)^m. \quad \square$$

Theorem 7.38 (Derivative test for multiple roots). *Let $f \in \mathbb{F}[X]$ be nonzero and $\alpha \in \mathbb{F}$. Then α is a repeated root of f (multiplicity ≥ 2) if and only if $f(\alpha) = 0$ and $f'(\alpha) = 0$. Consequently f has a repeated root in $\mathbb{F}$ only if $\gcd(f, f') \neq 1$; and f has no repeated roots in any field containing $\mathbb{F}$ (it is separable) if and only if $\gcd(f, f') = 1$ in $\mathbb{F}[X]$.*

Proof. Write $\text{mult}_\alpha(f) = m \geq 1$, so $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$. By the product and power rules,

$$f' = m(X - \alpha)^{m-1}g + (X - \alpha)^m g' = (X - \alpha)^{m-1} [mg + (X - \alpha)g'].$$

Evaluate the bracket at α : it equals $mg(\alpha) + 0 = mg(\alpha)$.

If $m = 1$: then $f' = g + (X - \alpha)g'$, so $f'(\alpha) = g(\alpha) \neq 0$. Thus a simple root is *not* a root of f' .

If $m \geq 2$: then $(X - \alpha)^{m-1}$ with $m - 1 \geq 1$ divides f' , so $f'(\alpha) = 0$; and of course $f(\alpha) = 0$.

Hence $f(\alpha) = f'(\alpha) = 0$ exactly when $m \geq 2$, proving the first claim. For the gcd statements: a common root $\alpha \in \mathbb{F}$ of f and f' forces $(X - \alpha) \mid \gcd(f, f')$, so a repeated root makes the gcd nontrivial. The full separability statement requires care over extensions: one shows $\gcd(f, f')$, computed in $\mathbb{F}[X]$, is unchanged under field extension (the Euclidean algorithm uses only field operations in $\mathbb{F}$), and that over a splitting field a repeated factor $(X - \alpha)^m$, $m \geq 2$, contributes a common factor $(X - \alpha)$ to f and f' by the computation above, while distinct simple linear factors contribute none. Thus $\gcd(f, f') = 1$ in $\mathbb{F}[X]$ if and only if f has no repeated root in any extension. $\square$

Remark 7.39 (The pitfall of positive characteristic). The integer coefficient i in $f' = \sum_i i a_i X^{i-1}$ is interpreted in $\mathbb{F}$, and may vanish there. Over a field of characteristic p (meaning $p = 0$ in $\mathbb{F}$, with p prime — the setting of $\mathbb{F}_p$ and $\mathbb{F}_q$), the derivative of X^p is $pX^{p-1} = 0$. Thus $f = X^p - 1$ over $\mathbb{F}_p$ has $f' = 0$, and $\gcd(f, f') = f \neq 1$; indeed $X^p - 1 = (X - 1)^p$ in $\mathbb{F}_p[X]$, a single root of multiplicity p . This characteristic- p behavior is not a pathology to avoid but a structural feature one must respect when working over the finite fields of Orchard.

7.9 Polynomials as formal objects versus functions

This section sharpens the distinction noted at the outset, and extracts from it the probabilistic principle on which the remainder of this book rests.

Definition 7.40 (Polynomial function). Every $f \in \mathbb{F}[X]$ induces a *polynomial function* $\tilde{f}: \mathbb{F} \rightarrow \mathbb{F}$, $\alpha \mapsto f(\alpha)$. The assignment $f \mapsto \tilde{f}$ is a ring homomorphism from $\mathbb{F}[X]$ to the ring $\mathbb{F}^{\mathbb{F}}$ of all functions $\mathbb{F} \rightarrow \mathbb{F}$ (with pointwise operations), by Proposition 7.16 applied at every point.

The central question is whether two *distinct* polynomials can induce the *same* function. Over an infinite field they cannot; over a finite field they can.

Proposition 7.41 (When formal equals functional). *Let $f, g \in \mathbb{F}[X]$.*

1. *If $\mathbb{F}$ is infinite, then $\tilde{f} = \tilde{g}$ (equal as functions) implies $f = g$ (equal as polynomials). Equivalently, the map $f \mapsto \tilde{f}$ is injective, and one may identify polynomials with polynomial functions.*
2. *If $\mathbb{F}$ is finite, the map $f \mapsto \tilde{f}$ is not injective. For instance, over $\mathbb{F}_p$ the nonzero polynomial $X^p - X$ induces the zero function, since by Fermat's little theorem $\alpha^p = \alpha$ for every $\alpha \in \mathbb{F}_p$.*

Proof. (1) Suppose $\tilde{f} = \tilde{g}$. Then $h := f - g$ satisfies $h(\alpha) = 0$ for every $\alpha \in \mathbb{F}$. If h were nonzero it would have some finite degree d and thus at most d roots by Theorem 7.18; but $\mathbb{F}$ is infinite, so h has infinitely many roots — a contradiction. Hence $h = 0$ and $f = g$.

(2) The polynomial $X^p - X \in \mathbb{F}_p[X]$ is nonzero (its coefficients are -1 at X^1 and 1 at X^p), yet by Fermat's little theorem $\alpha^p = \alpha$, so $(X^p - X)(\alpha) = 0$ for all $\alpha \in \mathbb{F}_p$. Thus $X^p - X$ and the zero polynomial induce the same (zero) function while being distinct formal objects, so the map is not injective. $\square$

Remark 7.42 (Why the formal viewpoint is the correct one). Proposition 7.41(2) is precisely the reason we define polynomials as coefficient sequences rather than as functions. In cryptography one works over finite fields $\mathbb{F}_p, \mathbb{F}_q$, where the functional viewpoint loses information; the *degree* of a polynomial — a property of the formal object, invisible to the induced function once the field is finite — is what the security of polynomial commitment schemes rests on. A prover commits to a low-degree formal polynomial; the verifier tests it by evaluation. The gap between agreement as a function at the tested points and identity to the claimed formal polynomial is precisely the gap that the next principle quantifies.

Theorem 7.43 (Univariate Schwartz–Zippel). *Let $f \in \mathbb{F}[X]$ be a nonzero polynomial of degree at most d , and let $S \subseteq \mathbb{F}$ be a finite, nonempty set. If α is chosen uniformly at random from S , then*

$$\Pr_{\alpha \sim S}[f(\alpha) = 0] \leq \frac{d}{|S|}. \quad (8)$$

More generally, if $f, g \in \mathbb{F}[X]$ are distinct polynomials each of degree at most d , then $\Pr_{\alpha \sim S}[f(\alpha) = g(\alpha)] \leq d/|S|$.

Proof. By Theorem 7.18, the nonzero polynomial f of degree at most d has at most d roots in $\mathbb{F}$, hence at most d roots inside S . The number of $\alpha \in S$ with $f(\alpha) = 0$ is therefore at most d , and since α is uniform on S ,

$$\Pr_{\alpha \sim S}[f(\alpha) = 0] = \frac{|\{\alpha \in S : f(\alpha) = 0\}|}{|S|} \leq \frac{d}{|S|}.$$

For the general statement, apply this to the nonzero polynomial $h = f - g$, which has degree at most d ; $f(\alpha) = g(\alpha)$ iff $h(\alpha) = 0$. $\square$

Remark 7.44 (The principle stated plainly, and its application). Inequality (8) formalizes the statement that *two low-degree polynomials that agree at many points must be equal*. The deterministic version is Corollary 7.20: agreement at $d + 1$ points forces equality. Schwartz–Zippel is its probabilistic refinement: even a *single* random evaluation point detects unequal degree- $\leq d$ polynomials except with probability at most $d/|S|$. If $|S|$ is exponentially large (as for $S = \mathbb{F}_q$ with $q \approx 2^{255}$, the field sizes in Orchard) and d is polynomially bounded, this failure probability is negligible. This constitutes the foundational soundness argument for polynomial identity testing, and through it for the polynomial commitment schemes and arithmetization checks at the heart of Halo 2: to verify a purported polynomial identity, one evaluates both sides at a random challenge and compares. We invoke this principle repeatedly in the analysis of those protocols.

Remark 7.45 (Looking ahead: multivariate generalization). The full Schwartz–Zippel lemma extends (8) to multivariate polynomials $f \in \mathbb{F}[X_1, \dots, X_n]$: a nonzero polynomial of total degree d vanishes at a uniformly random point of S^n with probability at most $d/|S|$. The univariate case proved here is both the base case of its inductive proof and the version most directly used in the analysis of Halo 2; we develop the multivariate ring $\mathbb{F}[X_1, \dots, X_n]$ and this generalization when required.

8 Roots of unity and evaluation domains

The polynomial-based proof systems underlying Halo 2 and Zcash Orchard operate over a single, highly particular kind of object: a finite multiplicative subgroup of a field, generated by a *root of unity*. Nearly every operation such a system performs—committing to a polynomial, checking that two polynomials agree, enforcing that a constraint vanishes everywhere it should, multiplying or interpolating efficiently—is ultimately a computation indexed by such a subgroup. This section develops the theory of these subgroups from first principles. We assume only the basic theory of fields, polynomials over a field, and finite cyclic groups, and we recall these as needed.

Throughout, $\mathbb{F}$ denotes a field. When $\mathbb{F}$ is to be finite the notation $\mathbb{F}_q$ denotes the field with q elements (so q is a prime power); when we intend a prime field specifically we use the notation $\mathbb{F}_p$. We write the multiplicative group of nonzero elements as $\mathbb{F}^\times = \mathbb{F} \setminus \{0\}$. Set $\mathbb{N} = \{1, 2, 3, \dots\}$, and reserve the letter n for the order of the domain we shall construct.

8.1 Roots of unity

The starting point is the equation $X^n = 1$. Its solutions in a field are the roots of unity of order dividing n .

Definition 8.1. Let $\mathbb{F}$ be a field and $n \in \mathbb{N}$. An element $\omega \in \mathbb{F}$ is an *n -th root of unity* if $\omega^n = 1$. We write

$$\mu_n(\mathbb{F}) = \{\omega \in \mathbb{F} : \omega^n = 1\}$$

for the set of all n -th roots of unity in $\mathbb{F}$.

Note that $0 \notin \mu_n(\mathbb{F})$, since $0^n = 0 \neq 1$. Thus every n -th root of unity is a unit, i.e. lies in $\mathbb{F}^\times$. The set $\mu_n(\mathbb{F})$ is not merely a set; it is a group.

Proposition 8.2. $\mu_n(\mathbb{F})$ is a finite subgroup of $\mathbb{F}^\times$ of order at most n .

Proof. The constant 1 satisfies $1^n = 1$, so $1 \in \mu_n(\mathbb{F})$. If $\omega, \zeta \in \mu_n(\mathbb{F})$ then $(\omega\zeta)^n = \omega^n\zeta^n = 1 \cdot 1 = 1$ because $\mathbb{F}^\times$ is abelian, so $\omega\zeta \in \mu_n(\mathbb{F})$. If $\omega \in \mu_n(\mathbb{F})$ then $(\omega^{-1})^n = (\omega^n)^{-1} = 1^{-1} = 1$, so $\omega^{-1} \in \mu_n(\mathbb{F})$. Hence $\mu_n(\mathbb{F})$ is a subgroup of $\mathbb{F}^\times$. Every element of $\mu_n(\mathbb{F})$ is a root of the polynomial $X^n - 1 \in \mathbb{F}[X]$, which has degree n ; a nonzero polynomial of degree n over a field has at most n roots, so $|\mu_n(\mathbb{F})| \leq n$. $\square$

The bound $|\mu_n(\mathbb{F})| \leq n$ need not be attained. Whether it is attained is precisely the question of the existence of a *primitive* n -th root of unity.

Definition 8.3. An element $\omega \in \mathbb{F}$ is a *primitive n -th root of unity* if $\omega^n = 1$ and $\omega^k \neq 1$ for every integer k with $1 \leq k < n$. Equivalently, ω is primitive of order n if its multiplicative order $\text{ord}(\omega)$ equals n .

Here $\text{ord}(\omega)$, the *order* of ω , is the least positive integer d with $\omega^d = 1$; it exists for any $\omega \in \mu_n(\mathbb{F})$ because n itself is such an exponent and the positive integers are well-ordered. The two formulations in [theorem 8.3](#) agree: $\omega^k \neq 1$ for $1 \leq k < n$ together with $\omega^n = 1$ states exactly that the least positive exponent killing ω is n .

Lemma 8.4. Let $\omega \in \mathbb{F}^\times$ have order $d = \text{ord}(\omega)$. Then for any integer m we have $\omega^m = 1$ if and only if $d \mid m$. In particular ω is an n -th root of unity if and only if $d \mid n$, and ω is a primitive n -th root of unity if and only if $d = n$.

Proof. If $d \mid m$, write $m = dk$; then $\omega^m = (\omega^d)^k = 1^k = 1$. Conversely suppose $\omega^m = 1$. Divide with remainder: $m = qd + r$ with $0 \leq r < d$. Then $1 = \omega^m = (\omega^d)^q \omega^r = \omega^r$. Minimality of d forces $r = 0$, so $d \mid m$. The remaining claims are immediate: $\omega^n = 1 \iff d \mid n$, and primitivity is the case $d = n$. $\square$

A primitive n -th root of unity, when one exists, generates all the others.

Proposition 8.5. *If $\omega \in \mathbb{F}$ is a primitive n -th root of unity, then*

$$\mu_n(\mathbb{F}) = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$$

is a cyclic group of order exactly n , and these n powers are pairwise distinct.

Proof. The powers $1, \omega, \dots, \omega^{n-1}$ are pairwise distinct: if $\omega^i = \omega^j$ with $0 \leq i \leq j \leq n-1$, then $\omega^{j-i} = 1$ with $0 \leq j-i \leq n-1 < n$, so by primitivity (theorem 8.4, with $d = n$) we must have $j-i = 0$. Thus $\langle \omega \rangle$ has exactly n elements, all of which lie in $\mu_n(\mathbb{F})$ since $(\omega^k)^n = (\omega^n)^k = 1$. But $|\mu_n(\mathbb{F})| \leq n$ by theorem 8.2, so the inclusion $\langle \omega \rangle \subseteq \mu_n(\mathbb{F})$ is an equality of n -element sets. $\square$

Over the complex numbers a primitive n -th root of unity always exists, namely $\omega = e^{2\pi i/n}$; this is the classical picture of n equally spaced points on the unit circle. Over a finite field the situation is more delicate, and it is exactly the finite case that matters for cryptography.

8.2 Existence over finite fields: the condition $n \mid q-1$

The decisive structural fact about finite fields is that their multiplicative group is cyclic. We recall and prove this fact here, since the existence theorem for roots of unity rests on it.

Theorem 8.6 (Multiplicative group of a finite field is cyclic). *Let $\mathbb{F}_q$ be a finite field with q elements. Then $\mathbb{F}_q^\times$ is a cyclic group of order $q-1$.*

Proof. $\mathbb{F}_q^\times$ is a finite abelian group of order $q-1$. Let $N = \mathbb{F}_q^\times$ and let e be the exponent of N , i.e. the least common multiple of the orders of all its elements; equivalently the least positive integer with $g^e = 1$ for every $g \in N$. In a finite abelian group there exists an element whose order equals the exponent e . (This is the standard fact: write $e = \prod_i p_i^{a_i}$; for each prime power $p_i^{a_i}$ dividing e some element g_i has order divisible by $p_i^{a_i}$, so a suitable power of g_i has order exactly $p_i^{a_i}$; in an abelian group the product of elements of pairwise coprime orders has order equal to the product of the orders, giving an element of order $\prod_i p_i^{a_i} = e$.) Now every element of N satisfies $x^e = 1$, hence is a root of $X^e - 1 \in \mathbb{F}_q[X]$. This polynomial has at most e roots, so $q-1 = |N| \leq e$. On the other hand the exponent of a finite group never exceeds its order by Lagrange's theorem, since each element's order divides $|N|$, so $e \leq q-1$. Therefore $e = q-1$, and the element of order e found above has order $q-1 = |N|$; it generates N , which is thus cyclic. $\square$

A generator of $\mathbb{F}_q^\times$ is classically called a *primitive element* or *primitive root* of $\mathbb{F}_q$; it is a primitive $(q-1)$ -th root of unity in the sense of theorem 8.3.

Theorem 8.7 (Existence and count of primitive n -th roots of unity). *Let $\mathbb{F}_q$ be a finite field and $n \in \mathbb{N}$ with $\gcd(n, q) = 1$ (automatic when $n < p$, where $p = \text{char } \mathbb{F}_q$; equivalently, in characteristic p , exactly the condition $p \nmid n$). Then:*

1. $\mathbb{F}_q$ contains a primitive n -th root of unity if and only if $n \mid q-1$.
2. When $n \mid q-1$, the group $\mu_n(\mathbb{F}_q)$ is cyclic of order exactly n , and the number of primitive n -th roots of unity in $\mathbb{F}_q$ is $\varphi(n)$, where φ is Euler's totient function.

Proof. Let g be a generator of $\mathbb{F}_q^\times$, which is cyclic of order $q-1$ by theorem 8.6.

($\Rightarrow$) Suppose ω is a primitive n -th root of unity in $\mathbb{F}_q$. Then $\omega \in \mathbb{F}_q^\times$ has order n by theorem 8.3, and by Lagrange's theorem the order of any element divides the group order $q-1$. Hence $n \mid q-1$.

($\Leftarrow$) Suppose $n \mid q-1$, say $q-1 = nm$. Set $\omega = g^m$. Then $\omega^n = g^{mn} = g^{q-1} = 1$, so $\omega \in \mu_n(\mathbb{F}_q)$. Its order is

$$\text{ord}(\omega) = \text{ord}(g^m) = \frac{\text{ord}(g)}{\gcd(m, \text{ord}(g))} = \frac{q-1}{\gcd(m, q-1)} = \frac{q-1}{m} = n,$$

using the standard formula for the order of a power in a cyclic group together with $m \mid q - 1$ (so $\gcd(m, q - 1) = m$). Thus ω is a primitive n -th root of unity. By [theorem 8.5](#), $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$ is cyclic of order exactly n .

Count. In the cyclic group $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$ of order n , the element ω^k has order $n / \gcd(k, n)$. This equals n precisely when $\gcd(k, n) = 1$. The number of $k \in \{0, 1, \dots, n - 1\}$ with $\gcd(k, n) = 1$ is by definition $\varphi(n)$. Hence there are exactly $\varphi(n)$ primitive n -th roots of unity. (The coprimality hypothesis $\gcd(n, q) = 1$ is what guarantees $X^n - 1$ is separable and that μ_n has the full n elements; if $p = \text{char } \mathbb{F}_q$ divided n , then $X^n - 1$ would have repeated roots and $\mu_n(\mathbb{F}_q)$ would be strictly smaller. But $n \mid q - 1$ already forces $\gcd(n, q) = 1$, since $\gcd(q - 1, q) = 1$, so the hypothesis is in fact subsumed.) $\square$

Remark 8.8. The proof shows that for a finite field the clean statement is simply: $\mathbb{F}_q$ has a primitive n -th root of unity $\iff n \mid q - 1$, with no separate coprimality side-condition, because $n \mid q - 1$ automatically makes n coprime to q . We stated the condition $\gcd(n, q) = 1$ explicitly only to flag the role of the characteristic.

Example 8.9. Take $\mathbb{F}_q = \mathbb{F}_p$ with $p = 7$. Here $q - 1 = 6$, and 3 is a generator of $\mathbb{F}_p^\times$ (its powers are 3, 2, 6, 4, 5, 1, all six nonzero residues). For $n = 3 \mid 6$, set $m = 6/3 = 2$ and $\omega = 3^2 = 2$. Indeed $2^1 = 2$, $2^2 = 4$, $2^3 = 8 \equiv 1$, so $\omega = 2$ has order 3 and $\mu_3(\mathbb{F}_7) = \{1, 2, 4\}$. There are $\varphi(3) = 2$ primitive cube roots, namely 2 and 4. By contrast $n = 4 \nmid 6$, so $\mathbb{F}_7$ has *no* primitive 4-th root of unity: $X^4 - 1 = (X^2 - 1)(X^2 + 1)$ and $X^2 + 1$ is irreducible over $\mathbb{F}_7$ (it has no root since -1 is not a square mod 7).

Remark 8.10 (Two-adicity and FFT-friendly primes). For the fast algorithms below it is desirable that n be a power of two (or have many factors of two and other small primes). [theorem 8.7](#) shows this is possible exactly when $2^k \mid q - 1$. The largest k with $2^k \mid q - 1$ is called the *two-adicity* of the field. Designers deliberately choose primes p with high two-adicity—so that $p - 1 = 2^k \cdot c$ for large k —as scalar fields in proof systems precisely so that large smooth evaluation domains exist. The Pallas and Vesta curves underlying Halo 2/Orchard have scalar fields whose orders are such high-two-adicity primes, which is what makes the entire machinery of this section available.

8.3 The evaluation domain H and its vanishing polynomial

We now fix the central object of this section.

Definition 8.11. Let $\mathbb{F}$ be a field containing a primitive n -th root of unity ω . The *evaluation domain* of order n is the multiplicative subgroup

$$H = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\} = \mu_n(\mathbb{F}) \subseteq \mathbb{F}^\times,$$

a cyclic group of order n by [theorem 8.5](#). We frequently index its elements as $\omega^0, \omega^1, \dots, \omega^{n-1}$, and abbreviate ω^i by h_i when convenient.

Because H is a multiplicative *group*, it is closed under products and inverses, and $1 \in H$. These elementary facts are exactly what make H substantially more useful than an arbitrary set of n interpolation points: cosets, products, and shifts of H interact algebraically. The single most important polynomial attached to H is the one that vanishes precisely on it.

Definition 8.12. The *vanishing polynomial* (or *zerofier*) of H is

$$Z_H(X) = \prod_{h \in H} (X - h) \in \mathbb{F}[X].$$

Theorem 8.13 (Closed form and factorization of the vanishing polynomial). *With $H = \mu_n(\mathbb{F})$ of order n generated by a primitive n -th root of unity ω ,*

$$Z_H(X) = X^n - 1 = \prod_{i=0}^{n-1} (X - \omega^i).$$

Moreover Z_H is separable: it has n distinct roots and $\gcd(Z_H, Z'_H) = 1$.

Proof. Every $h \in H$ satisfies $h^n = 1$, hence is a root of $X^n - 1$. The n elements of H are distinct (theorem 8.5), so $X^n - 1$ has n distinct roots, namely all of H . A degree- n polynomial with n distinct roots h is $c \prod_{h \in H} (X - h)$ for some constant c ; comparing leading coefficients (both $X^n - 1$ and the product are monic) gives $c = 1$. Hence $X^n - 1 = \prod_{h \in H} (X - h) = Z_H(X)$. Separability: the derivative is $Z'_H(X) = nX^{n-1}$. Since $n \mid q - 1$ implies $n \neq 0$ in $\mathbb{F}$ (the characteristic does not divide n , by theorem 8.8), nX^{n-1} is nonzero and its only root is 0, which is not a root of $X^n - 1$. Thus Z_H and Z'_H share no root, so $\gcd(Z_H, Z'_H) = 1$ and Z_H is separable. $\square$

The closed form $Z_H(X) = X^n - 1$ is the workhorse of polynomial proof systems: testing whether a polynomial f vanishes on all of H reduces to checking divisibility by $X^n - 1$, which one can verify by exhibiting a quotient $q(X)$ with $f(X) = q(X)(X^n - 1)$. Moreover, evaluating Z_H at any point costs a single exponentiation X^n , not n multiplications. We record several structural consequences below.

Proposition 8.14 (Cosets and shifts). *Let $\gamma \in \mathbb{F}^\times$ and let $\gamma H = \{\gamma h : h \in H\}$ be the coset. Then the polynomial vanishing on γH is*

$$Z_{\gamma H}(X) = \prod_{h \in H} (X - \gamma h) = X^n - \gamma^n.$$

In particular, distinct cosets of H in $\mathbb{F}^\times$ have vanishing polynomials $X^n - c$ for distinct constants $c = \gamma^n$, and these cosets partition (a subgroup of) $\mathbb{F}^\times$ into translates of H on which X^n is constant.

Proof. Substitute $Y = X/\gamma$: $\prod_h (X - \gamma h) = \gamma^n \prod_h (Y - h) = \gamma^n (Y^n - 1) = \gamma^n ((X/\gamma)^n - 1) = X^n - \gamma^n$, using theorem 8.13. The constant is $c = \gamma^n$, and $\gamma_1 H = \gamma_2 H \iff \gamma_1/\gamma_2 \in H \iff (\gamma_1/\gamma_2)^n = 1 \iff \gamma_1^n = \gamma_2^n$, so distinct cosets give distinct c . $\square$

Proposition 8.15 (Sum of roots of unity). *If $n > 1$ then $\sum_{i=0}^{n-1} \omega^i = 0$, and more generally for any integer k ,*

$$\sum_{i=0}^{n-1} \omega^{ki} = \begin{cases} n & \text{if } n \mid k, \\ 0 & \text{if } n \nmid k. \end{cases}$$

Proof. If $n \mid k$ then $\omega^{ki} = (\omega^n)^{ki/n} = 1$ for each i , so the sum is n . If $n \nmid k$ then $\zeta := \omega^k \neq 1$ (since ω has order n , so $\omega^k = 1 \iff n \mid k$), and the finite geometric series gives

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1} = \frac{(\omega^n)^k - 1}{\zeta - 1} = \frac{1 - 1}{\zeta - 1} = 0,$$

the denominator being nonzero because $\zeta \neq 1$. The first claim is the case $k = 1$, valid for $n > 1$ since then $n \nmid 1$. $\square$

Remark 8.16 (Cyclotomic refinement). Over $\mathbb{Q}$ (or any field of characteristic not dividing n) the polynomial $X^n - 1$ refines further as $X^n - 1 = \prod_{d \mid n} \Phi_d(X)$, where Φ_d is the d -th cyclotomic polynomial whose roots are exactly the primitive d -th roots of unity; $\deg \Phi_d = \varphi(d)$, consistent with $\sum_{d \mid n} \varphi(d) = n$. The evaluation-domain story does not need cyclotomic polynomials—there the only factorization we use is the complete splitting of theorem 8.13—but the refinement explains why a field with $n \mid q - 1$ contains all n -th roots: $X^n - 1$ splits completely over $\mathbb{F}_q$.

8.4 The Lagrange basis on H

Attention now passes from the group H to the polynomials we study on it. The space of polynomials of degree less than n has dimension n over $\mathbb{F}$; since H has exactly n points, their values on H determine such polynomials. The basis adapted to this situation is the Lagrange basis.

Definition 8.17. Let $\mathbb{F}[X]_{<n}$ denote the $\mathbb{F}$ -vector space of polynomials of degree strictly less than n , together with the zero polynomial. It has the monomial basis $\{1, X, \dots, X^{n-1}\}$ and so $\dim_{\mathbb{F}} \mathbb{F}[X]_{<n} = n$.

Definition 8.18. For each $i \in \{0, 1, \dots, n-1\}$, the i -th *Lagrange basis polynomial* for $H = \{\omega^0, \dots, \omega^{n-1}\}$ is

$$L_i(X) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} \frac{X - \omega^j}{\omega^i - \omega^j} \in \mathbb{F}[X]_{<n}.$$

Proposition 8.19 (Interpolation property). *The polynomials $L_0, \dots, L_{n-1}$ satisfy the cardinality conditions*

$$L_i(\omega^j) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

they form a basis of $\mathbb{F}[X]_{<n}$, and they give the unique interpolant: for any data $(v_0, \dots, v_{n-1}) \in \mathbb{F}^n$ the polynomial

$$f(X) = \sum_{i=0}^{n-1} v_i L_i(X)$$

is the unique element of $\mathbb{F}[X]_{<n}$ with $f(\omega^i) = v_i$ for all i .

Proof. Each L_i has degree $n-1$ (a product of $n-1$ linear factors) and is well-defined because the ω^j are distinct, so the denominators $\omega^i - \omega^j$ are nonzero for $j \neq i$. Evaluation at $X = \omega^j$ with $j \neq i$ makes the factor $(X - \omega^j)$ vanish, giving $L_i(\omega^j) = 0$; at $X = \omega^i$ every factor is $(\omega^i - \omega^j)/(\omega^i - \omega^j) = 1$, giving $L_i(\omega^i) = 1$. This is $L_i(\omega^j) = \delta_{ij}$.

For the interpolation claim, $f = \sum_i v_i L_i$ lies in $\mathbb{F}[X]_{<n}$ and $f(\omega^j) = \sum_i v_i \delta_{ij} = v_j$. Uniqueness follows because, if $f, \tilde{f} \in \mathbb{F}[X]_{<n}$ agree on all n points of H , then $f - \tilde{f} \in \mathbb{F}[X]_{<n}$ has n distinct roots but degree $< n$, hence is the zero polynomial. Finally the L_i are linearly independent—if $\sum_i c_i L_i = 0$, evaluation at ω^j yields $c_j = 0$ —and there are n of them in the n -dimensional space $\mathbb{F}[X]_{<n}$, so they form a basis. $\square$

On a generic point set the Lagrange polynomials admit no closed form simpler than the defining product. On the *group* H they collapse to a compact closed form involving the vanishing polynomial.

Theorem 8.20 (Closed form of the Lagrange basis on H). *For $H = \mu_n(\mathbb{F})$ with primitive root ω , and for each i ,*

$$L_i(X) = \frac{\omega^i}{n} \cdot \frac{X^n - 1}{X - \omega^i} = \frac{\omega^i (X^n - 1)}{n(X - \omega^i)}.$$

In particular, dividing out one linear factor and scaling yields all n Lagrange polynomials from the single polynomial $Z_H(X) = X^n - 1$.

Proof. Fix i . From [theorem 8.13](#), $Z_H(X) = \prod_{j=0}^{n-1} (X - \omega^j)$, so

$$\frac{Z_H(X)}{X - \omega^i} = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (X - \omega^j),$$

which is exactly the numerator of L_i in [theorem 8.18](#). It remains to evaluate the denominator $\prod_{j \neq i} (\omega^i - \omega^j)$. This equals $\lim_{X \rightarrow \omega^i} Z_H(X)/(X - \omega^i) = Z'_H(\omega^i)$, the value of the derivative; concretely $Z'_H(X) = nX^{n-1}$, so

$$Z'_H(\omega^i) = n(\omega^i)^{n-1} = n\omega^{in}\omega^{-i} = n \cdot 1 \cdot \omega^{-i} = \frac{n}{\omega^i},$$

using $\omega^{in} = (\omega^n)^i = 1$. Therefore

$$L_i(X) = \frac{Z_H(X)/(X - \omega^i)}{Z'_H(\omega^i)} = \frac{(X^n - 1)/(X - \omega^i)}{n/\omega^i} = \frac{\omega^i(X^n - 1)}{n(X - \omega^i)},$$

as claimed. (The cancellation $\prod_{j \neq i} (\omega^i - \omega^j) = Z'_H(\omega^i)$ is the standard fact that, for a polynomial split into distinct linear factors, the value at a root of the product with that factor removed equals the derivative at the root.) $\square$

Remark 8.21 (Barycentric form). The closed form renders the *barycentric* interpolation formula on H especially clean. Writing $f(X) = \sum_i v_i L_i(X)$ and substituting [theorem 8.20](#),

$$f(X) = \frac{X^n - 1}{n} \sum_{i=0}^{n-1} \frac{\omega^i v_i}{X - \omega^i}.$$

Evaluating f at a point $z \notin H$ thus costs one evaluation of $z^n - 1$ and n terms of the sum, with no per-point polynomial reconstruction. This is precisely the formula a verifier uses to evaluate a committed polynomial's Lagrange representation at a random challenge z .

Proposition 8.22 (Lagrange polynomials sum to one). $\sum_{i=0}^{n-1} L_i(X) = 1$ *identically*.

Proof. The polynomial $g(X) = \sum_i L_i(X) - 1$ lies in $\mathbb{F}[X]_{<n}$ (degree $< n$) and vanishes at every ω^j because $\sum_i L_i(\omega^j) = \sum_i \delta_{ij} = 1$. A polynomial of degree $< n$ with n distinct roots is zero, so $g \equiv 0$. $\square$

Example 8.23 (The all-ones value vector). The constant polynomial 1 interpolates the data $v_i = 1$ for all i ; [theorem 8.22](#) is the statement $\sum_i 1 \cdot L_i = 1$. Likewise the values of X on H are $v_i = \omega^i$, and [theorem 8.19](#) gives $\sum_i \omega^i L_i(X) = X$ (both sides agree on H and have degree $< n$). We invoke these identities constantly when rewriting monomials in the Lagrange basis.

8.5 Evaluation and interpolation as mutually inverse linear maps

An element of $\mathbb{F}[X]_{<n}$ admits two natural descriptions: by its n coefficients, or by its n values on H . Both are coordinate systems on the same n -dimensional vector space, and passage between them is a linear isomorphism.

Definition 8.24 (Evaluation map). The *evaluation map* on H is the $\mathbb{F}$ -linear map

$$\text{ev}_H: \mathbb{F}[X]_{<n} \longrightarrow \mathbb{F}^n, \quad \text{ev}_H(f) = (f(\omega^0), f(\omega^1), \dots, f(\omega^{n-1})).$$

Linearity is immediate from $(af + bg)(\omega^i) = af(\omega^i) + bg(\omega^i)$.

Theorem 8.25 (Evaluation and interpolation are mutually inverse). *The evaluation map ev_H is a vector-space isomorphism. Its inverse is the interpolation map*

$$\text{int}_H: \mathbb{F}^n \longrightarrow \mathbb{F}[X]_{<n}, \quad \text{int}_H(v_0, \dots, v_{n-1}) = \sum_{i=0}^{n-1} v_i L_i(X).$$

That is, $\text{int}_H \circ \text{ev}_H = \text{id}_{\mathbb{F}[X]_{<n}}$ and $\text{ev}_H \circ \text{int}_H = \text{id}_{\mathbb{F}^n}$.

Proof. Both maps are linear and act between spaces of the same finite dimension n , so it suffices to verify that they compose to the identity in one order; we check both for clarity. For $v \in \mathbb{F}^n$, $\text{int}_H(v) = \sum_i v_i L_i$ has value $\sum_i v_i L_i(\omega^j) = v_j$ at ω^j by [theorem 8.19](#), so $\text{ev}_H(\text{int}_H(v)) = v$. For $f \in \mathbb{F}[X]_{<n}$, $\text{int}_H(\text{ev}_H(f)) = \sum_i f(\omega^i) L_i$ is, by the uniqueness clause of [theorem 8.19](#), the unique degree- $< n$ polynomial taking the values $f(\omega^i)$ on H —which is f itself. Hence the two maps are inverse, and in particular ev_H is bijective and linear, i.e. an isomorphism. $\square$

In coordinates relative to the monomial basis, ev_H is multiplication by a Vandermonde matrix, which we name here because it is the bridge to the Fourier transform.

Definition 8.26 (Vandermonde / DFT matrix). Let $f(X) = \sum_{k=0}^{n-1} c_k X^k$ have coefficient vector $c = (c_0, \dots, c_{n-1})$. Then $\text{ev}_H(f)$ has i -th entry $\sum_k c_k \omega^{ik}$. Thus, relative to the monomial basis on the domain and the standard basis on $\mathbb{F}^n$, the *Vandermonde matrix*

$$V = (\omega^{ik})_{0 \leq i, k \leq n-1} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & & & & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)^2} \end{pmatrix},$$

represents ev_H , so that $\text{ev}_H(f) = Vc$. We call V the (discrete Fourier) transform matrix on H .

Theorem 8.27 (Invertibility and the inverse transform). *The matrix V of [theorem 8.26](#) is invertible, with*

$$(V^{-1})_{k,i} = \frac{1}{n} \omega^{-ik}.$$

Equivalently, $V^{-1} = \frac{1}{n} \bar{V}$ where $\bar{V} = (\omega^{-ik})_{i,k}$ is the transform matrix built from ω^{-1} (which is itself a primitive n -th root of unity). Consequently the inversion formula relates the coefficient and value representations by

$$c_k = \frac{1}{n} \sum_{i=0}^{n-1} f(\omega^i) \omega^{-ik}, \quad f(\omega^i) = \sum_{k=0}^{n-1} c_k \omega^{ik}.$$

Proof. It suffices to show $\bar{V}V = nI$. The (k, k') entry of $\bar{V}V$ is

$$\sum_{i=0}^{n-1} \omega^{-ik} \omega^{ik'} = \sum_{i=0}^{n-1} \omega^{i(k'-k)}.$$

By [theorem 8.15](#) this sum is n if $n \mid (k' - k)$ and 0 otherwise. For $k, k' \in \{0, \dots, n-1\}$ one has $n \mid (k' - k) \iff k = k'$. Hence $\bar{V}V = nI$, giving $V^{-1} = \frac{1}{n} \bar{V}$ (we may divide by n since $n \neq 0$ in $\mathbb{F}$). The component formulas write out the matrix identities $\text{ev}_H(f) = Vc$ and $c = V^{-1} \text{ev}_H(f)$. $\square$

Remark 8.28. [theorem 8.25](#) and [theorem 8.27](#) are two faces of one statement. The abstract face: evaluation on H is an isomorphism whose inverse is Lagrange interpolation. The concrete face: that isomorphism is the matrix V , and its inverse is, up to the scalar $1/n$, the same matrix with ω replaced by ω^{-1} . The near-symmetry between a transform and its inverse—differing only by conjugating $\omega \mapsto \omega^{-1}$ and a scale $1/n$ —is the algebraic reason the fast inverse transform costs exactly as much as the forward one.

8.6 The number-theoretic Fourier transform

The map ev_H , read as the matrix V , is the discrete Fourier transform, but over a finite field rather than over $\mathbb{C}$. Because it involves no analytic structure (no complex exponentials, no convergence), we call it the *number-theoretic transform*.

Definition 8.29 (Number-theoretic transform). Let ω be a primitive n -th root of unity in $\mathbb{F}$. The *number-theoretic transform* (NTT) of a vector $a = (a_0, \dots, a_{n-1}) \in \mathbb{F}^n$ is the vector $\hat{a} = (\hat{a}_0, \dots, \hat{a}_{n-1})$ with

$$\hat{a}_j = \sum_{k=0}^{n-1} a_k \omega^{jk}, \quad j = 0, \dots, n-1.$$

The *inverse number-theoretic transform* is

$$a_k = \frac{1}{n} \sum_{j=0}^{n-1} \hat{a}_j \omega^{-jk}.$$

In the language above: if a is the coefficient vector of $f \in \mathbb{F}[X]_{<n}$, then $\hat{a} = \text{ev}_H(f)$ is its value vector on H ; the NTT is evaluation, and the inverse NTT is interpolation.

That the inverse formula indeed inverts the forward one is exactly [theorem 8.27](#). The NTT enjoys the same convolution property as the classical DFT, which is the mechanism underlying fast polynomial multiplication.

Theorem 8.30 (Convolution theorem). *For $f, g \in \mathbb{F}[X]$ with $\deg f + \deg g < n$, let $c = fg$ be their product (degree $< n$). Then for every $h \in H$,*

$$c(h) = f(h)g(h),$$

*i.e. pointwise multiplication of value vectors corresponds to polynomial multiplication. Equivalently, writing a, b for the coefficient vectors of f, g and $a * b$ for the (cyclic) convolution defining the coefficients of the product modulo $X^n - 1$,*

$$\widehat{a * b}_j = \hat{a}_j \hat{b}_j \quad (j = 0, \dots, n-1),$$

so the NTT turns convolution into entrywise multiplication.

Proof. The first identity is the statement that evaluation is a ring homomorphism $\mathbb{F}[X] \rightarrow \mathbb{F}$, $f \mapsto f(h)$: $(fg)(h) = f(h)g(h)$ for any h . Since $\deg(fg) < n$, interpolation recovers the product $c = fg$ from its values on the n -point set H ([theorem 8.19](#)), so the value vectors multiply entrywise and uniquely determine c . For the convolution statement: multiplication in the quotient ring $\mathbb{F}[X]/(X^n - 1)$ corresponds to cyclic convolution of coefficient vectors, and since $H = \mu_n$ is exactly the set of roots of $X^n - 1$, the map $p \mapsto \text{ev}_H(p)$ is a ring isomorphism $\mathbb{F}[X]/(X^n - 1) \xrightarrow{\sim} \mathbb{F}^n$ (with $\mathbb{F}^n$ under entrywise operations). It sends the product (convolution) of a, b to the entrywise product of $\hat{a}, \hat{b}$. This is the displayed identity. When $\deg f + \deg g < n$ the cyclic and ordinary products coincide, so the same statement computes the genuine product fg . $\square$

Remark 8.31 (Computational significance). [theorem 8.30](#) reduces multiplying two degree- $< n/2$ polynomials—a quadratic-cost operation by the schoolbook method—to: transform both (NTT), multiply the value vectors entrywise (n multiplications), and transform back (inverse NTT). If the NTT runs in $O(n \log n)$ operations, the whole multiplication costs $O(n \log n)$. The fast Fourier transform supplies that conditional.

8.7 The fast Fourier transform

Computing $\hat{a} = Va$ directly from [theorem 8.26](#) costs $\Theta(n^2)$ field multiplications: each of the n outputs is a sum of n products. The fast Fourier transform (FFT)—here a number-theoretic FFT—computes the same vector in $O(n \log n)$ operations by a divide-and-conquer recursion, provided n is a power of two (or more generally smooth). This is where the requirement that H be a *smooth* subgroup, [theorem 8.10](#), becomes indispensable.

Theorem 8.32 (Cooley–Tukey radix-2 step). *Suppose $n = 2m$ and ω is a primitive n -th root of unity in $\mathbb{F}$. Given $f(X) = \sum_{k=0}^{n-1} a_k X^k$, split its coefficients by parity into*

$$f_{\text{even}}(Y) = \sum_{k=0}^{m-1} a_{2k} Y^k, \quad f_{\text{odd}}(Y) = \sum_{k=0}^{m-1} a_{2k+1} Y^k,$$

so that $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$. Then $\eta := \omega^2$ is a primitive m -th root of unity, and for $j = 0, \dots, m-1$,

$$f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j), \quad (9)$$

$$f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j). \quad (10)$$

Thus the length- n NTT of a reduces to two length- m NTTs (of the even and odd subsequences, over the domain $\langle \eta \rangle$) plus $O(n)$ extra operations.

Proof. The identity $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$ regroups the sum $\sum_k a_k X^k$ into even-index and odd-index terms; in the odd terms we pull out one factor of X , leaving X^{2k} . That $\eta = \omega^2$ is a primitive m -th root of unity follows from $\text{ord}(\omega^2) = n / \gcd(2, n) = n/2 = m$ (theorem 8.4 and the order-of-a-power formula), using $2 \mid n$. It remains to evaluate. For (9), put $X = \omega^j$; then $X^2 = \omega^{2j} = \eta^j$, so $f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j)$. For (10), put $X = \omega^{j+m}$; then $X^2 = \omega^{2j+2m} = \omega^{2j} \omega^n = \eta^j \cdot 1 = \eta^j$ (since $2m = n$ and $\omega^n = 1$), while the linear factor is $\omega^{j+m} = \omega^j \omega^m = -\omega^j$. Here $\omega^m = -1$ because ω^m has order $n / \gcd(m, n) = n/m = 2$, and -1 is the unique element of order 2 in a field (it satisfies $x^2 = 1$, $x \neq 1$, so $x = -1$; note $-1 \neq 1$ since $\text{char } \mathbb{F} \nmid n$ and n is even forces $\text{char } \mathbb{F} \neq 2$). Substituting, $f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j)$. $\square$

Remark 8.33 (The butterfly). Equations (9)–(10) constitute the *butterfly*: from the two subtransform outputs $E_j = f_{\text{even}}(\eta^j)$ and $O_j = f_{\text{odd}}(\eta^j)$ one forms the *twiddle* $t_j = \omega^j O_j$ and reads off the two outputs $E_j + t_j$ and $E_j - t_j$. Pictorially, E_j and O_j sit as two nodes on the left; two crossing edges (forming an $\times$, the shape that names the operation) carry them to the two outputs on the right, one edge weighted by $+\omega^j$ and the other by $-\omega^j$. Each butterfly costs one multiplication (the twiddle) and two additions, and there are $n/2$ of them per level.

Theorem 8.34 ($O(n \log n)$ cost of the FFT). *Let $n = 2^t$ and let $T(n)$ be the number of field additions and multiplications the radix-2 recursion of theorem 8.32 uses to compute a length- n NTT. Then*

$$T(n) = 2T(n/2) + cn \quad (n > 1), \quad T(1) = O(1),$$

for a constant c , and therefore $T(n) = O(n \log n)$. The inverse NTT has the same cost, since it is the forward NTT with ω replaced by ω^{-1} followed by scaling each entry by $1/n$ (theorem 8.27).

Proof. The recursion is exactly theorem 8.32: a length- n transform calls two length- $n/2$ transforms (on the even and odd coefficient subsequences) and then combines them with $n/2$ butterflies, each of $O(1)$ operations, for the cn overhead. Unrolling, with $n = 2^t$,

$$T(2^t) = 2T(2^{t-1}) + c2^t.$$

Dividing by 2^t , and letting $S(t) = T(2^t)/2^t$, gives $S(t) = S(t-1) + c$, so $S(t) = S(0) + ct = O(1) + ct$. Hence $T(2^t) = 2^t S(t) = O(2^t) + ct2^t = O(t2^t)$. With $n = 2^t$, $t = \log_2 n$, this is $T(n) = O(n \log n)$. (Equivalently: the recursion tree has $\log_2 n$ levels, each doing $\Theta(n)$ total work across all subproblems at that level.) The inverse-transform claim is theorem 8.27: the same Cooley–Tukey recursion with ω^{-1} in place of ω , and a final pass of n divisions by n , all within $O(n \log n)$. $\square$

Remark 8.35 (Mixed radix and smoothness). The radix-2 split requires $2 \mid n$ at every level, i.e. $n = 2^t$. More generally, if $n = p \cdot m$ one may split f into p subsequences by residue mod p and reduce a length- n NTT to p length- m NTTs plus $O(pn)$ combination work; recursing through *any* factorization $n = p_1 p_2 \cdots p_r$ into small primes yields cost $O(n \sum_\ell p_\ell)$, which is $O(n \log n)$ when all p_ℓ are bounded. The hypothesis that enables this is that n be *smooth*—a product of small primes—and, crucially, that the field actually *contain* a primitive n -th root of unity, i.e. $n \mid q - 1$ ([theorem 8.7](#)). A power-of-two n dividing $q - 1$ is the optimal case: pure radix-2 with the cleanest butterfly.

Example 8.36 (A length-4 transform). Take $n = 4$ with primitive 4-th root ω (so $\omega^2 = -1$, $\omega^4 = 1$, $\eta = \omega^2 = -1$ the primitive 2nd root). For $f = a_0 + a_1X + a_2X^2 + a_3X^3$, set $E = (a_0 + a_2, a_0 - a_2)$ and $O = (a_1 + a_3, a_1 - a_3)$, the two length-2 transforms over $\{1, -1\}$. Then the butterflies give

$$\hat{a}_0 = E_0 + \omega^0 O_0, \quad \hat{a}_2 = E_0 - \omega^0 O_0, \quad \hat{a}_1 = E_1 + \omega^1 O_1, \quad \hat{a}_3 = E_1 - \omega^1 O_1.$$

This uses 2 sub-transforms of 2 additions each plus 4 butterfly operations—far fewer than the 16 products of the direct 4×4 matrix multiply, and the gap widens as n grows.

8.8 The role of smooth multiplicative domains in polynomial proof systems

This section closes by assembling the pieces into the reason these domains form the substrate of Halo 2 and Orchard. A proof system of the polynomial type encodes a computation as a system of polynomial identities and convinces a verifier that those identities hold. The evaluation domain $H = \mu_n$ is where the encoding lives, and every property proved above is load-bearing.

Remark 8.37 (The roles played by H , Z_H , the Lagrange basis, and the FFT).

- **Witnesses as evaluations on H .** A prover lays out the trace of a computation as the values of one or more polynomials on the n points of H . By [theorem 8.25](#) this is the same data as a polynomial of degree $< n$; the prover may freely move between the value (Lagrange) representation, natural for stating constraints row by row, and the coefficient representation, natural for committing and for low-degree testing. The change of representation is the (inverse) NTT.
- **Constraints as divisibility by Z_H .** “Constraint polynomial C holds on every row” means C vanishes on all of H , which by [theorem 8.13](#) means exactly that $Z_H = X^n - 1$ divides C . The prover demonstrates this by exhibiting a quotient t with $C(X) = t(X)(X^n - 1)$. The inexpensive closed form $Z_H(X) = X^n - 1$ —one exponentiation to evaluate—is what makes this check efficient at a random point. [theorem 8.14](#) extends the same technique to cosets γH , used to separate quotient and remainder domains.
- **Lagrange selectors.** The Lagrange polynomials L_i of [theorem 8.18](#) are *selectors*: L_i is 1 on row i and 0 elsewhere ([theorem 8.19](#)), so boundary/instance constraints (e.g. “the first row equals the public input”) become $L_0 \cdot (\text{something})$. The closed form and barycentric formula ([theorem 8.20](#), [theorem 8.21](#)) let the verifier evaluate these at a random challenge in $O(n)$ or even $O(\log n)$ field operations rather than reconstructing polynomials.
- **Group structure for permutations and shifts.** Because H is a cyclic *group*, multiplication by ω is the cyclic “next-row” shift $\omega^i \mapsto \omega^{i+1}$. Constraints relating a row to its neighbour become $f(\omega X)$ versus $f(X)$; permutation arguments (the heart of Halo 2’s copy constraints) rest on the action of the group on itself and on its cosets. An arbitrary set of n points offers none of this—it requires the multiplicative *group* H of [theorem 8.11](#).
- **Smoothness for speed.** Provers manipulate polynomials of size n (often after blowing the domain up by a small constant factor for low-degree testing). Every transform between

representations, every multiplication, every coset evaluation is an NTT. By [theorem 8.34](#) these cost $O(n \log n)$ instead of $O(n^2)$ —the difference between feasible and infeasible at the sizes used in practice—*provided* n is smooth and $\mathbb{F}_q$ contains a primitive n -th root of unity. By [theorem 8.7](#) that requires $n \mid q - 1$; by [theorem 8.10](#) this is why we choose high-two-adicity primes for the scalar fields of the Pallas and Vesta curves.

In summary: the existence condition $n \mid q - 1$ ([theorem 8.7](#)) supplies the group H ; the group supplies the clean vanishing polynomial ([theorem 8.13](#)), the closed-form Lagrange basis ([theorem 8.20](#)), and the evaluation/interpolation isomorphism ([theorem 8.25](#)); and smoothness of n supplies the $O(n \log n)$ FFT ([theorem 8.34](#)). Together these are exactly the ingredients a polynomial proof system requires, which is why such systems rest on smooth multiplicative evaluation domains and on the fields that contain them.

9 Linear algebra over a field

Linear algebra is the study of *vector spaces* — collections of objects that can be added together and scaled by elements of a field — together with the structure-preserving maps between them. It is the computational backbone of the cryptography this monograph develops: commitments to vectors, the linear constraints that encode arithmetic circuits, the polynomial-evaluation arguments at the heart of the inner-product argument, and the Pedersen-style multiscalar multiplications that Halo 2 and Orchard use are all, at bottom, statements about linear maps and (bi)linear forms over a finite field. This section develops the theory from the ground up, assuming only that the reader knows what a field is (a set $\mathbb{F}$ with addition and multiplication in which every nonzero element is invertible; see the preceding section). The development then specialises to the two settings that matter here: the coordinate space $\mathbb{F}^n$ over a finite field $\mathbb{F} = \mathbb{F}_q$, and the “mixed” pairing between scalar vectors and vectors of group elements.

Throughout, $\mathbb{F}$ denotes an arbitrary field. We write its additive identity 0, its multiplicative identity 1, and call elements of $\mathbb{F}$ *scalars*. We use the field axioms freely: addition and multiplication are commutative and associative, multiplication distributes over addition, and every $a \neq 0$ has an inverse a^{-1} .

9.1 Vector spaces and subspaces

Definition 9.1 (Vector space). Let $\mathbb{F}$ be a field. A *vector space over $\mathbb{F}$* (or an $\mathbb{F}$ -*vector space*) is a set V together with two operations,

$$+ : V \times V \rightarrow V \quad (\text{vector addition}), \quad \cdot : \mathbb{F} \times V \rightarrow V \quad (\text{scalar multiplication}),$$

satisfying the following axioms for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$. We call the elements of V *vectors*.

1. (*Additive abelian group.*) V is an abelian group under $+$: addition is associative and commutative, there is a zero vector $\mathbf{0} \in V$ with $v + \mathbf{0} = v$, and every v has an additive inverse $-v$ with $v + (-v) = \mathbf{0}$.
2. (*Compatibility.*) $a \cdot (b \cdot v) = (ab) \cdot v$.
3. (*Identity.*) $1 \cdot v = v$.
4. (*Distributivity over vector addition.*) $a \cdot (u + v) = a \cdot u + a \cdot v$.
5. (*Distributivity over scalar addition.*) $(a + b) \cdot v = a \cdot v + b \cdot v$.

We usually write the scalar product $a \cdot v$ as av , and we drop the distinction between the scalar $0 \in \mathbb{F}$ and the vector $\mathbf{0} \in V$ when context makes it clear.

A few immediate consequences of the axioms recur so frequently that we record them once and for all.

Proposition 9.2. *Let V be an $\mathbb{F}$ -vector space, $v \in V$, and $a \in \mathbb{F}$. Then:*

1. $0 \cdot v = \mathbf{0}$;
2. $a \cdot \mathbf{0} = \mathbf{0}$;
3. $(-1) \cdot v = -v$;
4. *if $av = \mathbf{0}$ then $a = 0$ or $v = \mathbf{0}$.*

Proof. (1) By distributivity over scalar addition, $0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v$. Adding $-(0 \cdot v)$ to both sides gives $\mathbf{0} = 0 \cdot v$.

(2) Similarly $a \cdot \mathbf{0} = a \cdot (\mathbf{0} + \mathbf{0}) = a \cdot \mathbf{0} + a \cdot \mathbf{0}$, and cancellation gives $a \cdot \mathbf{0} = \mathbf{0}$.

(3) Using (1) and distributivity, $v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 + (-1)) \cdot v = 0 \cdot v = \mathbf{0}$, so $(-1) \cdot v$ is the additive inverse of v , i.e. $(-1) \cdot v = -v$.

(4) Suppose $av = \mathbf{0}$ and $a \neq 0$. Since $\mathbb{F}$ is a field, a^{-1} exists, and by the compatibility and identity axioms $v = 1 \cdot v = (a^{-1}a) \cdot v = a^{-1} \cdot (av) = a^{-1} \cdot \mathbf{0} = \mathbf{0}$ using (2). $\square$

Example 9.3 (The coordinate space $\mathbb{F}^n$). The single most important example is the *coordinate space*

$$\mathbb{F}^n = \{ (x_1, \dots, x_n) : x_i \in \mathbb{F} \},$$

the set of n -tuples of scalars, with componentwise operations

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \quad a \cdot (x_1, \dots, x_n) = (ax_1, \dots, ax_n).$$

The zero vector is $(0, \dots, 0)$. All the axioms follow from the field axioms applied coordinatewise. When $n = 0$ the result is the *trivial* (or *zero*) vector space $\{\mathbf{0}\}$, consisting of the empty tuple alone. We typically write vectors of $\mathbb{F}^n$ as columns and in boldface, $\mathbf{x} = (x_1, \dots, x_n)^T$.

Example 9.4 (Further examples). 1. *Polynomials*. The set $\mathbb{F}[X]$ of all polynomials in one variable with coefficients in $\mathbb{F}$ is an $\mathbb{F}$ -vector space under ordinary addition and scalar multiplication. So is the subset $\mathbb{F}[X]_{<n}$ of polynomials of degree less than n ; this is exactly the setting in which polynomial commitment schemes operate.

2. *Functions*. For any set S , the set $\mathbb{F}^S$ of all functions $S \rightarrow \mathbb{F}$ is an $\mathbb{F}$ -vector space under pointwise operations. Example 9.3 is the special case $S = \{1, \dots, n\}$.

3. *Matrices*. The set $\mathbb{F}^{m \times n}$ of $m \times n$ matrices with entries in $\mathbb{F}$ is an $\mathbb{F}$ -vector space under entrywise addition and scaling.

4. *Field extensions*. If $\mathbb{F} \subseteq \mathbb{K}$ is a subfield of a larger field $\mathbb{K}$, then $\mathbb{K}$ is an $\mathbb{F}$ -vector space. For instance $\mathbb{C}$ is a 2-dimensional $\mathbb{R}$ -vector space, and the finite field $\mathbb{F}_q$ with $q = p^k$ is a k -dimensional $\mathbb{F}_p$ -vector space — a fact we use below when reasoning about embedding-degree and field arithmetic.

A vector space often contains smaller vector spaces sitting inside it. These are the subspaces.

Definition 9.5 (Subspace). A subset $W \subseteq V$ of an $\mathbb{F}$ -vector space V is a (*linear*) *subspace* if it is itself a vector space under the operations inherited from V . Equivalently, W is a subspace iff it is nonempty and closed under the vector-space operations.

Proposition 9.6 (Subspace criterion). *A subset $W \subseteq V$ is a subspace if and only if all three of the following hold:*

1. $\mathbf{0} \in W$;
2. W is closed under addition: $u, v \in W \implies u + v \in W$;
3. W is closed under scalar multiplication: $a \in \mathbb{F}, v \in W \implies av \in W$.

Equivalently, $W \neq \emptyset$ and for all $a, b \in \mathbb{F}$ and $u, v \in W$ we have $au + bv \in W$.

Proof. If W is a subspace, the three conditions hold because they are part of (or follow from) the vector-space axioms restricted to W . Conversely, suppose (1)–(3) hold. Closure under $+$ and $\cdot$ makes the operations well defined on W . The additive inverse of $v \in W$ is $(-1)v \in W$ by (3) and Proposition 9.2(3). The space V supplies the remaining axioms (associativity, commutativity, distributivity, etc.) to W , because they are universally quantified identities that hold for all elements of V , in particular for elements of W . The “equivalently” clause follows by taking $(a, b) = (1, 1)$ and $(a, b) = (a, 0)$ to recover (2) and (3), and noting $\mathbf{0} = 0 \cdot v$ for any v in the nonempty W . $\square$

Example 9.7. In $\mathbb{F}^3$, the set $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$ is a subspace (the solution set of a homogeneous linear equation), whereas $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1\}$ is not, since it does not contain $\mathbf{0}$. More generally, the solution set of any homogeneous linear system $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{F}^n$; we revisit this in the study of kernels.

Proposition 9.8 (Intersections and sums). *Let $\{W_i\}_{i \in I}$ be a family of subspaces of V .*

1. *The intersection $\bigcap_{i \in I} W_i$ is a subspace.*
2. *The sum $W_1 + \cdots + W_k = \{w_1 + \cdots + w_k : w_j \in W_j\}$ of finitely many subspaces is a subspace, and it is the smallest subspace containing all the W_j .*

In general the union of two subspaces is not a subspace.

Proof. (1) Each W_i contains $\mathbf{0}$, so the intersection does too. If u, v lie in every W_i and $a, b \in \mathbb{F}$, then $au + bv \in W_i$ for every i by Proposition 9.6, hence in the intersection.

(2) Closure: given $\sum_j w_j$ and $\sum_j w'_j$ in the sum and scalars a, b , we have $a \sum_j w_j + b \sum_j w'_j = \sum_j (aw_j + bw'_j)$ with $aw_j + bw'_j \in W_j$. It contains $\mathbf{0} = \sum_j \mathbf{0}$ and each W_j (take all other summands zero). Any subspace containing every W_j must, by closure under addition, contain all such sums; hence the sum is the smallest such subspace. An example shows the union claim: in $\mathbb{F}^2$ with $|\mathbb{F}| > 1$, the two axes $\{(x, 0)\}$ and $\{(0, y)\}$ are subspaces whose union omits $(1, 1) = (1, 0) + (0, 1)$. $\square$

9.2 Linear combinations, span, and linear independence

Definition 9.9 (Linear combination and span). Let $S \subseteq V$. A *linear combination* of vectors in S is any finite sum

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k, \quad a_i \in \mathbb{F}, v_i \in S.$$

The *span* of S , written $\text{span}(S)$ or $\langle S \rangle$, is the set of all linear combinations of finitely many vectors in S . By convention, $\text{span}(\emptyset) = \{\mathbf{0}\}$. S *spans* (or *generates*) V if $\text{span}(S) = V$, and V is *finitely generated* (or *finite-dimensional*) if some finite set spans it.

Proposition 9.10. *For any $S \subseteq V$, $\text{span}(S)$ is the smallest subspace of V containing S ; that is, it is a subspace, it contains S , and it is contained in every subspace that contains S .*

Proof. A sum of two linear combinations of vectors in S is again such a linear combination, as is a scalar multiple of one, and $\mathbf{0}$ is the empty combination; hence $\text{span}(S)$ is a subspace by Proposition 9.6, and $S \subseteq \text{span}(S)$. If W is any subspace containing S , then W contains every linear combination of elements of S (by closure under addition and scaling), so $\text{span}(S) \subseteq W$. $\square$

Definition 9.11 (Linear independence). A finite list of vectors $v_1, \dots, v_k \in V$ is *linearly independent* if the only expression of the zero vector as a linear combination of them is the trivial one:

$$a_1 v_1 + \cdots + a_k v_k = \mathbf{0} \implies a_1 = \cdots = a_k = 0.$$

Otherwise the list is *linearly dependent*; we call a nontrivial relation $\sum_i a_i v_i = \mathbf{0}$ with some $a_i \neq 0$ a *dependence relation*. An arbitrary (possibly infinite) set S is linearly independent if every finite sublist of distinct vectors from S is independent.

Remark 9.12. Linear independence captures non-redundancy. A single vector v is independent iff $v \neq \mathbf{0}$. Two vectors are dependent iff one is a scalar multiple of the other. In general, a list is dependent iff some vector in it lies in the span of the others, as the next lemma makes precise.

Lemma 9.13 (Dependence and redundancy). *A list $v_1, \dots, v_k$ with $k \geq 1$ is linearly dependent if and only if some v_j is a linear combination of the others. Moreover, if $v_1, \dots, v_k$ are independent but $v_1, \dots, v_k, w$ are dependent, then $w \in \text{span}(v_1, \dots, v_k)$, and the coefficients expressing w are unique.*

Proof. If $v_j = \sum_{i \neq j} a_i v_i$, then $\sum_{i \neq j} a_i v_i + (-1)v_j = \mathbf{0}$ is a nontrivial relation. Conversely, given a dependence relation $\sum_i a_i v_i = \mathbf{0}$ with $a_j \neq 0$, one solves $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$.

For the second statement, a dependence relation $\sum_i a_i v_i + bw = \mathbf{0}$ must have $b \neq 0$ (otherwise it would be a nontrivial relation among the independent v_i), so $w = -b^{-1} \sum_i a_i v_i \in \text{span}(v_1, \dots, v_k)$. Uniqueness: if $w = \sum_i c_i v_i = \sum_i c'_i v_i$, then $\sum_i (c_i - c'_i) v_i = \mathbf{0}$, forcing $c_i = c'_i$ by independence. $\square$

9.3 Bilinear forms and inner products on $\mathbb{F}^n$

The maps considered so far are linear in a single argument. Commitment and proof systems are pervaded by expressions that are linear in *each* of two arguments — most importantly the inner product. We treat these next. Because the fields of interest are finite, there is no notion of “positive length” or “angle”; the appropriate level of generality is the *bilinear form*.

Definition 9.14 (Bilinear form). A *bilinear form* on an $\mathbb{F}$ -vector space V is a function $\beta: V \times V \rightarrow \mathbb{F}$ that is linear in each argument separately:

$$\beta(au + a'u', v) = a\beta(u, v) + a'\beta(u', v), \quad \beta(u, bv + b'v') = b\beta(u, v) + b'\beta(u, v').$$

It is *symmetric* if $\beta(u, v) = \beta(v, u)$ for all u, v , and *nondegenerate* if $\beta(u, v) = 0$ for all v implies $u = \mathbf{0}$.

Proposition 9.15 (Bilinear forms are matrices). Fix a basis $b_1, \dots, b_n$ of V . A unique matrix $M \in \mathbb{F}^{n \times n}$ with $M_{ij} = \beta(b_i, b_j)$ represents every bilinear form β , via

$$\beta(u, v) = [u]_B^T M [v]_B = \sum_{i,j} M_{ij} u_i v_j,$$

where $[u]_B = (u_i)$, $[v]_B = (v_j)$. The form is symmetric iff $M = M^T$, and nondegenerate iff M is invertible.

Proof. Expanding $u = \sum_i u_i b_i$, $v = \sum_j v_j b_j$ and using bilinearity, $\beta(u, v) = \sum_{i,j} u_i v_j \beta(b_i, b_j) = \sum_{i,j} u_i M_{ij} v_j = [u]_B^T M [v]_B$. Uniqueness and the matrix recover M_{ij} by evaluating on basis vectors. Symmetry of β is equivalent to $M_{ij} = M_{ji}$. For nondegeneracy: $\beta(u, \cdot) \equiv 0$ means $[u]^T M = \mathbf{0}^T$, i.e. $M^T [u] = \mathbf{0}$; this has a nonzero solution $[u]$ iff M^T (equivalently M) is singular, so nondegeneracy is exactly invertibility of M . $\square$

Definition 9.16 (Standard inner product on $\mathbb{F}^n$). The *standard inner product* (or *dot product*) on $\mathbb{F}^n$ is the symmetric bilinear form

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i = \mathbf{a}^T \mathbf{b}, \quad \mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n).$$

Its matrix in the standard basis is the identity I_n ; it is symmetric and nondegenerate.

Remark 9.17 (No positivity over finite fields). Over $\mathbb{R}$ the standard inner product is *positive definite*: $\langle \mathbf{a}, \mathbf{a} \rangle = \sum a_i^2 \geq 0$, vanishing only at $\mathbf{0}$, which yields the Euclidean norm $\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$ and the Cauchy–Schwarz inequality. Over a finite field $\mathbb{F}_q$ there is no ordering, so “positivity” is meaningless: one may have $\mathbf{a} \neq \mathbf{0}$ with $\langle \mathbf{a}, \mathbf{a} \rangle = 0$ (an *isotropic* vector). For example over $\mathbb{F}_5$, $\mathbf{a} = (1, 2)$ gives $1 + 4 = 5 \equiv 0$. The algebraically robust properties that survive are bilinearity, symmetry, and nondegeneracy; these are all that the arguments in proof systems rely on.

Remark 9.18 (Functionals via the inner product). Nondegeneracy yields a clean dictionary between vectors and linear functionals. Each $\mathbf{a} \in \mathbb{F}^n$ gives a functional $\mathbf{b} \mapsto \langle \mathbf{a}, \mathbf{b} \rangle$, and the map $\mathbf{a} \mapsto \langle \mathbf{a}, \cdot \rangle$ is an isomorphism from $\mathbb{F}^n$ onto its *dual space* $\text{Hom}(\mathbb{F}^n, \mathbb{F})$ (both have dimension n , and the map is injective by nondegeneracy). Thus “a linear functional on $\mathbb{F}^n$ ” and “a vector in $\mathbb{F}^n$ ” are interchangeable. Evaluating a committed polynomial p of coefficient vector $\mathbf{c}$ at a point z is the inner product $\langle \mathbf{c}, (1, z, z^2, \dots, z^{n-1}) \rangle$; this is precisely the shape an inner-product argument proves.

9.4 The mixed inner product with group elements

The commitments used in Halo 2 and Orchard pair a vector of *scalars* with a vector of *group elements*. We isolate the algebraic structure here; the section on elliptic curves introduces the relevant group fully, but only its abelian-group structure matters at present.

Let $(\mathbb{G}, +)$ be a finite abelian group, written additively, of prime order r with identity $\mathcal{O}$. For an integer a and $G \in \mathbb{G}$ write $[a]G$ for the a -fold sum $G + \dots + G$ (and $[0]G = \mathcal{O}$, $[-a]G = -([a]G)$; this is the *scalar multiplication* of the group). Because $\mathbb{G}$ has order r with r prime, $[a]G$ depends only on $a \bmod r$, so the operation descends to scalars from the field $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$: for $a \in \mathbb{F}_r$ and $G \in \mathbb{G}$, $[a]G$ is well defined. With this action $\mathbb{G}$ becomes a one-dimensional vector space over $\mathbb{F}_r$ whenever $\mathbb{G}$ is nontrivial — a viewpoint we exploit repeatedly.

Proposition 9.19 ($\mathbb{G}$ as an $\mathbb{F}_r$ -vector space). *Let $\mathbb{G}$ be a group of prime order r . Then the scalar multiplication $(a, G) \mapsto [a]G$ makes $\mathbb{G}$ into a vector space over $\mathbb{F}_r$ of dimension 1; any nonidentity element G is a basis, and every element is uniquely $[a]G$ for some $a \in \mathbb{F}_r$.*

Proof. The vector-space axioms are the standard laws of group scalar multiplication: $[a]([b]G) = [ab]G$, $[1]G = G$, $[a](G + H) = [a]G + [a]H$, and $[a + b]G = [a]G + [b]G$, all of which hold in any abelian group and respect reduction mod r . For a nonidentity G , Lagrange’s theorem in a group of prime order r forces G to have order r , so the r multiples $[0]G, [1]G, \dots, [r - 1]G$ are distinct and exhaust $\mathbb{G}$ (which has r elements). Hence $\{G\}$ spans and is independent ($[a]G = \mathcal{O}$ iff $a \equiv 0$), a basis; uniqueness of a is unique representation in this basis. $\square$

Definition 9.20 (Mixed inner product / multiscalar multiplication). Let $\mathbf{G} = (G_1, \dots, G_n) \in \mathbb{G}^n$ be a fixed tuple of group elements and $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{F}_r^n$ a vector of scalars. Their *mixed inner product* (also called a *multiscalar multiplication*, MSM, or *Pedersen-style inner product*) is the group element

$$\langle \mathbf{a}, \mathbf{G} \rangle = \sum_{i=1}^n [a_i]G_i \in \mathbb{G}.$$

We use the same angle-bracket notation as for the scalar inner product; whether the second argument is a scalar vector or a vector of group elements determines which one we mean.

Proposition 9.21 (Bilinearity of the mixed product). *Fix $\mathbf{G} \in \mathbb{G}^n$. The map $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is $\mathbb{F}_r$ -linear from $\mathbb{F}_r^n$ to $\mathbb{G}$:*

$$\langle c\mathbf{a} + c'\mathbf{a}', \mathbf{G} \rangle = [c]\langle \mathbf{a}, \mathbf{G} \rangle + [c']\langle \mathbf{a}', \mathbf{G} \rangle.$$

Symmetrically, for fixed $\mathbf{a}$ the map $\mathbf{G} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is a group homomorphism $\mathbb{G}^n \rightarrow \mathbb{G}$ that is $\mathbb{F}_r$ -linear when one views $\mathbb{G}^n$ as an $\mathbb{F}_r$ -vector space. Thus $\langle \cdot, \cdot \rangle: \mathbb{F}_r^n \times \mathbb{G}^n \rightarrow \mathbb{G}$ is bilinear over $\mathbb{F}_r$.

Proof. Using the vector-space laws on $\mathbb{G}$ (Proposition 9.19) componentwise,

$$\langle c\mathbf{a} + c'\mathbf{a}', \mathbf{G} \rangle = \sum_i [ca_i + c'a'_i]G_i = \sum_i ([c][a_i]G_i + [c'][a'_i]G_i) = [c] \sum_i [a_i]G_i + [c'] \sum_i [a'_i]G_i,$$

which is the first claim. Linearity in $\mathbf{G}$ is identical by symmetry of the defining sum and the law $[a](G + H) = [a]G + [a]H$. $\square$

Remark 9.22 (Why this is the right abstraction). The mixed inner product is the computational heart of Pedersen commitments and of the Halo 2 inner-product argument. Reading $\langle \mathbf{a}, \mathbf{G} \rangle$ as a *commitment* to the scalar vector $\mathbf{a}$ under the public “basis” $\mathbf{G}$, bilinearity is exactly what endows the commitment with its *homomorphic* property: a commitment to $c\mathbf{a} + c'\mathbf{a}'$ is the same linear combination $[c]$ - and $[c']$ -scaled of the individual commitments. Provided one chooses the G_i so that the prover knows no nontrivial relation $\sum_i [a_i]G_i = \mathcal{O}$ (the discrete-logarithm hardness assumption, treated later), the commitment is binding: the map $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is computationally injective. This map *cannot* be injective as a set map once $n \geq 2$, since by rank-nullity its kernel

(a subspace of $\mathbb{F}_r^n$ mapping into the one-dimensional $\mathbb{G}$) has dimension at least $n - 1$; binding is therefore a *computational*, not information-theoretic, statement. A typical Pedersen commitment to a single value a with randomness ρ is the special case $[a]G + [\rho]H = \langle (a, \rho), (G, H) \rangle$, which is moreover *perfectly hiding* because the $[\rho]H$ term is uniform in $\mathbb{G}$.

9.5 How this machinery appears in commitment and proof systems

We now draw the threads together; each point below is developed rigorously in later sections, but the linear-algebraic content is already in hand.

Remark 9.23 (Witnesses, constraints, and assignments). An arithmetic circuit over $\mathbb{F}_q$ is, after *arithmetisation*, a system of polynomial constraints in the wire values. Matrices over $\mathbb{F}_q$ acting on the assignment vector $\mathbf{z} \in \mathbb{F}_q^N$ capture the *linear* part of such a system — the wiring/copy constraints of PLONK, or the linear combinations $A\mathbf{z}, B\mathbf{z}, C\mathbf{z}$ of a rank-1 constraint system. Satisfying the linear constraints means $\mathbf{z}$ lies in an affine subspace (an affine subspace); the remaining (multiplicative) constraints are handled separately. Rank-nullity quantifies the residual degrees of freedom, i.e. the dimension of the witness space.

Remark 9.24 (Commitments as linear maps). A Pedersen vector commitment $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle = \sum_i [a_i]G_i$ is a linear map $\mathbb{F}_r^n \rightarrow \mathbb{G}$ (Proposition 9.21). Its linearity is the source of every algebraic manipulation a verifier performs: it can take known linear combinations of commitments, “fold” two commitment keys $\mathbf{G}, \mathbf{G}'$ into $\mathbf{G} + [x]\mathbf{G}'$ under a challenge x , and check claimed openings by re-evaluating the same linear map. A polynomial commitment is the same map applied to a coefficient vector, and an *evaluation* of that polynomial at z is the scalar inner product $\langle \mathbf{c}, (1, z, \dots, z^{n-1}) \rangle$ (Remark 9.18).

Remark 9.25 (The inner-product relation). The core statement an inner-product argument proves has the shape: “I know vectors $\mathbf{a}, \mathbf{b} \in \mathbb{F}_r^n$ such that $P = \langle \mathbf{a}, \mathbf{G} \rangle + \langle \mathbf{b}, \mathbf{H} \rangle + [\langle \mathbf{a}, \mathbf{b} \rangle]U$ for public $\mathbf{G}, \mathbf{H} \in \mathbb{G}^n$ and $U \in \mathbb{G}$.” Here three of the constructions meet at once: two mixed inner products $\langle \mathbf{a}, \mathbf{G} \rangle, \langle \mathbf{b}, \mathbf{H} \rangle$ (commitments to the witness vectors), one scalar inner product $\langle \mathbf{a}, \mathbf{b} \rangle$ (the value being argued), and the bilinearity of all three (which permits the prover and verifier to fold length- n instances into length- $n/2$ ones, using the bilinearity of these inner products). The argument’s soundness ultimately reduces to the nondegeneracy of these forms together with the hardness of finding kernel elements of the commitment map — that is, to exactly the linear-algebraic and computational facts assembled in this section.

Remark 9.26 (Summary). The objects constructed above — vector spaces and their dimensions, linear maps and their matrices, kernels and images governed by rank-nullity, invertibility, bilinear forms and the two flavours of inner product (scalar-with-scalar and scalar-with-group), and the tensor product that linearises bilinearity — constitute the complete linear-algebraic vocabulary needed for the cryptography ahead. Every commitment is a linear map; every opening is an inner product; every soundness argument rests on (non)degeneracy and dimension counting. With this foundation in place, we can develop the number theory of finite fields and the geometry of elliptic curves on which the group $\mathbb{G}$ is realised.

10 Elliptic curves

Having developed the theory of finite fields, we now turn attention to the geometric objects that supply the cyclic groups underlying the cryptography of Halo 2 and Zcash Orchard: *elliptic curves*. An elliptic curve is, concretely, the solution set of a certain cubic equation in two variables, together with one extra “point at infinity”. The notable feature of elliptic curves is that this solution set carries the structure of an abelian group, defined by a strikingly geometric rule, and that this group is hard to compute in: solving the discrete logarithm problem in a well-chosen elliptic curve group appears to require exponential time. This combination of a clean algebraic structure with a hard computational problem is exactly what cryptography requires.

Throughout this section we work over a field K whose characteristic is neither 2 nor 3. In the applications K will be a finite prime field $\mathbb{F}_p$ with p a large prime, but it is cleaner to develop the basic theory over a general field and specialize when needed. The reader should keep in mind the two *Pasta curves*

$$E_p : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \quad E_q : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_q,$$

where p and q are the two 255-bit primes

$$\begin{aligned} p &= 2^{254} + 0x224698fc094cf91b992d30ed00000001, \\ q &= 2^{254} + 0x224698fc0994a8dd8c46eb2100000001, \end{aligned}$$

introduced in the field-theory section. These are the curves *Pallas* (E_p , defined over $\mathbb{F}_p$) and *Vesta* (E_q , defined over $\mathbb{F}_q$); together they form a *cycle*, in that the number of $\mathbb{F}_p$ -points of Pallas equals q and the number of $\mathbb{F}_q$ -points of Vesta equals p . The equation $y^2 = x^3 + 5$ serves as the running example, with a smaller toy curve developed in parallel for hand computation.

10.1 Weierstrass equations

Definition 10.1 (short Weierstrass equation). Let K be a field of characteristic $\neq 2, 3$, and let $a, b \in K$. The *short Weierstrass equation* with coefficients a, b is the equation

$$y^2 = x^3 + ax + b. \tag{11}$$

The elements a and b are the *Weierstrass coefficients*.

Over a field of characteristic $\neq 2, 3$, an invertible affine change of variables always reduces the most general cubic curve $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ to the form (11): completing the square in y (possible since $2 \neq 0$) removes the xy and y terms, and then a shift $x \mapsto x - a_2/3$ (possible since $3 \neq 0$) removes the x^2 term. This is precisely why we exclude characteristics 2 and 3; those cases require the longer Weierstrass form and are not relevant to the Pasta curves, which live over large prime fields.

Not every choice of a, b gives a usable curve. We must exclude the cubics whose graph has a self-intersection or a cusp, because the group law (defined below) breaks down at such a singular point.

Definition 10.2 (discriminant and nonsingularity). The *discriminant* of the short Weierstrass equation (11) is

$$\Delta = -16(4a^3 + 27b^2) \in K. \tag{12}$$

The equation, and the cubic $f(x) = x^3 + ax + b$, is *nonsingular* (or *smooth*) if $\Delta \neq 0$, equivalently $4a^3 + 27b^2 \neq 0$.

Proposition 10.3. *Let $f(x) = x^3 + ax + b \in K[x]$. The following are equivalent.*

1. $4a^3 + 27b^2 \neq 0$.

2. f has no repeated root in the algebraic closure $\overline{K}$.
3. f and its derivative f' are coprime in $\overline{K}[x]$.
4. There is no point (x_0, y_0) in $\overline{K}^2$ satisfying both $y^2 = f(x)$ and the two partial-derivative equations $2y = 0$ and $f'(x) = 0$.

Proof. (2) $\Leftrightarrow$ (3): a polynomial has a repeated root iff it shares a root with its derivative, i.e. iff $\gcd(f, f') \neq 1$ in $\overline{K}[x]$. (1) $\Leftrightarrow$ (2): the discriminant of the cubic $x^3 + ax + b$ equals, up to the nonzero factor -16 , $4a^3 + 27b^2$; a univariate polynomial has a repeated root iff its discriminant vanishes. One may verify the identity directly: if $f(x) = (x - r_1)(x - r_2)(x - r_3)$ then $\prod_{i < j} (r_i - r_j)^2 = -(4a^3 + 27b^2)$, using $r_1 + r_2 + r_3 = 0$ (the coefficient of x^2 is zero). (4) $\Leftrightarrow$ (2): a singular point of the affine curve $y^2 = f(x)$ is a point where both partial derivatives of $g(x, y) = y^2 - f(x)$ vanish, namely $\partial g / \partial y = 2y = 0$ and $\partial g / \partial x = -f'(x) = 0$. Since $\text{char}(K) \neq 2$, the first forces $y_0 = 0$, and then the curve equation gives $f(x_0) = 0$; combined with $f'(x_0) = 0$ this says x_0 is a repeated root of f . Conversely a repeated root x_0 yields the singular point $(x_0, 0)$. $\square$

Definition 10.4 (elliptic curve, affine points). An *elliptic curve* over K is a short Weierstrass equation (11) with nonzero discriminant, $4a^3 + 27b^2 \neq 0$. For any field extension $L \supseteq K$ the set of L -rational affine points is

$$E^{\text{aff}}(L) = \{ (x, y) \in L \times L : y^2 = x^3 + ax + b \},$$

and the set of L -rational points is

$$E(L) = E^{\text{aff}}(L) \cup \{\mathcal{O}\},$$

where $\mathcal{O}$ is a single formal symbol called the *point at infinity*. When $L = K$ one writes simply E for the curve and $E(K)$ for its rational points.

Remark 10.5. For the Pasta curves $a = 0$, $b = 5$, so $4a^3 + 27b^2 = 27 \cdot 25 = 675$. As long as the characteristic does not divide $675 = 3^3 \cdot 5^2$, i.e. the prime is not 3 or 5, the curve is nonsingular. The Pasta primes p, q are enormous, so this is automatic. More generally any curve $y^2 = x^3 + b$ with $b \neq 0$ is nonsingular away from characteristics 2, 3.

10.2 The point at infinity, geometrically

To motivate the point at infinity, view the affine plane as sitting inside the *projective plane*. A point of the projective plane $\mathbb{P}^2(K)$ is an equivalence class $[X : Y : Z]$ of nonzero triples $(X, Y, Z) \in K^3 \setminus \{0\}$, where $(X, Y, Z) \sim (\lambda X, \lambda Y, \lambda Z)$ for every $\lambda \in K^\times$. The affine points (x, y) embed as $[x : y : 1]$; the points with $Z = 0$, the *line at infinity*, are the extra “directions”. Homogenizing (11) by setting $x = X/Z$, $y = Y/Z$ and clearing denominators gives the projective Weierstrass equation

$$Y^2 Z = X^3 + aXZ^2 + bZ^3. \tag{13}$$

Setting $Z = 0$ forces $X^3 = 0$, hence $X = 0$, leaving the single projective point $[0 : 1 : 0]$. This is the point at infinity $\mathcal{O}$: it lies “infinitely far up” in the vertical direction, and every vertical line $x = \text{const}$ passes through it. As we shall see, $\mathcal{O}$ is the identity element of the group law. The reader may safely picture $\mathcal{O}$ as a single point at the top (and bottom) of the y -axis where all vertical lines meet.

10.3 The chord-and-tangent group law: geometry

We now describe the procedure for “adding” two points of $E(K)$. The construction is purely geometric and rests on a single fact: a line meets a cubic curve in exactly three points, counted with multiplicity. The point at infinity is engineered precisely so that this counting is always exact.

Definition 10.6 (the group law, geometric form). Let $P, Q \in E(K)$. Define $P + Q$ as follows.

- If $P = \mathcal{O}$, set $P + Q = Q$; if $Q = \mathcal{O}$, set $P + Q = P$.
- Otherwise $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are affine. Draw the line ℓ through P and Q ; if $P = Q$, let ℓ be the tangent line to the curve at P . The line ℓ meets the curve in a third point R' (counted with multiplicity), where if ℓ is vertical the third point is $\mathcal{O}$. Define $P + Q$ to be the *reflection of R' across the x -axis*: if $R' = (x_3, -y_3)$ then $P + Q = (x_3, y_3)$, and if $R' = \mathcal{O}$ then $P + Q = \mathcal{O}$.

The reflection step is what converts “the three intersection points sum to a fixed thing” into a genuine group law with $\mathcal{O}$ as identity. Concretely, the rule is: the three intersection points of any line with the curve add up to $\mathcal{O}$. The reflection across the x -axis sends an affine point (x, y) to $(x, -y)$, which (as we shall see) is its additive inverse; and it sends $\mathcal{O}$ to $\mathcal{O}$.

To convert this geometry into formulas one must (i) find the third intersection point of a line with the cubic and (ii) handle the degenerate cases. The key algebraic device is Vieta’s formula relating the roots of a cubic to its coefficients.

10.4 Explicit affine formulas

Fix an elliptic curve $E : y^2 = x^3 + ax + b$ over K . Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be affine points.

Negation. Define

$$-\mathcal{O} = \mathcal{O}, \quad -(x_1, y_1) = (x_1, -y_1).$$

Note $(x_1, -y_1)$ is again on the curve since the equation depends on y only through y^2 . The line through P and $-P$ is the vertical line $x = x_1$, which meets the curve at P , $-P$, and $\mathcal{O}$.

Addition (the generic chord). Suppose $P, Q \neq \mathcal{O}$ and $x_1 \neq x_2$. The line through P and Q is $y = \lambda x + \mu$ with

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \quad \mu = y_1 - \lambda x_1.$$

Substituting into the curve equation, $(\lambda x + \mu)^2 = x^3 + ax + b$, gives

$$x^3 - \lambda^2 x^2 + (a - 2\lambda\mu)x + (b - \mu^2) = 0.$$

This cubic in x has the three roots x_1, x_2, x_3 , where x_3 is the x -coordinate of the third intersection point R' . By Vieta’s formula the sum of the roots equals the negative of the x^2 -coefficient, so $x_1 + x_2 + x_3 = \lambda^2$. Hence

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1, \tag{14}$$

where the y -formula already incorporates the reflection across the x -axis: the third intersection point is $(x_3, \lambda x_3 + \mu)$, and reflecting gives $y_3 = -(\lambda x_3 + \mu) = -(\lambda x_3 + y_1 - \lambda x_1) = \lambda(x_1 - x_3) - y_1$. Then $P + Q = (x_3, y_3)$.

Doubling (the tangent). Suppose $P = Q = (x_1, y_1)$ with $y_1 \neq 0$. Implicit differentiation of $y^2 = x^3 + ax + b$ yields the slope of the tangent line: $2y dy = (3x^2 + a) dx$, so

$$\lambda = \frac{3x_1^2 + a}{2y_1}.$$

The same Vieta computation now uses a double root at x_1 , giving $x_1 + x_1 + x_3 = \lambda^2$, hence

$$x_3 = \lambda^2 - 2x_1, \quad y_3 = \lambda(x_1 - x_3) - y_1, \quad [2]P = (x_3, y_3). \quad (15)$$

Edge cases. The remaining cases are exactly where the line is vertical:

- If $P = \mathcal{O}$, then $P + Q = Q$; symmetrically if $Q = \mathcal{O}$.
- If $x_1 = x_2$ but $y_1 = -y_2$ (so $Q = -P$, including the subcase $y_1 = y_2 = 0$), the line is vertical, the third point is $\mathcal{O}$, and $P + Q = \mathcal{O}$.
- If $P = Q$ and $y_1 = 0$, the tangent is vertical (the derivative slope formula has zero denominator), so $[2]P = \mathcal{O}$. These are exactly the points of order two.

Every case is covered: either one input is $\mathcal{O}$; or both are affine with $x_1 \neq x_2$ (use (14)); or both are affine with $x_1 = x_2$, in which case either $y_1 = -y_2$ (result $\mathcal{O}$) or $y_1 = y_2 \neq 0$ (use (15)).

Proposition 10.7 (closure). *For all $P, Q \in E(K)$, the point $P + Q$ computed above lies in $E(K)$. In particular $E(K)$ is closed under $+$.*

Proof. If either input is $\mathcal{O}$ the result is the other input, which lies in $E(K)$ by hypothesis; if the result is $\mathcal{O}$ it lies in $E(K)$ by definition. Otherwise (x_3, y_3) follows from (14) or (15). By construction $(x_3, -y_3)$ (the third intersection point R') satisfies both the line equation and the curve equation, hence lies on the curve; since the curve equation is invariant under $y \mapsto -y$, the reflected point (x_3, y_3) is on the curve too, so $P + Q \in E^{\text{aff}}(K) \subseteq E(K)$. All coordinates are rational expressions in x_i, y_i, a with denominators $x_2 - x_1$ or $2y_1$ that are nonzero in the relevant cases, so $x_3, y_3 \in K$. $\square$

10.5 Identity, inverses, commutativity, and associativity

Proposition 10.8 (identity and inverses). *The operation $+$ on $E(K)$ has the following properties.*

1. $\mathcal{O}$ is a two-sided identity: $P + \mathcal{O} = \mathcal{O} + P = P$ for all P .
2. Every P has a two-sided inverse $-P$: $P + (-P) = \mathcal{O}$, where $-\mathcal{O} = \mathcal{O}$ and $-(x, y) = (x, -y)$.
3. The operation is commutative: $P + Q = Q + P$.

Proof. (1) follows immediately from the definition. (2): for $P = (x, y)$ with $y \neq 0$, the line through P and $-P = (x, -y)$ is vertical and meets the curve at $\mathcal{O}$ as the third point, so $P + (-P) = \mathcal{O}$; for $y = 0$, $P = -P$ and $[2]P = \mathcal{O}$ directly; and $\mathcal{O} + \mathcal{O} = \mathcal{O}$. (3): the chord through P and Q does not depend on their order, and neither does the tangent in the doubling case; algebraically, formulas (14) are symmetric under the swap $(x_1, y_1) \leftrightarrow (x_2, y_2)$ (the slope λ is unchanged, and $x_3 = \lambda^2 - x_1 - x_2$ is symmetric). $\square$

Theorem 10.9 (group law). *Let E be an elliptic curve over a field K . The set $E(K)$ with the operation $+$ of Definition 10.6 is an abelian group with identity $\mathcal{O}$.*

The only group axiom not yet established is *associativity*, $(P + Q) + R = P + (Q + R)$. This is the deep and surprising part of the theory: the definition gives no reason why reflecting third intersection points should be an associative operation, and a brute-force verification through the explicit formulas requires splitting into a large number of coordinate cases and checking lengthy rational-function identities. We sketch below the conceptual proof, which exhibits the reason associativity holds.

Proof of associativity (sketch). Sketch. The proof uses the geometry of cubic curves. The key tool is the following *Cayley–Bacharach*-type fact: if two cubic curves C_1, C_2 in $\mathbb{P}^2$ meet in exactly nine points, and a third cubic C passes through eight of those nine points, then C passes through the ninth as well. (This follows from a dimension count: cubics form a 9-dimensional projective space of coefficients, and passing through a point is one linear condition; eight general points impose independent conditions, pinning down the pencil through them.)

To prove $(P + Q) + R = P + (Q + R)$, one writes both sides via the chord-and-tangent construction and arranges eight common intersection points among three suitably chosen lines (a degenerate cubic, namely a product of three lines) and the curve E ; the Cayley–Bacharach principle then forces the two candidate sums to share their final coordinate, hence to be equal. Translating “reflect across the x -axis” into “the three points on a line sum to $\mathcal{O}$ ” renders the bookkeeping uniform. We obtain a complete, case-free treatment by identifying $E(K)$ with the degree-zero part $\text{Pic}^0(E)$ of the divisor class group of the curve, where the group law is manifestly associative because it is inherited from addition of divisors modulo principal divisors; the map $P \mapsto [P] - [\mathcal{O}]$ is a bijection $E(K) \rightarrow \text{Pic}^0(E)$ intertwining the two operations. Either route yields a genuine proof; both lie beyond the elementary scope here, so we take associativity as established. $\square$

Remark 10.10. For $n \in \mathbb{Z}$ and $P \in E(K)$ we write $[n]P$ for the n -fold sum $P + \cdots + P$ (with $[0]P = \mathcal{O}$ and $[-n]P = -([n]P)$). This makes $E(K)$ a module over $\mathbb{Z}$, i.e. its abelian-group structure written multiplicatively-by-integers. Note that $[n]P$ is the *scalar multiplication* of the next subsection, not a coordinate-wise product.

10.6 Projective and Jacobian coordinates

The affine formulas (14) and (15) each require a field *inversion* (dividing by $x_2 - x_1$ or $2y_1$). In a large prime field $\mathbb{F}_p$, an inversion costs roughly as much as dozens of multiplications (the implementation typically computes it via the extended Euclidean algorithm or as z^{p-2} by Fermat). When a long chain of additions arises, as in scalar multiplication, it is advantageous to defer all inversions to the very end by carrying points in a redundant coordinate system where the group law uses only additions, subtractions, and multiplications.

Definition 10.11 (projective coordinates). Any triple $(X : Y : Z)$ with $Z \neq 0$ and

$$x = X/Z, \quad y = Y/Z,$$

represents a point $P \in E(K)$ other than $\mathcal{O}$ with affine coordinates (x, y) in (*homogeneous*) *projective coordinates*, i.e. by the projective point $[X : Y : Z]$ on the curve (13). The point at infinity is $\mathcal{O} = [0 : 1 : 0]$.

Definition 10.12 (Jacobian coordinates). Any triple $(X : Y : Z)$ with $Z \neq 0$ and

$$x = X/Z^2, \quad y = Y/Z^3,$$

represents a point $P \neq \mathcal{O}$ with affine coordinates (x, y) in *Jacobian coordinates*, under the equivalence $(X, Y, Z) \sim (\lambda^2 X, \lambda^3 Y, \lambda Z)$ for $\lambda \in K^\times$. Substituting into (11) gives the *Jacobian curve equation*

$$Y^2 = X^3 + aXZ^4 + bZ^6.$$

The point at infinity is $[1 : 1 : 0]$.

In both systems we can rewrite the addition and doubling formulas with no divisions. For instance, in Jacobian coordinates the doubling of $(X_1 : Y_1 : Z_1)$ on $y^2 = x^3 + ax + b$ is

$$\begin{aligned} S &= 4X_1Y_1^2, & M &= 3X_1^2 + aZ_1^4, \\ X_3 &= M^2 - 2S, & Y_3 &= M(S - X_3) - 8Y_1^4, & Z_3 &= 2Y_1Z_1, \end{aligned}$$

which uses only field multiplications and additions. The single recovery of affine coordinates at the end of a computation, $(X, Y, Z) \mapsto (X/Z^2, Y/Z^3)$, costs one inversion (compute Z^{-1} once, then Z^{-2}, Z^{-3} by multiplication). The trade-off is that each group operation now uses more multiplications; however, for curves like $y^2 = x^3 + b$ with $a = 0$ the term aZ_1^4 vanishes, making Jacobian doubling especially cheap, which is one reason the Pasta curves take $a = 0$.

Remark 10.13. Projective coordinates are simplest to derive (one merely homogenizes), but Jacobian coordinates usually yield faster doubling and are the standard choice in cryptographic libraries, including the implementations used by Halo 2. Mixed strategies (e.g. adding an affine point to a Jacobian point) save further multiplications. We do not need the detailed operation counts here; the essential point is that *one can carry out a long sequence of group operations with a single final inversion*.

10.7 Scalar multiplication and double-and-add

The fundamental operation in elliptic-curve cryptography is *scalar multiplication*: given a point P and an integer $n \geq 0$, compute

$$[n]P = \underbrace{P + P + \cdots + P}_{n \text{ times}}.$$

Computing this by $n - 1$ successive additions is infeasible when n has hundreds of bits. Instead one uses the binary expansion of n , exactly as in fast modular exponentiation.

Definition 10.14 (double-and-add). Write $n = \sum_{i=0}^{k-1} n_i 2^i$ with $n_i \in \{0, 1\}$ and $n_{k-1} = 1$. The *left-to-right double-and-add* algorithm computes $[n]P$ as follows: set $R \leftarrow \mathcal{O}$; for $i = k - 1$ down to 0, set $R \leftarrow [2]R$, and if $n_i = 1$ set $R \leftarrow R + P$; return R .

Proposition 10.15. *Double-and-add returns $[n]P$ using at most $k - 1$ doublings and at most $k - 1$ additions, where $k = \lfloor \log_2 n \rfloor + 1$ is the bit length of n .*

Proof. Let R_j be the value of R after processing bits $n_{k-1}, \dots, n_j$. We claim $R_j = \lfloor n/2^j \rfloor P$, and prove this by downward induction on j . For the top bit, after $i = k - 1$ one has $R = [2]\mathcal{O} + n_{k-1}P = [n_{k-1}]P = [1]P = \lfloor n/2^{k-1} \rfloor P$. Assuming $R_{j+1} = \lfloor n/2^{j+1} \rfloor P$, the next iteration computes $[2]R_{j+1} + [n_j]P = [2\lfloor n/2^{j+1} \rfloor + n_j]P = \lfloor n/2^j \rfloor P$, since $2\lfloor n/2^{j+1} \rfloor + n_j = \lfloor n/2^j \rfloor$. At $j = 0$ one obtains $R_0 = [n]P$. The loop runs $k - 1$ times after the initial top bit is consumed, giving the stated counts (the doubling of $\mathcal{O}$ and addition for the top bit are free/trivial). $\square$

Thus scalar multiplication by an n -bit scalar costs $O(\log n)$ group operations, i.e. $O(\log^3 n)$ or better field operations. This efficiency in the “forward” direction, contrasted with the apparent hardness of the inverse problem (§10.10), is the foundation of the cryptography.

Remark 10.16. The naive algorithm performs an addition only on 1-bits, so its operation pattern leaks the bits of n through timing. Cryptographic implementations use constant-time variants (e.g. the Montgomery ladder, or always performing a dummy addition) so that the sequence of operations is independent of the secret scalar. We note this to indicate that the simple algorithm above is pedagogical, not directly deployable.

10.8 The group structure of $E(\mathbb{F}_p)$

Now specialize $K = \mathbb{F}_p$, the finite field with p elements (p prime, $p > 3$). Since $\mathbb{F}_p$ is finite, so is $E(\mathbb{F}_p)$: it has at most $2p + 1$ points (for each of the p values of x , at most two values of y , plus $\mathcal{O}$). Theorem 10.9 makes $E(\mathbb{F}_p)$ a finite abelian group. One of the central theorems of the subject governs its order.

Theorem 10.17 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_p$. Write*

$$\#E(\mathbb{F}_p) = p + 1 - t,$$

which defines the integer t , the trace of Frobenius. Then

$$|t| \leq 2\sqrt{p}.$$

Equivalently, $\#E(\mathbb{F}_p)$ lies in the Hasse interval $[p + 1 - 2\sqrt{p}, p + 1 + 2\sqrt{p}]$.

Proof (sketch). Sketch. Heuristically, as x ranges over $\mathbb{F}_p$, the value $x^3 + ax + b$ is a nonzero square about half the time (giving two points), a nonsquare about half the time (giving none), and zero occasionally (giving one). Using the Legendre symbol $\left(\frac{\cdot}{p}\right)$, the number of affine points is

$$\#E^{\text{aff}}(\mathbb{F}_p) = \sum_{x \in \mathbb{F}_p} \left(1 + \left(\frac{x^3 + ax + b}{p} \right) \right) = p + \sum_{x \in \mathbb{F}_p} \left(\frac{x^3 + ax + b}{p} \right),$$

so $t = -\sum_x \left(\frac{x^3 + ax + b}{p} \right)$. The sum is a character sum that exhibits square-root cancellation; the sharp bound $|t| \leq 2\sqrt{p}$ is Hasse's theorem. The conceptual proof realizes t as the trace of the p -power Frobenius endomorphism $\pi: (x, y) \mapsto (x^p, y^p)$ acting on the curve. One shows π satisfies the characteristic equation $\pi^2 - [t]\pi + [p] = 0$ in the endomorphism ring, and that the associated quadratic form (the degree) is positive definite; the discriminant condition $t^2 - 4p \leq 0$ for the (complex) roots then yields $|t| \leq 2\sqrt{p}$. The full argument requires the theory of isogenies and the Weil pairing, beyond the present scope. $\square$

Hasse's theorem states that the order of $E(\mathbb{F}_p)$ is *very close* to p : it cannot deviate from $p + 1$ by more than $2\sqrt{p}$, a relative error of order $p^{-1/2}$. For cryptography this is crucial: it establishes that the group is large (about p elements, so about 255 bits for the Pasta curves) without any brute-force point count, and point-counting algorithms (Schoof–Elkies–Atkin) compute the *exact* order in polynomial time.

Theorem 10.18 (structure of $E(\mathbb{F}_p)$). *There exist unique integers n_1, n_2 with $n_1 \mid n_2$ (and $n_1 \mid p - 1$) such that*

$$E(\mathbb{F}_p) \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$$

as abelian groups. In particular $E(\mathbb{F}_p)$ is either cyclic ($n_1 = 1$) or a product of two cyclic groups.

Proof (sketch). Sketch. Every finite abelian group is a product of cyclic groups of prime-power order (the structure theorem for finite abelian groups), and one can rewrite it in invariant-factor form $\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}$ with $n_1 \mid \cdots \mid n_r$. For elliptic curves one shows the rank r is at most 2: the m -torsion $E[m]$ over the algebraic closure is $(\mathbb{Z}/m\mathbb{Z})^2$ for m coprime to p , so at most two invariant factors can survive over $\mathbb{F}_p$. The divisibility $n_1 \mid p - 1$ comes from the Weil pairing, which embeds the n_1 -torsion's pairing values into the n_1 -th roots of unity in $\mathbb{F}_p$, forcing $\mu_{n_1} \subseteq \mathbb{F}_p^\times$. $\square$

For the cryptographic application one wants $E(\mathbb{F}_p)$ to contain a large *prime-order* subgroup. The Pasta curves are designed so that $\#E(\mathbb{F}_p)$ is itself prime, hence $E(\mathbb{F}_p)$ is cyclic of prime order, the optimal case.

10.9 Torsion, the cofactor, and prime-order subgroups

Definition 10.19 (order of a point, torsion). The *order* of a point $P \in E(\mathbb{F}_p)$ is the least integer $m \geq 1$ with $[m]P = \mathcal{O}$ (it exists and divides $\#E(\mathbb{F}_p)$ by Lagrange's theorem). For an integer $m \geq 1$, the *m -torsion subgroup* is

$$E[m] = \{ P \in E(\overline{\mathbb{F}_p}) : [m]P = \mathcal{O} \},$$

and $E(\mathbb{F}_p)[m] = E[m] \cap E(\mathbb{F}_p)$ is its rational part. For m coprime to p , $E[m] \cong (\mathbb{Z}/m\mathbb{Z})^2$.

Definition 10.20 (cofactor). Suppose $\#E(\mathbb{F}_p) = h \cdot r$ where r is the (large) prime to be used and h is the remaining factor. The integer h is the *cofactor*, and r the *order* of the prime-order subgroup. A point P of order r generates a cyclic subgroup $\langle P \rangle \cong \mathbb{Z}/r\mathbb{Z}$, the *prime-order subgroup* used in cryptography.

A nontrivial cofactor $h > 1$ is undesirable: it admits “small-subgroup” points (of order dividing h) that can leak information or break protocol assumptions, so implementations either clear the cofactor by multiplying inputs by h , or use prime-order curves where $h = 1$. The Pasta curves have *cofactor 1*: $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ are themselves prime, so $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ and $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$ are cyclic of prime order and no cofactor clearing is needed. This is one of the principal design advantages of the Pasta cycle for proof systems.

Proposition 10.21 (two-torsion). *The affine points of order 2 in $E(\mathbb{F}_p)$ are exactly the points $(x_0, 0)$ with $f(x_0) = x_0^3 + ax_0 + b = 0$. Hence $E(\mathbb{F}_p)$ has 0, 1, or 3 points of order 2 according as f has 0, 1, or 3 roots in $\mathbb{F}_p$, and $\#E(\mathbb{F}_p)$ is even iff f has a root in $\mathbb{F}_p$.*

Proof. $P = (x, y)$ has order 2 iff $P = -P = (x, -y)$ and $P \neq \mathcal{O}$, i.e. $y = -y$, i.e. $y = 0$ (as $\text{char} \neq 2$); the curve equation then forces $f(x) = 0$. Together with $\mathcal{O}$ these form the 2-torsion subgroup $E(\mathbb{F}_p)[2]$, which is trivial, $\mathbb{Z}/2\mathbb{Z}$, or $(\mathbb{Z}/2\mathbb{Z})^2$; its size is even iff a root exists, and by Lagrange evenness of the size is equivalent to evenness of $\#E(\mathbb{F}_p)$. $\square$

Remark 10.22. For $y^2 = x^3 + 5$ over $\mathbb{F}_p$: the curve has a point of order 2 iff $x^3 + 5 = 0$ has a solution, i.e. iff -5 is a cube in $\mathbb{F}_p$. For the Pasta primes $\#E_p(\mathbb{F}_p) = q$ is odd (in fact prime), so there are *no* rational 2-torsion points; consistently, -5 is not a cube in $\mathbb{F}_p$.

10.10 The elliptic-curve discrete logarithm problem

Definition 10.23 (ECDLP). Let $P \in E(\mathbb{F}_p)$ be a point of large prime order r , generating $G = \langle P \rangle$. The *elliptic-curve discrete logarithm problem* (ECDLP) is: given P and a point $Q \in G$, find the unique integer $n \in \{0, 1, \dots, r-1\}$ with $Q = [n]P$. One writes $n = \log_P Q$.

The map $n \mapsto [n]P$ is a group isomorphism $\mathbb{Z}/r\mathbb{Z} \rightarrow G$, so a discrete logarithm always exists and is unique; the content lies in its *computational* difficulty. Scalar multiplication (the forward map) is fast by double-and-add, but no efficient algorithm is known to invert it on a well-chosen elliptic curve.

Theorem 10.24 (generic hardness). *The best known algorithms for ECDLP on a general curve of prime order r (Pollard’s rho and the baby-step giant-step method) run in time $O(\sqrt{r})$ group operations and, for Pollard’s rho, $O(1)$ space. For r of cryptographic size (e.g. $r \approx 2^{255}$ for the Pasta curves) this is $\approx 2^{128}$ operations, infeasible.*

Proof (sketch). Sketch. Baby-step giant-step writes $n = i\lceil\sqrt{r}\rceil + j$ with $0 \leq i, j < \lceil\sqrt{r}\rceil$, precomputes the “baby steps” $[j]P$ in a table, and matches against the “giant steps” $Q - [i\lceil\sqrt{r}\rceil]P$; a collision reveals n , in $O(\sqrt{r})$ time and space. Pollard’s rho achieves the same time with negligible space by a pseudo-random walk on G and the birthday paradox: a self-collision in the walk yields a relation $[a]P + [b]Q = [a']P + [b']Q$, from which $n \equiv (a - a')(b' - b)^{-1} \pmod{r}$. Both are exponential in the bit length $\log_2 r$. Crucially, on curves with special structure one can beat these: if $\#E(\mathbb{F}_p) = p$ (anomalous curves) there is a polynomial-time attack via the formal group, and the MOV/Frey–Rück attack transfers the ECDLP to a finite-field DLP when the *embedding degree* is small. Cryptographically secure curves, including the Pasta curves, are chosen to have a large prime subgroup order, large embedding degree, and $\#E(\mathbb{F}_p) \neq p$, defeating all known subexponential attacks. $\square$

Remark 10.25. The contrast with multiplicative groups $\mathbb{F}_p^\times$ is essential: in $\mathbb{F}_p^\times$ the index-calculus method solves the discrete logarithm in subexponential time $L_p[1/3]$, forcing those groups to be

very large. On a well-chosen elliptic curve no index calculus is known, so the generic $O(\sqrt{r})$ bound stands and much smaller groups suffice (about 256 bits for 128-bit security, versus thousands of bits for $\mathbb{F}_p^\times$). Smaller groups mean faster arithmetic and shorter keys, which is why elliptic curves dominate modern protocols, including Halo 2.

10.11 The j -invariant and the GLV endomorphism

Definition 10.26 (j -invariant). The j -invariant of the curve $y^2 = x^3 + ax + b$ (with $4a^3 + 27b^2 \neq 0$) is

$$j = 1728 \frac{4a^3}{4a^3 + 27b^2} \in K.$$

Proposition 10.27. *Two elliptic curves over $\overline{K}$ are isomorphic (over $\overline{K}$) if and only if they have the same j -invariant. The admissible coordinate changes preserving short Weierstrass form are $(x, y) \mapsto (u^2x, u^3y)$ for $u \in \overline{K}^\times$, under which $(a, b) \mapsto (u^{-4}a, u^{-6}b)$; the combination j is invariant.*

Proof (sketch). Sketch. A direct computation shows the only isomorphisms between short Weierstrass curves fixing $\mathcal{O}$ are the scalings $(x, y) \mapsto (u^2x, u^3y)$; substituting into the equation gives $a \mapsto u^{-4}a$, $b \mapsto u^{-6}b$. Then $4a^3/(4a^3 + 27b^2)$ is unchanged because numerator and denominator both scale by u^{-12} . Conversely, given equal j -invariants one solves for a suitable u to construct an isomorphism (with separate easy cases $j = 0$ and $j = 1728$). $\square$

Remark 10.28. For $y^2 = x^3 + b$ one has $a = 0$, so $j = 0$. The Pasta curves both have $j = 0$. Curves with $j = 0$ are special: they admit an extra automorphism of order 3, which underlies the efficient endomorphism described next. (At the other extreme, $b = 0$ gives $j = 1728$ and an order-4 automorphism.)

The vanishing $j = 0$ is not accidental in the Pasta design; it is what makes the GLV (Gallant–Lambert–Vanstone) speedup available, halving the cost of scalar multiplication.

Proposition 10.29 (the GLV endomorphism for $y^2 = x^3 + b$). *Suppose $\mathbb{F}_p$ contains a primitive cube root of unity ω , i.e. $p \equiv 1 \pmod{3}$, so $\omega \in \mathbb{F}_p$ with $\omega^2 + \omega + 1 = 0$. On $E : y^2 = x^3 + b$ the map*

$$\phi(x, y) = (\omega x, y), \quad \phi(\mathcal{O}) = \mathcal{O},$$

is a group endomorphism of $E(\mathbb{F}_p)$. On the prime-order subgroup $G = \langle P \rangle$ of order r it acts as multiplication by a scalar: there is $\lambda \in \mathbb{Z}/r\mathbb{Z}$ with $\lambda^2 + \lambda + 1 \equiv 0 \pmod{r}$ and

$$\phi(Q) = [\lambda]Q \quad \text{for all } Q \in G.$$

Proof. First, ϕ maps the curve to itself: if $y^2 = x^3 + b$ then $(y)^2 = (\omega x)^3 + b$ because $\omega^3 = 1$, so $(\omega x, y) \in E$. One can check that ϕ is a homomorphism from the addition formulas (14): scaling all x -coordinates by ω and leaving y -coordinates fixed is compatible with the slope $\lambda_{\text{line}} = (y_2 - y_1)/(\omega x_2 - \omega x_1) = \omega^{-1} \lambda_{\text{line}}^{\text{old}}$ and with $x_3 = \lambda^2 - x_1 - x_2 \mapsto \omega^{-2} \lambda^2 - \omega x_1 - \omega x_2 = \omega \lambda^2 - \omega x_1 - \omega x_2 = \omega(\dots)$, using $\omega^{-2} = \omega$; a short computation confirms $\phi(P_1 + P_2) = \phi(P_1) + \phi(P_2)$. Now ϕ has order 3 (since $\omega^3 = 1$ and $\omega \neq 1$), and $\phi^2 + \phi + 1 = 0$ as an endomorphism because $\omega^2 + \omega + 1 = 0$. Restricted to the cyclic group $G \cong \mathbb{Z}/r\mathbb{Z}$, every endomorphism is multiplication by some $\lambda \in \mathbb{Z}/r\mathbb{Z}$ (as $\text{Hom}(\mathbb{Z}/r\mathbb{Z}, \mathbb{Z}/r\mathbb{Z}) \cong \mathbb{Z}/r\mathbb{Z}$); the relation $\phi^2 + \phi + 1 = 0$ becomes $\lambda^2 + \lambda + 1 \equiv 0 \pmod{r}$. Such λ exists precisely when $r \equiv 1 \pmod{3}$, which holds for the Pasta subgroup orders. $\square$

Remark 10.30 (how GLV speeds up scalar multiplication). Because computing ϕ takes a single field multiplication $x \mapsto \omega x$ (negligible compared to a group operation), one can decompose a scalar $n \in \mathbb{Z}/r\mathbb{Z}$ as

$$[n]P = [n_1]P + [n_2]\phi(P), \quad n \equiv n_1 + n_2\lambda \pmod{r},$$

where n_1, n_2 are each only about $\frac{1}{2} \log_2 r$ bits long (a lattice reduction in the lattice of (a, b) with $a + b\lambda \equiv 0 \pmod{r}$ finds them). The two half-length scalar multiplications, run simultaneously with shared doublings (Straus–Shamir’s trick), cost about half the doublings of a single full-length double-and-add. This is a substantial speedup, and its availability is one reason the Pasta curves take the form $y^2 = x^3 + b$ with $p \equiv 1 \pmod{3}$.

10.12 A worked toy example

We illustrate the entire machinery below on a curve small enough to compute by hand, followed by the corresponding facts for the Pasta curves.

Example 10.31 (the curve $y^2 = x^3 + 5$ over $\mathbb{F}_{11}$). Take $p = 11$ (note $11 > 3$, and $4 \cdot 0 + 27 \cdot 5^2 = 675 \equiv 675 - 61 \cdot 11 = 4 \not\equiv 0 \pmod{11}$), so the curve is nonsingular). The squares in $\mathbb{F}_{11}$ are

$$\{0^2, \dots, 10^2\} = \{0, 1, 4, 9, 5, 3\},$$

i.e. the nonzero quadratic residues are $\{1, 3, 4, 5, 9\}$. Tabulating $f(x) = x^3 + 5 \pmod{11}$ and reading off the y -values:

x	$x^3 \pmod{11}$	$f(x) = x^3 + 5$	y with $y^2 = f(x)$
0	0	5	± 4 ($4^2 = 16 \equiv 5$)
1	1	6	none ($6 \notin \text{QR}$)
2	8	2	none
3	5	10	none
4	9	3	± 5 ($5^2 = 25 \equiv 3$)
5	4	9	± 3
6	7	1	± 1
7	2	7	none
8	6	0	0
9	3	8	none
10	10	4	± 2

Counting: $x \in \{0, 4, 5, 6, 10\}$ each give 2 points (10 points), $x = 8$ gives 1 point $(8, 0)$, and adding $\mathcal{O}$ yields

$$\#E(\mathbb{F}_{11}) = 10 + 1 + 1 = 12.$$

Hasse’s bound: $p + 1 = 12$, so $t = 0$ and indeed $|t| = 0 \leq 2\sqrt{11} \approx 6.63$. The single point $(8, 0)$ has order 2 (here $-5 \equiv 6$ is a cube, namely $8^3 = 512 \equiv 6$), consistent with $\#E$ being even.

An addition. Let $P = (0, 4)$ and $Q = (5, 3)$. Since $x_1 = 0 \neq 5 = x_2$, use the chord:

$$\lambda = \frac{3 - 4}{5 - 0} = \frac{-1}{5} = (-1) \cdot 5^{-1} = (-1) \cdot 9 = -9 \equiv 2 \pmod{11},$$

since $5 \cdot 9 = 45 \equiv 1$. Then

$$x_3 = \lambda^2 - x_1 - x_2 = 4 - 0 - 5 = -1 \equiv 10, \quad y_3 = \lambda(x_1 - x_3) - y_1 = 2(0 - 10) - 4 = -24 \equiv 9 \pmod{11}.$$

So $P + Q = (10, 9)$, which is on the curve: $9^2 = 81 \equiv 4$ and $10^3 + 5 = 1005 \equiv 4 \pmod{11}$. ✓

A doubling. Take $P = (0, 4)$. With $a = 0$,

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{0}{8} = 0, \quad x_3 = \lambda^2 - 2x_1 = 0, \quad y_3 = \lambda(x_1 - x_3) - y_1 = -4 \equiv 7.$$

So $[2](0, 4) = (0, 7) = -(0, 4)$; hence $[3](0, 4) = \mathcal{O}$, so $P = (0, 4)$ has order 3.

The GLV endomorphism. Since $11 \equiv 2 \pmod{3}$, there is *no* primitive cube root of unity in $\mathbb{F}_{11}$, so the GLV map is not available over $\mathbb{F}_{11}$; it would require $p \equiv 1 \pmod{3}$. (The Pasta primes do satisfy $p, q \equiv 1 \pmod{3}$, so GLV applies there.) For a small GLV-friendly illustration one could instead take $p = 13 \equiv 1 \pmod{3}$, where $\omega = 3$ satisfies $3^2 + 3 + 1 = 13 \equiv 0$, and then $\phi(x, y) = (3x, y)$ is a nontrivial endomorphism of $y^2 = x^3 + b$ over $\mathbb{F}_{13}$.

Example 10.32 (the Pasta curves). For the running example $y^2 = x^3 + 5$:

- Over $\mathbb{F}_p$ (Pallas, with p the 255-bit Pasta prime): $\#E_p(\mathbb{F}_p) = q$, a prime; the group is cyclic, $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$; cofactor $h = 1$; the trace is $t = p + 1 - q$, with $|t| \leq 2\sqrt{p}$.
- Over $\mathbb{F}_q$ (Vesta): $\#E_q(\mathbb{F}_q) = p$, a prime; $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$; cofactor 1.
- Both have $j = 0$ (since $a = 0$), admit the order-3 endomorphism $\phi(x, y) = (\omega x, y)$ with ω a primitive cube root of unity (which exists since $p, q \equiv 1 \pmod{3}$), and so support the GLV speedup.
- The “cycle” property $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ means the scalar field of one curve is the base field of the other and vice versa; this is exactly what lets recursive proof systems chain proofs across the two curves without expensive non-native field arithmetic. The base field of Pallas is $\mathbb{F}_p$ and its scalar field (the order of the group) is $\mathbb{F}_q$, matching the base field of Vesta, and symmetrically.

The discrete logarithm problem in $E_p(\mathbb{F}_p)$ and $E_q(\mathbb{F}_q)$ is believed to require about 2^{128} operations (half the bit length of $r \approx 2^{255}$), the security level targeted by Halo 2 and Zcash Orchard.

10.13 Summary

We have constructed the elliptic curve group from the ground up: a nonsingular short Weierstrass cubic over a field of characteristic $\neq 2, 3$, its affine points together with a point at infinity, and the chord-and-tangent addition with explicit formulas (14)–(15) and all edge cases. The point at infinity is the identity, (x, y) and $(x, -y)$ are inverses, commutativity is immediate, and associativity, though geometrically subtle, makes $E(K)$ an abelian group. Projective and Jacobian coordinates remove inversions from the inner loop, and double-and-add computes $[n]P$ in $O(\log n)$ operations. Over $\mathbb{F}_p$ the group has order $p + 1 - t$ with $|t| \leq 2\sqrt{p}$ (Hasse) and structure $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$; choosing the order to be prime (cofactor 1, as for the Pasta curves) yields a clean cyclic group in which the discrete logarithm problem is hard. Finally, the $j = 0$ curves $y^2 = x^3 + b$ admit the efficiently computable GLV endomorphism. The Pasta curves $y^2 = x^3 + 5$ over $\mathbb{F}_p$ and $\mathbb{F}_q$ realize all of these features simultaneously and form the 2-cycle that carries the recursive proofs of Halo 2 and Zcash Orchard.

11 Probability, asymptotics, and computation

Cryptography is the discipline of rendering certain computations easy for legitimate parties and infeasible for adversaries. Stating and proving such guarantees requires three intertwined languages: the language of *probability*, which permits reasoning about random keys, random challenges, and the probability that an attack succeeds; the language of *asymptotics*, which permits precise description of how resources and success probabilities scale; and the language of *computation*, which fixes the notion of an efficient algorithm and of what it means for a problem to be hard. This section develops all three from elementary foundations. The only prerequisites are familiarity with finite sets, sums, and the elementary calculus of limits; we construct everything else here.

Throughout, all probability spaces are finite. This is not a serious restriction for the present purposes: keys, messages, group elements, and the random coins of any terminating algorithm all reside in finite sets. Working with finite spaces avoids measure theory entirely while sacrificing nothing we require.

11.1 Finite probability spaces and events

The basic object is a finite set of *outcomes* together with an assignment of weights summing to one.

Definition 11.1 (Finite probability space). A *finite probability space* is a pair $(\Omega, \Pr)$ where Ω is a nonempty finite set, called the *sample space*, and $\Pr: \Omega \rightarrow [0, 1]$ is a function, called the *probability mass function*, satisfying the normalisation condition

$$\sum_{\omega \in \Omega} \Pr[\omega] = 1.$$

We call an element $\omega \in \Omega$ an *outcome* or an *elementary event*.

Definition 11.2 (Event). An *event* is a subset $A \subseteq \Omega$. Its *probability* is

$$\Pr[A] = \sum_{\omega \in A} \Pr[\omega],$$

with the convention $\Pr[\emptyset] = 0$ (an empty sum). The event A *occurs* on outcome ω if $\omega \in A$.

Because Ω is finite, every subset is an event; we require no separate σ -algebra. The following elementary facts hold, and we use them freely.

Proposition 11.3 (Basic axioms). *For any finite probability space $(\Omega, \Pr)$ and events $A, B \subseteq \Omega$:*

1. $0 \leq \Pr[A] \leq 1$, $\Pr[\emptyset] = 0$, and $\Pr[\Omega] = 1$.
2. (Complement.) $\Pr[\Omega \setminus A] = 1 - \Pr[A]$.
3. (Monotonicity.) If $A \subseteq B$ then $\Pr[A] \leq \Pr[B]$.
4. (Inclusion–exclusion for two events.) $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$.
5. (Finite additivity.) If $A_1, \dots, A_n$ are pairwise disjoint then $\Pr[\bigcup_{i=1}^n A_i] = \sum_{i=1}^n \Pr[A_i]$.

Proof. Every claim follows by manipulating the defining sums. For (1), each summand $\Pr[\omega]$ lies in $[0, 1]$ and the total is 1 by normalisation; an event's probability is a partial sum of nonnegative terms, hence in $[0, 1]$. For (2), $\Pr[A] + \Pr[\Omega \setminus A] = \sum_{\omega \in \Omega} \Pr[\omega] = 1$ since A and $\Omega \setminus A$ partition Ω . For (3), $\Pr[B] = \sum_{\omega \in B} \Pr[\omega] = \sum_{\omega \in A} \Pr[\omega] + \sum_{\omega \in B \setminus A} \Pr[\omega] \geq \Pr[A]$, the last term being a sum of nonnegative numbers. For (5), disjointness implies that each ω in the union lies in exactly one A_i , so the double sum $\sum_i \sum_{\omega \in A_i} \Pr[\omega]$ counts each $\omega \in \bigcup_i A_i$ exactly once. For (4), write $A \cup B = A \sqcup (B \setminus A)$ (disjoint), so by (5) $\Pr[A \cup B] = \Pr[A] + \Pr[B \setminus A]$; also $B = (A \cap B) \sqcup (B \setminus A)$, giving $\Pr[B \setminus A] = \Pr[B] - \Pr[A \cap B]$. Substituting yields the claim. $\square$

The single most important example, ubiquitous in cryptography, is the uniform distribution.

Definition 11.4 (Uniform distribution). Let S be a nonempty finite set. The *uniform distribution* on S is the probability space $(S, \Pr)$ with $\Pr[\omega] = 1/|S|$ for every $\omega \in S$. The notation $x \leftarrow S$ (or $x \xleftarrow{\$} S$) denotes that we sample x according to the uniform distribution on S . Under the uniform distribution, for any event $A \subseteq S$,

$$\Pr[A] = \frac{|A|}{|S|}.$$

Thus, for uniform sampling, probability is exactly the ratio of favourable outcomes to total outcomes, recovering the classical "counting" notion of probability.

11.2 Conditional probability and independence

Frequently we learn that one event has occurred, and we must update the probability of another accordingly. Conditioning captures this.

Definition 11.5 (Conditional probability). Let A, B be events with $\Pr[B] > 0$. The *conditional probability of A given B* is

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

When $\Pr[B] = 0$ we leave the quantity undefined.

Conditioning on B restricts the sample space to B and renormalises so that B has probability 1: the map $A \mapsto \Pr[A \mid B]$ is itself a probability mass function on B . Two immediate and indispensable consequences follow.

Proposition 11.6 (Chain rule and total probability). *Let $A_1, \dots, A_n$ be events with $\Pr[A_1 \cap \dots \cap A_{n-1}] > 0$. Then*

$$\Pr[A_1 \cap \dots \cap A_n] = \Pr[A_1] \Pr[A_2 \mid A_1] \Pr[A_3 \mid A_1 \cap A_2] \cdots \Pr[A_n \mid A_1 \cap \dots \cap A_{n-1}].$$

Moreover, if $B_1, \dots, B_k$ partition Ω (pairwise disjoint with union Ω) and each $\Pr[B_i] > 0$, then for any event A ,

$$\Pr[A] = \sum_{i=1}^k \Pr[A \mid B_i] \Pr[B_i].$$

Proof. The chain rule follows by telescoping: each factor $\Pr[A_j \mid A_1 \cap \dots \cap A_{j-1}]$ equals $\Pr[A_1 \cap \dots \cap A_j] / \Pr[A_1 \cap \dots \cap A_{j-1}]$ by Definition 11.5, and the product collapses to $\Pr[A_1 \cap \dots \cap A_n] / 1$. (Monotonicity ensures all intermediate intersections have positive probability, so every factor is defined.) For total probability, the events $A \cap B_i$ are pairwise disjoint with union A (since the B_i partition Ω), so by finite additivity $\Pr[A] = \sum_i \Pr[A \cap B_i] = \sum_i \Pr[A \mid B_i] \Pr[B_i]$, using $\Pr[A \cap B_i] = \Pr[A \mid B_i] \Pr[B_i]$. $\square$

Theorem 11.7 (Bayes). *Let A, B be events with $\Pr[A] > 0$ and $\Pr[B] > 0$. Then*

$$\Pr[B \mid A] = \frac{\Pr[A \mid B] \Pr[B]}{\Pr[A]}.$$

Proof. Both sides equal $\Pr[A \cap B] / \Pr[A]$: the left by definition, and the right because $\Pr[A \mid B] \Pr[B] = \Pr[A \cap B]$ by definition of conditional probability. $\square$

Independence is the formal statement that one event carries no information about another.

Definition 11.8 (Independence of events). Two events A, B are *independent* if

$$\Pr[A \cap B] = \Pr[A] \Pr[B].$$

A family $\{A_i\}_{i \in I}$ of events is *mutually independent* if for every finite subset $J \subseteq I$,

$$\Pr\left[\bigcap_{i \in J} A_i\right] = \prod_{i \in J} \Pr[A_i].$$

Remark 11.9. When $\Pr[B] > 0$, independence of A and B is equivalent to $\Pr[A \mid B] = \Pr[A]$: conditioning on B does not change the probability of A . Mutual independence is strictly stronger than pairwise independence. The standard counterexample is the following: toss two fair coins, and let A = "first is heads", B = "second is heads", C = "the two coins differ". Each pair is independent, yet $\Pr[A \cap B \cap C] = 0 \neq \frac{1}{8} = \Pr[A] \Pr[B] \Pr[C]$, so the three are not mutually independent.

11.3 Random variables and expectation

A random variable assigns a value to each outcome; it summarises a random experiment by a number (or a vector, etc.).

Definition 11.10 (Random variable). Let $(\Omega, \Pr)$ be a finite probability space and T any set. A (T -valued) *random variable* is a function $X: \Omega \rightarrow T$. When $T \subseteq \mathbb{R}$, we call X *real-valued*. For $t \in T$,

$$\Pr[X = t] = \Pr[\{\omega \in \Omega : X(\omega) = t\}],$$

and more generally $\Pr[X \in B]$ denotes the probability that X takes a value in $B \subseteq T$. The function $t \mapsto \Pr[X = t]$ on the finite image of X is the *distribution* (or *law*) of X .

Definition 11.11 (Independent random variables). Random variables $X_1, \dots, X_n$ (with values in sets $T_1, \dots, T_n$) are *mutually independent* if for all $t_1 \in T_1, \dots, t_n \in T_n$,

$$\Pr[X_1 = t_1, \dots, X_n = t_n] = \prod_{i=1}^n \Pr[X_i = t_i].$$

Definition 11.12 (Expectation). Let $X: \Omega \rightarrow \mathbb{R}$ be a real-valued random variable on a finite space. Its *expectation* (or *mean*) is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \Pr[\omega] = \sum_{t \in \text{im}(X)} t \cdot \Pr[X = t],$$

where $\text{im}(X)$ denotes the (finite) image of X . The two expressions agree by grouping outcomes according to their X -value.

A convenient and frequently used special case is the following: the expectation of the *indicator* of an event is exactly its probability.

Definition 11.13 (Indicator). For an event A , its *indicator random variable* $\mathbf{1}_A: \Omega \rightarrow \{0, 1\}$ takes the value $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise.

Proposition 11.14 (Indicator expectation). *For any event A , $\mathbb{E}[\mathbf{1}_A] = \Pr[A]$.*

Proof. $\mathbb{E}[\mathbf{1}_A] = \sum_{\omega} \mathbf{1}_A(\omega) \Pr[\omega] = \sum_{\omega \in A} \Pr[\omega] = \Pr[A]$, since the indicator vanishes off A . $\square$

The principal property of expectation is its linearity, which holds with no independence assumption whatsoever. This renders expectation considerably more tractable than probability.

Theorem 11.15 (Linearity of expectation). *Let $X, Y: \Omega \rightarrow \mathbb{R}$ be random variables on a finite space and $a, b \in \mathbb{R}$. Then*

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y].$$

More generally, for random variables $X_1, \dots, X_n$ and scalars $a_1, \dots, a_n$,

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

This holds whether or not the X_i are independent.

Proof. Directly from the definition,

$$\mathbb{E}[aX + bY] = \sum_{\omega \in \Omega} (aX(\omega) + bY(\omega)) \Pr[\omega] = a \sum_{\omega} X(\omega) \Pr[\omega] + b \sum_{\omega} Y(\omega) \Pr[\omega] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

splitting the finite sum and extracting the constants. The n -term statement follows by induction on n . $\square$

Example 11.16 (Counting via indicators). Throw m balls independently and uniformly into n bins, and let X be the number of bins that remain empty. Computing the distribution of X directly is awkward, but linearity makes the mean immediate. Write $X = \sum_{j=1}^n \mathbf{1}_{E_j}$, where E_j is the event “bin j is empty”. A fixed ball misses bin j with probability $1 - 1/n$, and the m balls are independent, so $\Pr[E_j] = (1 - 1/n)^m$. By Proposition 11.14 and Theorem 11.15,

$$\mathbb{E}[X] = \sum_{j=1}^n \mathbb{E}[\mathbf{1}_{E_j}] = n \left(1 - \frac{1}{n}\right)^m.$$

The argument needs no independence among the E_j , even though they are in fact *not* independent.

For independent variables, expectation also factors over products; we seldom need this, but record it here.

Proposition 11.17 (Expectation of a product). *If X and Y are independent real-valued random variables, then $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$.*

Proof. Using the joint distribution and independence,

$$\mathbb{E}[XY] = \sum_{s \in \text{im}(X)} \sum_{t \in \text{im}(Y)} st \Pr[X = s, Y = t] = \sum_s s \Pr[X = s] \sum_t t \Pr[Y = t] = \mathbb{E}[X] \mathbb{E}[Y],$$

where the middle equality uses $\Pr[X = s, Y = t] = \Pr[X = s] \Pr[Y = t]$. $\square$

11.4 The union bound and a birthday calculation

The most frequently used inequality in cryptographic proofs is the union bound: the probability that at least one of several events occurs is at most the sum of their probabilities. It is crude but requires no independence and almost always suffices.

Theorem 11.18 (Union bound / Boole’s inequality). *For any events $A_1, \dots, A_n$ in a finite probability space,*

$$\Pr\left[\bigcup_{i=1}^n A_i\right] \leq \sum_{i=1}^n \Pr[A_i].$$

Proof. Define the disjoint "first occurrence" events $B_1 = A_1$ and $B_i = A_i \setminus (A_1 \cup \dots \cup A_{i-1})$ for $i \geq 2$. The B_i are pairwise disjoint, $B_i \subseteq A_i$, and $\bigcup_i B_i = \bigcup_i A_i$. By finite additivity and monotonicity (Proposition 11.3),

$$\Pr\left[\bigcup_i A_i\right] = \Pr\left[\bigcup_i B_i\right] = \sum_i \Pr[B_i] \leq \sum_i \Pr[A_i]. \quad \square$$

The complementary inequality for *conjunctions* is equally useful and follows by taking complements.

Corollary 11.19 (Union bound for the good event). *If each event A_i holds except with probability at most ε_i (that is, $\Pr[\Omega \setminus A_i] \leq \varepsilon_i$), then all of them hold simultaneously except with probability at most $\sum_i \varepsilon_i$:*

$$\Pr\left[\bigcap_{i=1}^n A_i\right] \geq 1 - \sum_{i=1}^n \varepsilon_i.$$

Proof. By De Morgan, $\Omega \setminus \bigcap_i A_i = \bigcup_i (\Omega \setminus A_i)$. Apply the union bound to the complements and take the complement of both sides. $\square$

We now derive the *birthday bound*. It governs how soon collisions appear when sampling uniformly and underlies the generic attacks against n -bit hash functions and the security loss in many reductions.

Proposition 11.20 (Birthday bound). *Sample $x_1, \dots, x_k$ independently and uniformly from a set of size N . Let C be the event that some two of them collide, i.e. $x_i = x_j$ for some $i < j$. Then*

$$\Pr[C] \leq \binom{k}{2} \frac{1}{N} = \frac{k(k-1)}{2N},$$

and, on the other side,

$$\Pr[C] \geq 1 - \prod_{i=0}^{k-1} \left(1 - \frac{i}{N}\right) \geq 1 - e^{-k(k-1)/(2N)}.$$

In particular a collision becomes likely once $k = \Theta(\sqrt{N})$.

Proof. Upper bound. For $i < j$ let A_{ij} be the event $x_i = x_j$. Since x_i, x_j are independent and uniform, $\Pr[A_{ij}] = \sum_v \Pr[x_i = v] \Pr[x_j = v] = N \cdot (1/N)^2 = 1/N$. Now $C = \bigcup_{i < j} A_{ij}$, a union of $\binom{k}{2}$ events, so the union bound (Theorem 11.18) gives $\Pr[C] \leq \binom{k}{2}/N$.

Lower bound. The complement $\overline{C}$ is the event that all k samples are distinct. Counting ordered tuples of distinct values, $\Pr[\overline{C}] = \prod_{i=0}^{k-1} (1 - i/N)$: having fixed i distinct values, the next sample avoids them with probability $1 - i/N$. Using $1 - x \leq e^{-x}$ for all $x \in \mathbb{R}$,

$$\Pr[\overline{C}] = \prod_{i=0}^{k-1} \left(1 - \frac{i}{N}\right) \leq \prod_{i=0}^{k-1} e^{-i/N} = \exp\left(-\frac{1}{N} \sum_{i=0}^{k-1} i\right) = e^{-k(k-1)/(2N)}.$$

Taking complements gives $\Pr[C] = 1 - \Pr[\overline{C}] \geq 1 - e^{-k(k-1)/(2N)}$. $\square$

Remark 11.21. The threshold $k \approx \sqrt{N}$ is the reason an n -bit hash function (so $N = 2^n$) offers only about $2^{n/2}$ collision resistance: an adversary expects a collision after roughly $2^{n/2}$ random evaluations. For this reason one chooses output lengths twice the desired security level. The same square-root phenomenon recurs in random-walk discrete-log attacks (Pollard's rho), so designers size the curves used in Halo 2 and Orchard so that $\sqrt{N}$ is itself astronomically large.

11.5 Statistical distance

Comparing distributions—for example a real key-generation procedure against an idealised uniform one—requires a notion of how far apart two distributions are. The appropriate notion for “no statistical test can distinguish them well” is statistical (total variation) distance.

Definition 11.22 (Statistical distance). Let P and Q be probability distributions on the same finite set S (i.e. probability mass functions $P, Q: S \rightarrow [0, 1]$). Their *statistical distance* (or *total variation distance*) is

$$\Delta(P, Q) = \frac{1}{2} \sum_{s \in S} |P(s) - Q(s)|.$$

For random variables X, Y on S , $\Delta(X, Y)$ denotes the statistical distance of their distributions.

The factor $\frac{1}{2}$ makes Δ range over $[0, 1]$ and gives it a clean interpretation as the maximum advantage of a distinguisher, which we establish next.

Theorem 11.23 (Properties of statistical distance). *Let P, Q, R be distributions on a finite set S . Then:*

1. $0 \leq \Delta(P, Q) \leq 1$; and $\Delta(P, Q) = 0$ iff $P = Q$.
2. (Metric.) Δ is symmetric and satisfies the triangle inequality $\Delta(P, R) \leq \Delta(P, Q) + \Delta(Q, R)$.
3. (Variational characterisation.) $\Delta(P, Q) = \max_{A \subseteq S} (P(A) - Q(A)) = \max_{A \subseteq S} |P(A) - Q(A)|$, where $P(A) = \sum_{s \in A} P(s)$.
4. (Data-processing / post-processing.) For any (possibly randomised) function f applied to a sample, $\Delta(f(P), f(Q)) \leq \Delta(P, Q)$. In particular no algorithm can increase statistical distance.

Proof. (1) Nonnegativity is clear. For the upper bound, let $S^+ = \{s : P(s) \geq Q(s)\}$ and $S^- = S \setminus S^+$. Since $\sum_s P(s) = \sum_s Q(s) = 1$, we have $\sum_s (P(s) - Q(s)) = 0$, so $\sum_{s \in S^+} (P(s) - Q(s)) = \sum_{s \in S^-} (Q(s) - P(s)) =: D \geq 0$. Hence $\Delta(P, Q) = \frac{1}{2} (\sum_{S^+} (P - Q) + \sum_{S^-} (Q - P)) = D$. As $D = \sum_{S^+} (P(s) - Q(s)) \leq \sum_{S^+} P(s) \leq 1$, we get $\Delta \leq 1$. If $\Delta = 0$ every $|P(s) - Q(s)| = 0$, so $P = Q$; the converse is immediate.

(2) Symmetry is evident. For the triangle inequality, $|P(s) - R(s)| \leq |P(s) - Q(s)| + |Q(s) - R(s)|$ termwise by the triangle inequality on $\mathbb{R}$; summing and halving gives the claim.

(3) With D as above, take $A = S^+$: then $P(A) - Q(A) = \sum_{S^+} (P - Q) = D = \Delta(P, Q)$, so the maximum is at least Δ . Conversely, for *any* A , $P(A) - Q(A) = \sum_{s \in A} (P(s) - Q(s)) \leq \sum_{s \in A, P(s) \geq Q(s)} (P(s) - Q(s)) \leq D = \Delta$, since dropping the negative terms only increases the sum and extending from $A \cap S^+$ to all of S^+ only increases it further. Thus the maximum over A of $P(A) - Q(A)$ is exactly Δ ; replacing A by its complement swaps the sign, giving the same value for $\max |P(A) - Q(A)|$.

(4) First take f deterministic, $f: S \rightarrow U$. For $u \in U$, $f(P)(u) = P(f^{-1}(u))$. Then

$$\Delta(f(P), f(Q)) = \frac{1}{2} \sum_u |P(f^{-1}(u)) - Q(f^{-1}(u))| \leq \frac{1}{2} \sum_u \sum_{s \in f^{-1}(u)} |P(s) - Q(s)| = \Delta(P, Q),$$

using $|\sum_s a_s| \leq \sum_s |a_s|$ on each fibre, and that the fibres $f^{-1}(u)$ partition S . A randomised f is a deterministic function of s together with independent fresh coins r ; applying the deterministic case to the joint variable (s, r) (whose two distributions $P \times U_r$ and $Q \times U_r$ have the same statistical distance $\Delta(P, Q)$, the shared coin distribution contributing nothing) yields the bound. $\square$

Remark 11.24 (Distinguishing interpretation). Part (3) states that $\Delta(P, Q)$ is exactly the best advantage of any (even computationally unbounded) test that, given a single sample, must guess whether it came from P or Q : the optimal test outputs “ P ” on the set $A = S^+$, and its advantage $\Pr_P[A] - \Pr_Q[A]$ equals $\Delta(P, Q)$. Part (4) is what makes statistical distance compose in proofs: applying the same procedure to two close inputs keeps the outputs close. If $\Delta(P, Q) \leq \varepsilon$, then P and Q are interchangeable in any analysis at the cost of an additive ε in every probability. When ε is negligible (defined below), we call P and Q *statistically indistinguishable*.

11.6 Uniform sampling and the bias of modular reduction

Cryptography requires uniform elements of a finite set throughout: a uniform scalar in $\mathbb{Z}/q\mathbb{Z}$, a uniform field element of $\mathbb{F}_p$, a uniform point in a group. The most tractable source of randomness, however, is a stream of independent uniform bits. We must therefore convert uniform bitstrings into uniform elements of S , and we must understand the bias this conversion introduces.

If $|S| = 2^\ell$ is a power of two, the conversion is exact: read ℓ uniform bits as a number in $\{0, \dots, 2^\ell - 1\}$ and index into S . The nontrivial case arises when $|S|$ is not a power of two—most importantly when $S = \mathbb{Z}/q\mathbb{Z}$ for a prime q , which is the scalar field of an elliptic curve.

A common and elementary method is *reduce-modulo*: sample a long uniform integer and reduce it mod q . The result is not perfectly uniform—some residues receive one more preimage than others—but the bias is negligible provided the integer is long enough. We make this precise below.

Proposition 11.25 (Bias of modular reduction). *Let $q \geq 2$ be an integer and let $K = 2^L$ for some integer L . Sample U uniformly from $\{0, 1, \dots, K - 1\}$ (equivalently, read L uniform bits as an integer), and set $R = U \bmod q \in \{0, 1, \dots, q - 1\}$. Let $\mathcal{U}_q$ denote the uniform distribution on $\{0, \dots, q - 1\}$. Then the statistical distance of R from uniform satisfies*

$$\Delta(R, \mathcal{U}_q) \leq \frac{q}{K} = \frac{q}{2^L}.$$

Consequently, if $q < 2^n$ and one takes $L = n + s$ bits, then $\Delta(R, \mathcal{U}_q) < 2^{-s}$.

Proof. Write $K = mq + t$ with $m = \lfloor K/q \rfloor$ and remainder $t = K \bmod q$, so $0 \leq t < q$. For a residue $r \in \{0, \dots, q - 1\}$, the number of integers in $\{0, \dots, K - 1\}$ congruent to r modulo q is $m + 1$ if $r < t$ and m if $r \geq t$. Hence

$$\Pr[R = r] = \begin{cases} (m + 1)/K, & r < t, \\ m/K, & r \geq t. \end{cases}$$

Comparing with the target probability $1/q$ and using $m/K \leq 1/q \leq (m + 1)/K$ (since $mq \leq K \leq (m + 1)q$),

$$\left| \Pr[R = r] - \frac{1}{q} \right| \leq \frac{m + 1}{K} - \frac{m}{K} = \frac{1}{K} \quad \text{for every } r.$$

Therefore

$$\Delta(R, \mathcal{U}_q) = \frac{1}{2} \sum_{r=0}^{q-1} \left| \Pr[R = r] - \frac{1}{q} \right| \leq \frac{1}{2} \cdot q \cdot \frac{1}{K} = \frac{q}{2K} \leq \frac{q}{K}.$$

(The computation yields the sharper bound $q/(2K)$; q/K is the usual stated form.) For the final claim, $q < 2^n$ and $K = 2^{n+s}$ give $q/K < 2^n/2^{n+s} = 2^{-s}$. $\square$

Remark 11.26. The consequence is as follows: to sample a scalar mod a ~ 256 -bit prime q with statistical distance below 2^{-128} from uniform, draw $256 + 128 = 384$ uniform bits and reduce. The bias 2^{-128} is negligible (Definition 11.33), so the reduce-modulo sample is statistically indistinguishable from a perfectly uniform scalar, and by the data-processing inequality (Theorem 11.23(4)) substituting one for the other anywhere costs at most 2^{-128} per use. An alternative,

rejection sampling—draw $\lceil \log_2 q \rceil$ bits, accept if the result is $< q$, else retry—gives *exactly* uniform output at the cost of a variable but, with overwhelming probability, small number of retries; for the present purposes the negligible bias of reduce-modulo is acceptable and simpler to analyse. This is precisely the kind of distribution that the Orchard specification employs when deriving scalars from hash outputs.

11.7 Asymptotic notation

The discussion now turns from probability to growth rates. To address efficiency and hardness, we must compare the manner in which functions of an integer parameter grow as that parameter tends to infinity. The Bachmann–Landau notation makes such comparisons precise. Throughout, functions map $\mathbb{N}$ (or a cofinite subset thereof) to the nonnegative reals $\mathbb{R}_{\geq 0}$, and “for sufficiently large n ” means “for all n beyond some threshold n_0 ”.

Definition 11.27 (Big-O, Omega, Theta, little-o, little-omega). Let $f, g: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ with g eventually positive. We define the following:

- $f = O(g)$ (read “ f is big-O of g ”, an asymptotic upper bound) if there exist constants $c > 0$ and n_0 with $f(n) \leq c g(n)$ for all $n \geq n_0$.
- $f = \Omega(g)$ (asymptotic lower bound) if there exist $c > 0$ and n_0 with $f(n) \geq c g(n)$ for all $n \geq n_0$. Equivalently, $f = \Omega(g)$ iff $g = O(f)$.
- $f = \Theta(g)$ (tight bound) if $f = O(g)$ and $f = \Omega(g)$; i.e. there are $c_1, c_2 > 0$ and n_0 with $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for $n \geq n_0$.
- $f = o(g)$ (read “little-o”, strictly smaller order) if for every $c > 0$ there is n_0 with $f(n) \leq c g(n)$ for all $n \geq n_0$; equivalently $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$.
- $f = \omega(g)$ (strictly larger order) if $g = o(f)$; equivalently $\lim_{n \rightarrow \infty} f(n)/g(n) = \infty$.

Remark 11.28. The “=” in “ $f = O(g)$ ” is traditional but denotes set membership: $O(g)$ is the class of all functions bounded above by a constant multiple of g , and $f = O(g)$ means $f \in O(g)$. One never reads such equations right to left. We follow the customary abuse of notation here.

Example 11.29. $3n^2 + 10n + 7 = \Theta(n^2)$; indeed for $n \geq 1$ one has $3n^2 \leq 3n^2 + 10n + 7 \leq 20n^2$. Also $n^2 = o(n^3)$ since $n^2/n^3 = 1/n \rightarrow 0$; $\log_2 n = o(n)$; $n^{100} = o(2^n)$ (any polynomial is little-o of any exponential with base > 1); and $n! = \omega(2^n)$. Note that $2^n = \Theta(2^{n+1})$ is false as a statement about *little-o* but true for Θ , since $2^{n+1} = 2 \cdot 2^n$ is within a constant factor.

Proposition 11.30 (Useful closure facts). *Let $f_1 = O(g_1)$ and $f_2 = O(g_2)$. Then $f_1 + f_2 = O(\max(g_1, g_2)) = O(g_1 + g_2)$ and $f_1 f_2 = O(g_1 g_2)$. The same closure statements hold with O replaced uniformly by Ω or Θ . Moreover $O(\cdot)$ is transitive: $f = O(g)$ and $g = O(h)$ imply $f = O(h)$.*

Proof. Choose c_1, n_1 and c_2, n_2 witnessing the two hypotheses, and let $n_0 = \max(n_1, n_2)$. For $n \geq n_0$: $f_1(n) + f_2(n) \leq c_1 g_1(n) + c_2 g_2(n) \leq (c_1 + c_2) \max(g_1(n), g_2(n))$, and $\max(g_1, g_2) \leq g_1 + g_2 \leq 2 \max(g_1, g_2)$, so the two right-hand sides agree up to a constant; this proves the sum rule. For products, $f_1(n) f_2(n) \leq c_1 c_2 g_1(n) g_2(n)$. Transitivity: $f \leq c g$ and $g \leq c' h$ eventually give $f \leq cc' h$ eventually. The Ω statements follow by the equivalence $f = \Omega(g) \Leftrightarrow g = O(f)$, and Θ by combining the two. $\square$

11.8 Polynomial, exponential, and negligible functions

We now classify the growth rates relevant to cryptography. Fix a distinguished variable $\lambda \in \mathbb{N}$, the *security parameter* (discussed in detail in Subsection 11.10); informally larger λ means more security and larger keys. Resources (running time, key length) and success probabilities are functions of λ .

Definition 11.31 (Polynomial and super-polynomial). A function $f: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *polynomially bounded*, written $f = \text{poly}(\lambda)$, if there is a constant c (and a threshold) with $f(\lambda) \leq \lambda^c + c$ for all large λ ; equivalently $f(\lambda) = O(\lambda^c)$ for some constant c . It is *super-polynomial* if it grows faster than every polynomial, i.e. $f(\lambda) = \omega(\lambda^c)$ for every constant c (equivalently $\lambda^c = o(f(\lambda))$ for all c).

Definition 11.32 (Exponential). A function f is *exponential* if $f(\lambda) = 2^{\Omega(\lambda^\epsilon)}$ for some constant $\epsilon > 0$; the prototypical case is $f(\lambda) = 2^{c\lambda}$ for a constant $c > 0$. Every exponential function is super-polynomial, but the converse fails: e.g. $\lambda^{\log \lambda} = 2^{(\log \lambda)^2}$ is super-polynomial yet sub-exponential.

The central cryptographic notion is that of a *negligible* function: one that shrinks faster than the reciprocal of every polynomial. A negligible probability is one we may ignore, because no efficient (polynomially many) repetition of an experiment can amplify it to a noticeable chance.

Definition 11.33 (Negligible function). A function $\nu: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *negligible*, written $\nu = \text{negl}(\lambda)$, if for every positive constant c there exists λ_0 such that

$$\nu(\lambda) < \frac{1}{\lambda^c} \quad \text{for all } \lambda \geq \lambda_0.$$

Equivalently, $\nu(\lambda) = o(\lambda^{-c})$ for every constant $c > 0$; equivalently, $\lambda^c \nu(\lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$ for every constant c . A function is *noticeable* (or non-negligible) if it is not negligible, i.e. $f(\lambda) \geq \lambda^{-c}$ infinitely often for some constant c ; it is *overwhelming* if $1 - f$ is negligible.

Example 11.34. $2^{-\lambda}$, $2^{-\sqrt{\lambda}}$, and $\lambda^{-\log \lambda}$ are negligible. Indeed for $2^{-\lambda}$: given any c , since $2^{-\lambda} \lambda^c = \lambda^c / 2^\lambda \rightarrow 0$ (a polynomial over an exponential), the function meets the definition. By contrast, $1/\lambda^{100}$ is *not* negligible—taking $c = 101$ shows $1/\lambda^{100} > 1/\lambda^{101}$ for all λ —it is noticeable, even though it tends to 0. This is the key subtlety: tending to zero does not suffice; a negligible function must beat every inverse polynomial.

The reason “negligible” is the correct closure of “ignorable” is the following stability under the operations that appear in security proofs.

Proposition 11.35 (Closure properties of negligible functions). *Let ν, μ be negligible and let p be a polynomially bounded function. Then:*

1. $\nu + \mu$ is negligible (the class is closed under addition, hence under any fixed finite sum).
2. $p \cdot \nu$ is negligible (a polynomial times a negligible function is negligible).
3. If $f(\lambda) \leq \nu(\lambda)$ for all large λ , then f is negligible (domination by a negligible function).
4. For any constant $k \geq 1$, summing $p(\lambda)$ negligible functions, each negligible uniformly, yields a negligible function; in particular a union bound over polynomially many negligible events is negligible.

Proof. (1) Fix c . There are thresholds beyond which $\nu(\lambda) < \frac{1}{2}\lambda^{-c}$ and $\mu(\lambda) < \frac{1}{2}\lambda^{-c}$ (apply the definition with the same c to each, possible since one may shrink the bound by any constant; enlarging c is not even needed—negligibility gives $\nu < \frac{1}{2}\lambda^{-c}$ by using exponent c and absorbing the constant for large λ). Adding, $\nu(\lambda) + \mu(\lambda) < \lambda^{-c}$ for large λ . More carefully: given c , negligibility of ν applied to constant c gives $\nu(\lambda) < \lambda^{-c}$ eventually; we need the sum below λ^{-c} , so apply the definition to $c + 1$: for large λ , $\nu(\lambda), \mu(\lambda) < \lambda^{-(c+1)} \leq \frac{1}{2}\lambda^{-c}$ once $\lambda \geq 2$, whence the sum is $< \lambda^{-c}$.

(2) Suppose $p(\lambda) \leq \lambda^d$ for large λ (and some constant d). Fix c . By negligibility applied to the exponent $c + d$, $\nu(\lambda) < \lambda^{-(c+d)}$ for large λ . Then $p(\lambda)\nu(\lambda) < \lambda^d \lambda^{-(c+d)} = \lambda^{-c}$ for large λ . Hence $p\nu$ is negligible.

(3) Immediate: if $f \leq \nu$ eventually and $\nu < \lambda^{-c}$ eventually, then $f < \lambda^{-c}$ eventually, for every c .

(4) Write the sum $\Sigma(\lambda) = \sum_{i=1}^{p(\lambda)} \nu_i(\lambda)$ where each $\nu_i \leq \nu$ for a single negligible ν (the "uniform" hypothesis; in security proofs all the ν_i arise from one bound). Then $\Sigma(\lambda) \leq p(\lambda) \nu(\lambda)$, which is negligible by (2); apply (3). $\square$

Remark 11.36 (Why negligible is the threshold of security). Suppose an adversary wins a game with negligible probability $\nu(\lambda)$. If it repeats the attack $p(\lambda)$ times (polynomially many, the most an efficient adversary can afford), the probability that some repetition succeeds is, by the union bound, at most $p(\lambda)\nu(\lambda)$ —still negligible by Proposition 11.35(2). Thus negligible success probability remains negligible under any efficient amplification: it is genuinely ignorable. Conversely a noticeable success probability $\geq \lambda^{-c}$ can be boosted to a constant by λ^c repetitions. This dichotomy is the formal backbone of the phrase "the scheme is secure except with negligible probability".

11.9 Algorithms, running time, and PPT

It remains to make "efficient algorithm" and "infeasible" precise. We adopt the standard model: algorithms are Turing machines (or, equivalently up to polynomial factors, programs in any reasonable model), inputs and outputs are finite binary strings, and the number of elementary steps as a function of input length is the resource we measure. Write $\{0,1\}^*$ for the set of all finite binary strings and $|x|$ for the length of $x \in \{0,1\}^*$.

Definition 11.37 (Deterministic polynomial-time algorithm). An algorithm M runs in (*deterministic*) *polynomial time* if there is a polynomial p such that, on every input $x \in \{0,1\}^*$, M halts within at most $p(|x|)$ steps and produces an output. The class of decision problems solvable by such machines is P .

Determinism is insufficient for cryptography: key generation, sampling, and many algorithms require randomness. We model randomness by granting the machine access to an auxiliary tape of independent uniform random bits (the "coins").

Definition 11.38 (Probabilistic algorithm and PPT). A *probabilistic* (or *randomised*) algorithm M is a Turing machine with an additional read-only *random tape* initialised with independent uniformly random bits $r \in \{0,1\}^*$. For a fixed input x , the output $M(x)$ is a random variable over the choice of r ; write $M(x;r)$ when one wishes to display the coins explicitly, and then $M(x;\cdot)$ is a deterministic function. The algorithm runs in *probabilistic polynomial time* (it is *PPT*) if there is a polynomial p such that, for every input x and every choice of coins r , M halts within $p(|x|)$ steps. (Bounding the time for every r , not merely in expectation, is the standard "strict" notion and ensures M ever reads only $p(|x|)$ coins.)

Remark 11.39 (Non-uniformity and circuits). There are two standard ways to formalise an efficient adversary. The *uniform* model is a single PPT machine that works for all input lengths, as above. The *non-uniform* model allows a separate algorithm (equivalently, a Boolean circuit) of size $\text{poly}(\lambda)$ for each λ , which may be hard-wired with an arbitrary polynomial-length "advice" string depending on λ . Non-uniform adversaries are at least as powerful as uniform ones and are often more convenient in reduction proofs because the advice can encode a "best" attack. The asymptotic theory below is identical for both; security definitions in the Halo 2/Orchard setting typically state the security against non-uniform PPT adversaries, which is the more conservative choice.

Definition 11.40 (Negligible-error decision and Monte Carlo). A probabilistic algorithm *decides* a language L with error $\varepsilon(\cdot)$ if for every x , $\Pr_r[M(x;r) \neq \mathbf{1}[x \in L]] \leq \varepsilon(|x|)$. When M is PPT and $\varepsilon = 1/3$ (a constant $< 1/2$) the corresponding class of languages is BPP ; by independent repetition and majority vote one can drive the error down to $2^{-\text{poly}}$ (negligible) while remaining PPT. We use such error reduction freely.

Proposition 11.41 (Error reduction by majority). *Let M be a PPT algorithm deciding L with error at most $\frac{1}{2} - \delta$ for a constant $\delta > 0$. Running M independently t times on the same input and outputting the majority answer yields a PPT algorithm with error at most $e^{-2\delta^2 t}$. In particular, taking $t = \Theta(\lambda/\delta^2)$ gives negligible error $2^{-\Omega(\lambda)}$.*

Proof. Sketch. Let $Y_i = 1$ if the i -th run is correct, so the Y_i are independent with $\mathbb{E}[Y_i] \geq \frac{1}{2} + \delta$. The majority is wrong only if $\sum_i Y_i \leq t/2$, i.e. the empirical mean falls at least δ below its expectation. The Hoeffding bound for sums of independent $[0, 1]$ -valued variables gives $\Pr[\frac{1}{t} \sum_i Y_i \leq \frac{1}{2}] \leq \Pr[\frac{1}{t} \sum_i Y_i - \mathbb{E} \leq -\delta] \leq e^{-2\delta^2 t}$. Since each run is PPT and $t = \text{poly}(\lambda)$, the repeated algorithm is PPT, and $e^{-2\delta^2 t}$ is negligible for $t = \Theta(\lambda/\delta^2)$ by Example 11.34. $\square$

With efficiency pinned down, we can state what it means for a computational task to be hard. The guiding principle is the Cobham–Edmonds thesis: “efficiently computable” = “computable in polynomial time”. Its negation gives infeasibility.

Definition 11.42 (Computational infeasibility). A computational task parameterised by the security parameter λ is (*computationally*) *feasible* if there is a PPT algorithm solving it (with overwhelming success probability). It is *infeasible* if for *every* PPT algorithm $\mathcal{A}$ the probability that $\mathcal{A}$ solves it is negligible in λ :

$$\Pr[\mathcal{A} \text{ solves the task on parameter } \lambda] = \text{negl}(\lambda),$$

where the probability is over the task’s randomness and $\mathcal{A}$ ’s coins. Equivalently, no efficient algorithm does noticeably better than the trivial strategy.

Remark 11.43 (Reading a hardness assumption). Cryptographic assumptions are statements of this form. For instance, the discrete-logarithm assumption in a family of groups $\{\mathbb{G}_\lambda\}$ of order $q(\lambda)$ asserts: for every PPT $\mathcal{A}$,

$$\Pr[\mathcal{A}(g, g^x) = x \mid g \leftarrow \text{generator}, x \leftarrow \mathbb{Z}/q\mathbb{Z}] = \text{negl}(\lambda).$$

The entire edifice of Halo 2 and Orchard rests on such infeasibility statements, which one makes true in practice by choosing λ (and hence the group size $\approx 2^\lambda$) large enough that the best known attack—running in time $\Theta(\sqrt{q}) = \Theta(2^{\lambda/2})$ by the birthday/rho phenomenon of Remark 11.21—is astronomically slow. The asymptotic language of this section is what permits the terms “negligible” and “polynomial” without committing to a particular λ ; the concrete-security viewpoint then instantiates λ (e.g. $\lambda = 128$) to read off actual bit-counts.

11.10 The security parameter

This section closes by making explicit the parameter that threads through every preceding definition.

Definition 11.44 (Security parameter). The *security parameter* is a positive integer $\lambda \in \mathbb{N}$ that one supplies to all algorithms of a cryptographic scheme, conventionally in unary as the string 1^λ (so that “polynomial in the input length” coincides with “polynomial in λ ”). It governs simultaneously:

- the *sizes* of objects (key lengths, group order $\approx 2^\lambda$, field size), each $\text{poly}(\lambda)$;
- the *running time* of honest algorithms, which must be $\text{poly}(\lambda)$; and
- the *security level*: every PPT adversary’s advantage must be $\text{negl}(\lambda)$.

Remark 11.45 (Why unary, and what λ buys). Passing 1^λ (a string of λ ones, of length λ) rather than the integer λ in binary (length $\log_2 \lambda$) is a deliberate convention: it forces "polynomial-time in the input length" to mean polynomial in λ itself, so an honest key generator may take time $\text{poly}(\lambda)$ to produce λ -bit keys. As λ grows, two things happen in lockstep: honest parties pay a polynomial cost, while every adversary's success probability falls negligibly and (concretely) the best attack's running time grows exponentially. Asymptotic security is the statement that this gap holds for all sufficiently large λ ; the practitioner then fixes a single value (commonly $\lambda = 128$, targeting 2^{128} -operation attacks) and verifies that the concrete sizes—a ~ 255 -bit Pallas/Vesta scalar field, hash outputs of ≥ 256 bits per the birthday bound of Proposition 11.20, and sampling slack of ~ 128 extra bits per Proposition 11.25—deliver it. Every "negligible" in the remainder of this monograph is implicitly a function of this λ .